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STUDY OF THE ROLE OF GAS IN THE FORMATION OF SMBH SEED, THROUGH SIMULATIONS OF NSC WITH GAS EMBEDDED

TESIS PARA OPTAR AL GRADO DE MAGÍSTER EN CIENCIAS, MENCIÓN ASTRONOMÍA

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Abstractd

Una frase de dedicatoria,
pueden ser dos líneas.

Saludos

Acknowledgments

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Chapter 1

Introduction

One of the most enigmatic phenomena in astronomy is the existence of black holes. Since our understanding of the universe was in principle through the observation of light, the discovery of objects with no emission is, at least, challenging. This lack of emission can be understood from a fundamental property of astronomical objects, the escape velocity. For any body enough massive, is possible to measure their gravitational acceleration:

$$\mathbf{a} = \frac{GM}{r^2} \ , \tag{1.1}$$

where M is the mass and r the distance from the object. With this data is possible to calculate the velocity for escaping from their gravity, the escape velocity:

$$v_{\rm esc} = \sqrt{\frac{2GM}{r}} \ . \tag{1.2}$$

When an object is dense enough such that the escape velocity of their surface crosses the threshold of the speed of light, is classified as a black hole. However, black holes are not totally invisible, as their effect on the surrounding is detected.

Black holes are classified according to their mass $M_{\rm BH}$. Stellar-mass black holes (BHs), typically with masses up to $\sim 100\ M_\odot$, are thought to form from the collapse of massive stars, either through supernova explosions or via direct collapse (Belczynski et al., 2010; O'Connor and Ott, 2011). On the other side are supermassive black holes (SMBHs), exhibiting masses in the range 10^6–$10^{10}\ M_\odot$ (Natarajan and Treister, 2009; Woods et al., 2019). However, the principal mechanism of formation is an area of active research in astronomy.

1.1. Brief history: from Quasars to SMBHs

By the decade of 1960, there was evidence accounting for star-like objects with radio sources. In 1962, three of the brightest radio sources were identified in the northern sky by Thomas Matthews and Allan Sandage: 3C 48, 3C 196 and 3C 286 (Garwin and Lincoln, 2019). The spectrum of 3C 48 was characterized by broad emission lines, and from photometry the object showed to be variable in a period of months, and also an excess of ultraviolet emission, compared with normal stars (Garwin and Lincoln, 2019; Shields, 1999). Those objects became known as quasi-stellar radio sources (QSRS), quasi-stellar sources (QSO), or quasars (Shields, 1999). In 1963, Greenstein and Matthews (1963) identified on 3C 48 emission lines of Mg II, [Ne III], [Ne V], and [O II], with a redshift of $z = 0.3675$. Interpreting this object as a galaxy core, they estimated a distance of 1.1×10^9 pc, and absolute magnitude (after redshift correction) of -26.3, making it at least ten times brighter than the brightest known elliptical galaxies. By the same year, Schmidt (1963) reported the observation of 3C 273. Schmidt noticed that the spectrum presented emission lines from hydrogen, shifted to longer wavelengths. He identified four emission bands as Balmer lines, $H\beta$, $H\gamma$, $H\delta$ and $H\varepsilon$, with a redshift of $z = 0.158$.

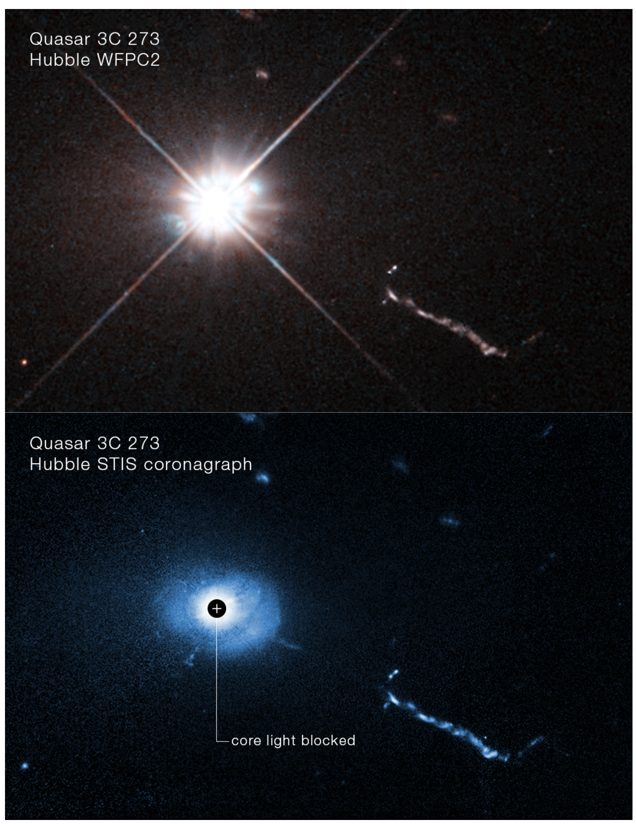

Figure 1.1: Image of 3C 273 from Hubble data. Credits: https://science.nasa.gov/asset/hubble/quasar-3c-273-2/.

If these objects were stars in our Galaxy, the large redshift would have to be gravitational in origin, requiring extremely high densities. Schmidt (1963) estimated a radius on the order of ~ 10 km for such a compact object. The second explanation, which Schmidt considered the most direct and least objectionable, was that 3C 273 was the nucleus of a distant galaxy with a cosmological redshift, with a relative velocity of 47 400 km/sec, and a distance of ~ 500 Mpc, and a diameter of $\lesssim 1$ kpc.

The first suggestion that quasars are fueled by SMBHs came with the work of Salpeter (1965). Salpeter proposed that extremely massive objects could accrete interstellar gas. Given an object of mass M, velocity v relative to a gas of density n (H atoms per cm^3), he defined a characteristic radius $s_0 = GM/v^2$ (from Hoyle and Lyttleton (1939)). Then, an accretion rate is defined:

$$\frac{dM}{dt} = 2\pi\alpha s_0^2 n v = \frac{\alpha M}{t_0} , \tag{1.3}$$

where α is a parameter that depends on the specific heat γ of the gas. Salpeter also estimated that for the total mass being accreted per unit time, a fraction f of 0.05–0.2 can escape as energy to infinity, while a fraction $1 - f > 0.8$ contribute to the increase in mass of the object. The optical luminosity is estimated as:

$$L_{\rm opt} = \left(\frac{25}{v}\right)^3 n \frac{\alpha}{0.25}\frac{f_{\rm opt}}{0.03}\left(\frac{M}{10^7}\right)^2 \times 2\times 10^9 L_\odot , \tag{1.4}$$

where $L_\odot$ is the sum's luminosity and v is in km/s. In solar units, $L_{\rm opt}/M$ can grow as large as 3×10^4, then radiation pressure lowers α, such that $L_{\rm opt}/M$ keeps near constant.

The model was further developed and popularized by Lynden-Bell (1969), who argued that the remnants of quasars should be common in the centres of nearby galaxies. He proposed that most quasars should have consumed their fuel, on a timescale of 10^{10} years (the age of the universe). Those dead quasars should then be common in galactic centers. This is explained by the fact that as quasars consume the accreted gas, the Schwarzschild radius (the event horizon, a sphere of radius $r = 2GM/c^2$ where the escape velocity equals the speed of light) becomes larger, consuming larger amounts of material. Also, in the past galaxies have larger amounts of gas than now.

One of the most important confirmations came with the discovery of a SMBH in the center of the Milky Way. In 1974, a compact radio source at the center of our galaxy, Sagittarius A* (Sgr A*), was discovered (Balick and Brown, 1974). In the late 1990s, Ghez et al. (1999) studied the proper motions of 90 stars near the center. They found that the movements indicated that a dark mass of $2.6(\pm 0.2)\times 10^6\ M_\odot$ was confined to a volume of 10^{-6} pc^3, coincident with Sgr A*. They concluded that our Galaxy harbours a SMBH at its center. The definitive proof came years later. In 2017, the Event Horizon Telescope (EHT; Doeleman et al. (2009)), an interferometer based on Very Long Baseline Interferometry (VLBI), observed Sgr A* (Collaboration et al., 2022). They found that images of Sgr A* were consistent with a Kerr black hole, with a mass of $\sim 4\times 10^6\ M_\odot$, and a bright ring with a diameter of 51.8 ± 2.3 μas, and a dim interior (see Figure 1.2).

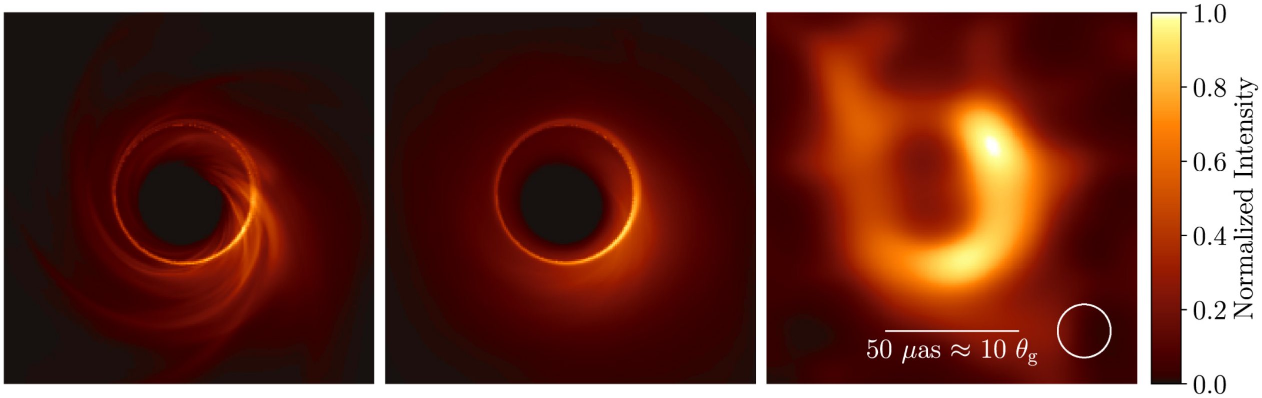

Figure 1.2: Simulated images of Sgr A*. The left panel is a snapshot of a numerical simulation of Sgr A*, and middle panel is the average of snapshots of the simulation. On the right panel, a representative image reconstruction, using the synthetic visibilities from the simulation. Credits: Collaboration et al. (2022).

1.2. Formation channels for SMBHs

The accumulated evidence for the existence of SMBHs suggests that they lie in the mass range of $\sim 10^6$–10^{10} $M_\odot$ (Gillessen et al., 2009; Latif and Ferrara, 2016; Natarajan and Treister, 2009; Volonteri, 2010; Woods et al., 2019). The origin of such dense and massive BHs is an active research topic in astronomy, as its principal formation mechanism remains unknown (Woods et al., 2019).

The challenge intensified with the discovery of quasars at redshifts $z > 6$. Early surveys, such as the Canada-France High-z Quasar Survey (CFHQS), identified the first quasars beyond $z = 6$ (Willott et al., 2007). Similarly, Jiang et al. (2015) reported eight quasars at $z \sim 6$ from the Sloan Digital Sky Survey (SDSS). More recently, Yang et al. (2026) reported the discovery of 31 new quasars at $6.6 < z < 7.8$ (Yang et al., 2026) from the Euclid Wide Survey. Among them, EUCL J1729 is the most distant quasar discovered to date, with a redshift of $z \sim 7.77$, surpassing the quasar reported by Mortlock et al. (2011) at $z \sim 7.085$, and J0313-1806 at $z \sim 7.6$ (Wang et al., 2021), which hosts a SMBH of $\sim 1.6 \times 10^9$ $M_\odot$. At $z \sim 7.77$, the age of the Universe was ~ 662 Myr (Yang et al., 2026), raising the natural question: how could SMBHs have formed at such early times?

To understand the formation pathways of SMBHs, it is important to know how they are seeded. Regan and Volonteri (2024) make the distinction between light and heavy seeds, where the former are objects with mass < 1000 $M_\odot$, and the latter are objects with mass > 1000 $M_\odot$. They estimate that light seeds accreting gas to a "standard" radiative efficiency of $\varepsilon_r \sim 0.1$, takes ~ 1 Gyr to grow up to a mass of $\sim 10^9$ $M_\odot$. However, those seeds are unlikely to grow efficiently enough. Alvarez et al. (2009) showed that for black hole remnant of Population III stars photoionization and heating affect the inflow of gas, resulting in negligible mass growth of the seed, then being unlikely to produce the observed quasars at $z > 6$. In agreement, another study of a simulation of 15000 black holes remnant of Population III stars, over 300 Myr, showed that less than 100 BH grew by more than 10^{-3} of their initial masses, most of them coming from dense regions (Smith et al., 2018). Heavy seeds, however, can be formed by runaway collisions of dense star clusters (Escala, 2021; Portegies Zwart et al., 2004), or the formation of a super massive star (SMS) of $\gtrsim 10^5\ M_\odot$ (Woods et al., 2017).

Although heavy seeds are more likely to grow into SMBHs, they require rarer environmental conditions, resulting in a lower number density. For a redshift of $z \sim 10$, the heaviest seeds, with masses up to $\sim 10^5\ M_\odot$, have number densities of $\sim 10^{-10}$–10^{-5} cMpc^{-3}, while other heavy seeds have number densities of $\sim 10^{-10}$–10^{-1} cMpc^{-3}. In contrast, light seeds are more common, with a number density of $\sim 10^{-1}$–10^{1} cMpc^{-3} (Regan and Volonteri, 2024).

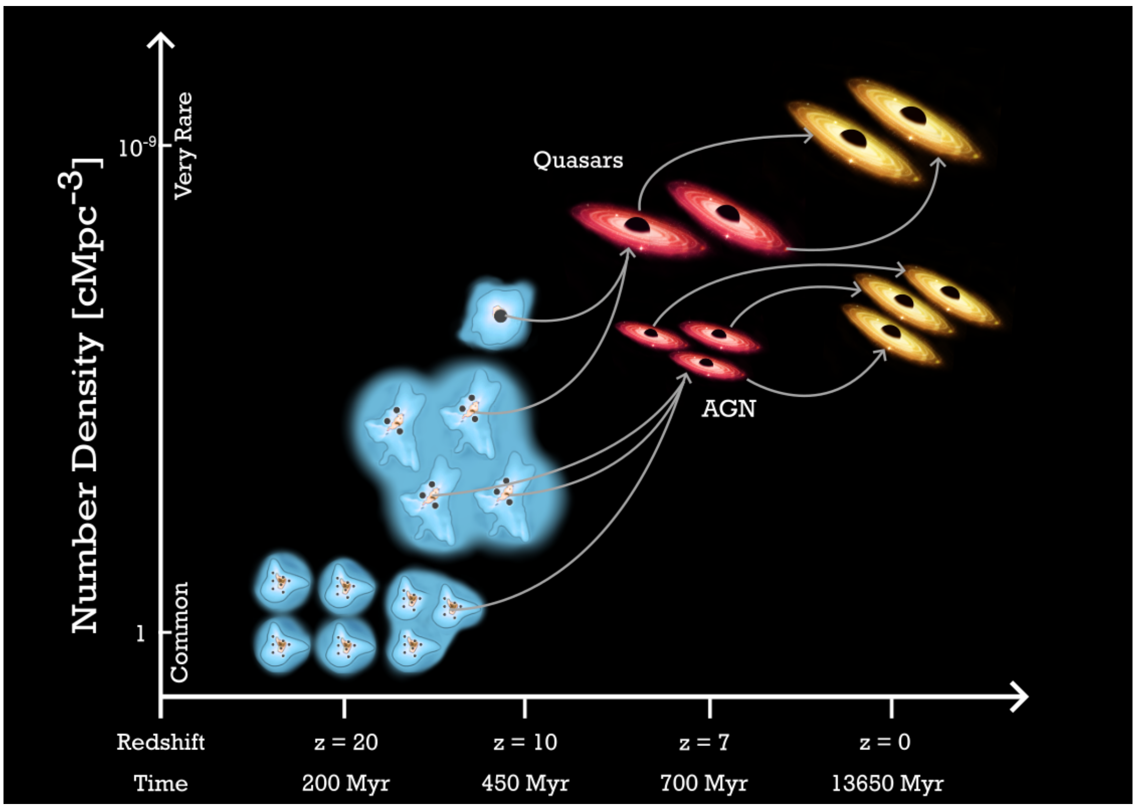

Figure 1.3: Number density in cMpc^{-3} (y-axis) vs redshift (x-axis). The y-axis, from top to bottom, goes from very rare ($\sim 10^{-9}$) to common ~ 1. Credits: Regan and Volonteri (2024).

There are many pathways for the formation of SMBHs (Volonteri, 2010). In the following, we describe three channels: (i) Population III stars remnant, (ii) direct collapse black hole, and (iii) runaway stellar collisions in dense star clusters.

1.2.1. Population III stars

One of the pathways of SMBH formation is through remnants of Population III stars. Simulations have shown that Population III stars form in halos of 10^5–10^6 $M_\odot$, and since fragmentation is typically suppressed under these conditions, they form in isolation at $z \sim 17$ (O'Shea and Norman, 2007, 2008). After ~ 3 Myr, those stars can end in a supernova explosion or via direct collapse, leaving a BH remnant of $\sim 10^2$ $M_\odot$ (Fryer et al., 2001). However, their subsequent growth into the SMBH mass range seems unlikely. In an optimistic scenario, Vesperini et al. (2010) showed that the gas ejected by first-generation stars can explain the chemical abundances in second generation (SG) stars, and that this dense gas can lead to significant accretion onto a pre-existing BH at the Eddington rate, increasing the mass by up to a factor of 100 ($\sim 10^4$ $M_\odot$). They estimated that a BH mass of $\gtrsim 10^4$ $M_\odot$ is required for the wind produced by the black hole to expel the accreting gas. However, simulations show that a 100 $M_\odot$ BH accreting metal-free gas exhibits intermittent accretion, which is on average reduced to $\sim 32\%$ of the Eddington rate (Milosavljević et al., 2009). In consequence, the mass increase is reduced to a factor of $\sim$ 2–5 (Vesperini et al., 2010). Other simulations also report inefficient accretion onto Pop III BH remnants, due to the erratic dynamics of these BHs within their host clusters (Pfister et al., 2019; Smith et al., 2018). However, if Pop III stars form in dense clusters rather than in isolation, runaway collisions can lead to the formation of heavier seeds ($\sim 10^3$–10^4 $M_\odot$), as we discuss later.

1.2.2. Direct collapse of primordial gas clouds

Another pathway is the formation of a heavy seed of $\sim 10^5$ $M_\odot$, via the direct collapse of primordial gas clouds. A set of simulations performed by Bromm and Loeb (2003) studied the evolution of a gas cloud with a total mass of 10^8 $M_\odot$ (including dark matter and baryons), varying the spin parameter λ, the inclusion of H_2 cooling, and the presence of a photo-dissociating Lyman-Werner (LW) background. They found that when H_2 cooling is suppressed, the gas collapses towards the centre via atomic hydrogen cooling, without fragmenting. Their simulations showed that $\gtrsim 10^6$ $M_\odot$ of the gas condenses into a region of $\lesssim 1$ pc, after which a supermassive star is likely to form. This SMS could subsequently collapse into a BH, retaining approximately 90% of its mass (Baumgarte and Shapiro, 1999; Shibata and Shapiro, 2002). However, the presence of H_2 cools the gas efficiently, leading to the formation of Population III stars. Their short lives (~ 3 Myr) produce supernova events (SNe) that enrich the gas, favouring subsequent star formation. To suppress H_2 cooling, the collapsing cloud must be exposed to a strong Lyman-Werner flux, typically provided by a neighbouring star-forming halo. This seeding mechanism requires very specific conditions, making it difficult to explain the observed number density of quasars at $z > 6$. Although Shang et al. (2010) found that the number of halos capable of suppressing H_2 formation may be larger, the subsequent mass growth of a $\sim 10^5$ $M_\odot$ seed faces significant challenges in sustaining the necessary accretion rates, as demonstrated by Chon et al. (2021).

1.2.3. Runaway collisions in dense star clusters

A promising channel is runaway collisions in nuclear star clusters (NSC)

1.3. Research Objectives and Thesis Outline

Chapter 2

A scenario for SMBH formation

In this section we describe the scenario proposed by Escala (2021) extended to include an external potential. The scenario explores the role of collisional and relaxation timescales ($t_{\rm coll}$ and $t_{\rm relax}$) in determining the global stability of stellar systems.

As stars move in a cluster, gravitational interactions redistribute kinetic and potential energy through two-body encounters. Strong encounters can lead to the formation or disruption of binaries, the ejection of stars, or stellar mergers when stars physically overlap (Binney and Tremaine, 2008; Spitzer, 1987). Mass segregation drives massive stars toward the centre, increasing the central density and accelerating core collapse. When the density becomes sufficiently high, stellar collisions become frequent, potentially triggering a runaway phase that can lead to a global instability (Escala, 2021).

Weak encounters produce small, gradual changes in stellar velocities, but their cumulative effect leads to two-body relaxation, which drives mass segregation and can bring the system toward a state of partial energy equipartition (Bianchini et al., 2016; Trenti and van der Marel, 2013). The relevant timescales for describing these processes are the collisional timescale and the relaxation timescale.

The collisional timescale $t_{\rm coll}$ provides an estimate of the collision rate between particles in a system. For a stellar system of identical stars, a first estimate of the collision rate $1/t_{\rm coll}$ is given by Binney and Tremaine (2008):

$$\frac{1}{t_{\rm coll}} = 16\sqrt{\pi} n \sigma r_\star^2 (1+\Theta) \; , \tag{2.1}$$

where $r_\star$ is the stellar radius, σ is the velocity dispersion of the system, n is the number density of stars, and

$$\Theta \equiv \frac{v_\star^2}{4\sigma^2} = 9.54 \frac{m}{M_\odot} \frac{R_\odot}{r_\star} \left(\frac{100 \text{ km s}^{-1}}{\sigma} \right)^2 , \tag{2.2}$$

is the Safronov number, with m the mass of each star and $v_\star = \sqrt{2Gm/r_\star}$ the escape speed from the stellar surface. This derivation assumes a Maxwellian velocity distribution with dispersion σ. Equation (2.1) gives the global collision rate for the system, from typical stellar properties (m, $r_\star$, and σ).

The other relevant timescale is the relaxation time. As a system evolves, two-body relaxation can alter its initial conditions. Close encounters and mass segregation redistribute energy and mass, changing the system's initial global properties (e.g., radius and velocity dispersion). Although strong encounters can significantly change a star's velocity, they are less frequent than weak encounters. The relaxation timescale $t_{\rm relax}$ is the time required for a typical star to change its velocity significantly through the cumulative effect of weak encounters. From Binney and Tremaine (2008),

$$t_{\rm relax} = n_{\rm relax} t_{\rm cross}, \tag{2.3}$$

where $t_{\rm cross}$ is the crossing time of a typical star, and $n_{\rm relax}$ is the number of crossings needed for the cumulative change in velocity Δv to become comparable to the star's velocity ($dv = \sum \delta v \sim v$). For a system of N identical stars, $n_{\rm relax} \sim 0.1N/\log N$ (see Appendix). Thus,

$$t_{\rm relax} = \frac{0.1N}{\log(N)} t_{\rm cross}. \tag{2.4}$$

2.1. Critical mass

The relevance of the relaxation and collision timescales depends on the characteristic timescale of the system, $t_{\rm H}$, typically taken as the age of the system. If $t_{\rm coll} \lesssim t_{\rm H}$, collisions are expected to be dynamically relevant. For a virialized system of total mass M and radius R, Equation (2.1) can be rearranged (Escala, 2021):

$$t_{\rm coll} = \frac{1}{n\Sigma_0} \sqrt{\frac{R}{GM}} , \tag{2.5}$$

where $\Sigma_0 = 16\sqrt{\pi} r_\star^2 (1+\Theta)$ is the effective cross-section, and $\sigma = \sqrt{GM/R}$ for a virialized system. The Safronov number can be simplified to $\Theta \sim 9.54(m/M_\odot)(R_\odot/r_\star)$ if we assume $\sigma \sim 100$ km s^{-1}. Assuming a homogeneous density distribution as a first approximation, the number density is $n = 3M/(4\pi R^3 m)$. Imposing $t_{\rm coll} \leq t_{\rm H}$ and solving for the mass yields a critical mass (Escala, 2021; Vergara et al., 2023):

$$M \geq \left(\frac{4\pi m}{3\Sigma_0 t_{\rm H} G^{1/2}}\right)^{2/3} R^{7/3} = M_{\rm crit} \ . \tag{2.6}$$

The condition $t_{\rm coll} \leq t_{\rm H}$ is equivalent to $M \geq M_{\rm crit}$. Thus, $M_{\rm crit}$ is the mass of a virialized system of radius R and equal-mass stars whose collision timescale equals $t_{\rm coll} = t_{\rm H}$. Including gravitational focusing and an external potential of mass $M_{\rm ext} = qM$, the critical mass can be written as (see Appendix):

$$M_{\rm crit} = \left(\frac{\beta R^{7/2}}{2} + \sqrt{\frac{\beta^2 R^7}{4} + \frac{\alpha^3 R^3}{27}}\right)^{2/3} + \left(\frac{\beta R^{7/2}}{2} - \sqrt{\frac{\beta^2 R^7}{4} + \frac{\alpha^3 R^3}{27}}\right)^{2/3} - \frac{2\alpha R}{3} \ , \tag{2.7}$$

with

$$\alpha = \frac{9.54 \ m R_\odot v_{100}^2}{r_\star G M_\odot (1+q)^2} \tag{2.8}$$

$$\beta = \frac{\sqrt{\pi} m}{12 G^{1/2} t_H r_\star^2 (1+q)} \ . \tag{2.9}$$

2.2. Relaxation mass

Similarly, we can obtain a condition using the relaxation time. From Appendix A.4, including an external potential, the relaxation timescale (Equation (A.41)) yields

$$t_{\rm relax} = \frac{0.1 N (1+q)^3}{\ln(N(1+q)^2)} \frac{R^{3/2}}{\sqrt{GM}} \ . \tag{2.10}$$

Replacing $N = M/m$, and solving $t_{\rm relax} \leq t_{\rm H}$ for the characteristic radius R:

$$R \leq \left(\frac{t_{\rm H} m}{0.1(1+q)^3} \ln\left(\frac{M(1+q)^2}{m}\right)\right)^{2/3} \left(\frac{G}{M}\right)^{1/3} \ . \tag{2.11}$$

This equation gives a condition for the radius, such that the system evolves changing due to two-body relaxation. We can rewrite Equation (2.11) by solving for the mass,

$$M \leq M_{\rm relax} = \frac{m}{(1+q)^2} \exp\left(-2W_{-1}\left(\frac{-1}{2\lambda}\right)\right) , \tag{2.12}$$

where $W_{-1}(z)$ is the lower branch of the Lambert function, and $\lambda = (t_{\rm H}\sqrt{Gm})/(0.1(1+q)^2 R^{3/2})$. In such systems, two-body relaxation does not significantly alter the initial conditions over the characteristic timescale $t_{\rm H}$. If, in addition, $t_{\rm coll} \ll t_{\rm H}$, the system remains approximately in virial equilibrium. The case where $t_{\rm coll} \leq t_{\rm H}$ despite $t_{\rm relax} \geq t_{\rm H}$ corresponds to a different regime, which we discuss in the following section.

2.3. Global instability scenario

In this section, we analyze the scenario for supermassive black hole (SMBH) formation from NSCs proposed by Escala (2021). The critical mass $M_{\rm crit}$ from Equation (2.7) gives a threshold for the condition $t_{\rm coll} = t_{\rm H}$. For stellar clusters where $t_{\rm coll} \leq t_{\rm H}$ ($M \geq M_{\rm crit}$), the onset of runaway collisions is expected, eventually forming a BH at the centre. On the other hand, when $t_{\rm relax} \leq t_{\rm H}$ (which is equivalent to Equation (2.12), where $M_{\rm relax}$ is the limiting case), two-body interactions could expand the cluster, modifying its initial conditions. Escala (2021) proposed a new regime of global instability, given by $t_{\rm coll} \leq t_{\rm H} \leq t_{\rm relax}$. The two important aspects that describe this regime are: as $t_{\rm coll} \leq t_{\rm H}$, collisions occur on a timescale $t_{\rm H}$, as described above; the condition $t_{\rm coll} \leq t_{\rm relax}$ implies that two-body relaxation is unable to reverse the global collapse, and the onset of runaway collisions leads to the formation of a massive black hole (MBH). There is a second scenario that could form a central BH. We define an intermediate regime, given by clusters where $t_{\rm relax} \leq t_{\rm coll} \leq t_{\rm H}$. In this regime, the system could begin to expand on a timescale comparable to $t_{\rm relax}$. However, this does not necessarily prevent the onset of a collisional runaway in the core. Since the central density is significantly higher than the global average, the local collision time $t_{\rm coll,core}$ is considerably shorter than the global collision time $t_{\rm coll}$. If $t_{\rm coll,core} \leq t_{\rm relax,core}$, the core could contract and decouple dynamically from the expanding envelope, which could be compatible with a gravothermal instability scenario (Cohn, 1980; Spitzer, 1987). Then, the core could undergo runaway collisions, forming a central BH coexisting with a surrounding star cluster, while the outer stars, heated by two-body relaxation and dynamical encounters, expand. Finally, a stable regime is defined by $t_{\rm relax} \leq t_{\rm H} \leq t_{\rm coll}$. In this stable regime, clusters can expand due to two-body relaxation, which further decreases the collision rate. Although a BH can form via collisions in the core, most of the mass could remain in the stellar cluster.

The final fate of the CMO is conveniently explained in terms of the black hole formation efficiency, defined as $\varepsilon_{\rm BH} = M_{\rm BH}/M_{\rm CMO}$ (Escala, 2021; Vergara et al., 2023), where $M_{\rm BH}$ is the mass

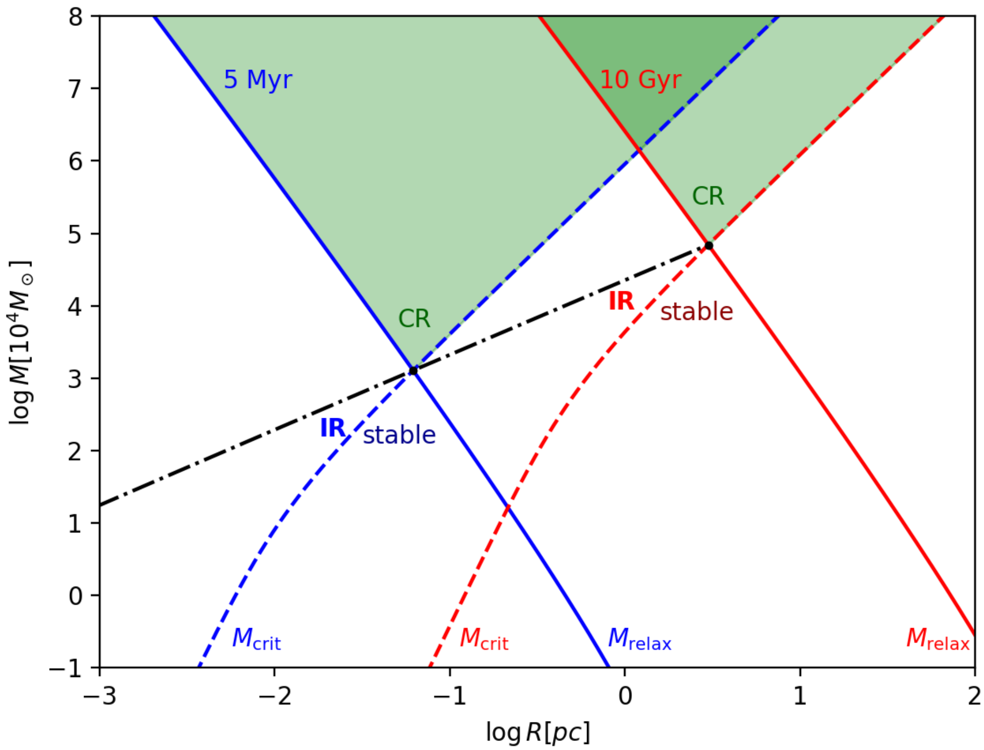

Figure 2.1: Collision and relaxation timescales in a mass versus radius diagram. Dashed lines correspond to the critical mass $M_{\rm crit}$, while solid lines correspond to the relaxation mass $M_{\rm relax}$, both calculated for solar-mass clusters. The colours represent two different characteristic timescales: blue for $t_{\rm H} = 5 \times 10^6$ yr, and red for $t_{\rm H} = 10^{10}$ yr. The black dash-dotted line represents $t_{\rm coll} = t_{\rm relax}$, with $t_{\rm H}$ varying from 10^2 yr to 10^{10} yr. Clusters above the black dash-dotted line satisfy $t_{\rm coll} \leq t_{\rm relax}$. For each color, the collisional regime (CR) is defined by $t_{\rm coll} \leq t_{\rm H} \leq t_{\rm relax}$ (green colored zone); the stable regime is where $t_{\rm relax} \leq t_{\rm H} \leq t_{\rm coll}$; and the intermediate regime (IR) fulfils $t_{\rm relax} \leq t_{\rm coll} \leq t_{\rm H}$.

of the central black hole, and $M_{\rm CMO} = M_{\rm BH} + M_{\rm NSC}$ is the mass of the central massive object, composed of a central BH and/or a surrounding nuclear stellar cluster. In Figure 2.1, critical mass and relaxation mass are displayed (dashed and solid curves, respectively), for solar-mass clusters. Two characteristic timescales are used, with $t_{\rm H} = 5 \times 10^6$ yr for blue curves, and $t_{\rm H} = 10^{10}$ yr for red curves. The stable and unstable regimes are separated at the intersection of the solid, dashed and black dot-dashed lines, where $t_{\rm coll} = t_{\rm relax} = t_{\rm H}$. The collisional regime (CR), characterized by $t_{\rm coll} \leq t_{\rm H} \leq t_{\rm relax}$, is in the zone where the mass of the system is simultaneously larger than the critical and the relaxation mass. In this regime, the system is unable to expand via two-body relaxation before collisions become relevant. Although binary heating can, in principle, oppose core contraction, in systems with masses larger than 10^7 $M_\odot$ binary heating cannot reverse core collapse, and the system can suffer a merger instability, which may lead to the formation of a central BH (Lee, 1987). The dissipative nature of collisions thus drives a net contraction of the cluster. Then, most of the mass could remain in the central BH, having efficiencies $\varepsilon_{\rm BH} \sim 1$. In contrast, in the stable regime the dynamics are dominated by two-body relaxation over collisions. In those systems, the mass occupies the region where $M \leq M_{\mathrm{crit}}$ and $M \leq M_{\mathrm{relax}}$ simultaneously. The expansion of the cluster moves it to the right in a M–R diagram, being possible to cross to the right side of the solid line. Although a central black hole could form, most of the mass remains in the stellar cluster, resulting in an NSC. Consequently, efficiencies are $\varepsilon_{\mathrm{BH}} \ll 1$. The intermediate regime (IR) corresponds to the region where $t_{\mathrm{relax}} \leq t_{\mathrm{coll}} \leq t_{\mathrm{H}}$. In the diagram, this is the area where $M_{\mathrm{crit}} \leq M \leq M_{\mathrm{relax}}$. As described above, the system could expand while the core undergoes a runaway collision process, forming a central BH. These systems eventually move to the right as their radius increases, crossing into the stable regime (right side of the dashed curve), with efficiencies spanning a wider range $\varepsilon_{\mathrm{BH}} \leq 1$. As noted by Escala (2021), the observed BH formation efficiency as a function of M_{CMO} shows a step-like form that clearly separates NSCs from SMBHs. A possible explanation for this observational gap is that systems in the intermediate regime may form IMBHs; however, an important fraction of the mass could remain in the surrounding stellar cluster. Consequently, their gravitational sphere of influence is small and challenging to resolve (Trenti, 2006), making them difficult to detect. This would naturally contribute to the lack of IMBH detections in NSC-dominated systems.

Chapter 3

Numerical Methods

In this chapter, we describe the numerical methods employed to model the coupled evolution of stars and gas in dense stellar clusters. We require three principal components: (i) an N-body code (`ph4`) to model stellar dynamics, (ii) a smoothed particle hydrodynamics (SPH) code (`Fi`) to simulate the gas, and (iii) a coupling scheme (`BRIDGE`) to handle their mutual gravitational interaction. The N-body and SPH codes used are accessed through the Astrophysical Multipurpose Software Environment (AMUSE; Zwart and McMillan (2018)), which provides a modular framework for solving multi-physics problems. We do not include gas accretion onto stars, as our focus is exclusively on the gravitational interaction between the two components.

3.1. The AMUSE Framework

The numerical codes used in this work are accessed through the Astrophysical Multipurpose Software Environment (AMUSE; Zwart and McMillan (2018)). The AMUSE project aims to provide a flexible and modular framework for solving multi-physics problems in astrophysics. It offers a variety of community codes, classified into four main domains: gravitational dynamics, hydrodynamics, stellar evolution, and radiative transfer (Figure 3.1). Some codes, such as GADGET-2 (Springel, 2005), can cover more than one domaine.

The structure of amuse is composed of three principal layers (Figure 3.2). The top layer is the user interface, where a Python script is written to set up and run a specific scientific problem. In this script, codes are initialized, and additional routines (e.g., for handling stellar collisions) can be implemented. This layer also contains the manager, whose primary functions are to provide data structures and methods for communication between the user script and the community codes, as well as to handle unit conversions. The middle layer provides the interfaces to the community codes. It facilitates bidirectional communication between the manager and the underlying code

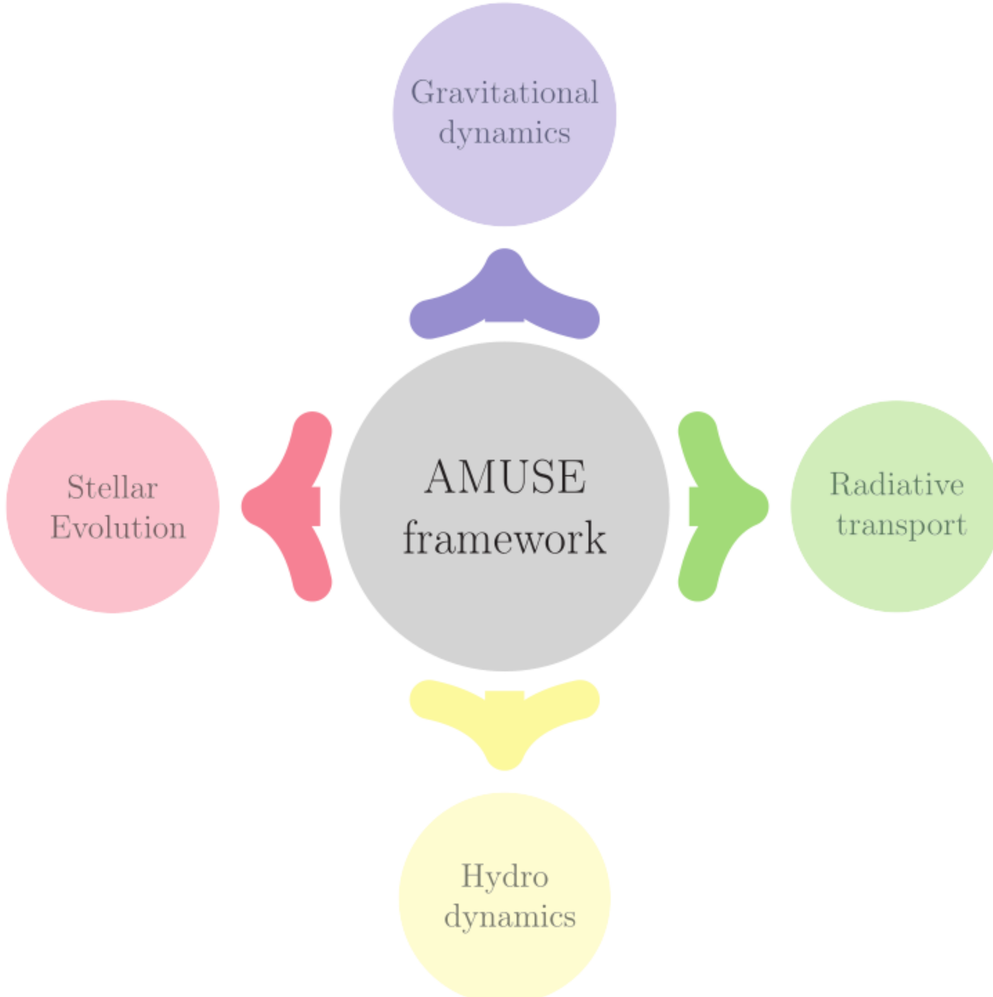

Figure 3.1: Principal domains in AMUSE framework. Multiple physical problems can be solved coupling codes from different domains. Credits: Zwart and McMillan (2018).

layer. Each interface consists of two components: a proxy, written in Python, which initializes the necessary functions and parameters, and a partner, written in the code's native language (e.g., C++, CUDA, or Fortran). The proxy and partner communicate via the Message Passing Interface (MPI). Finally, the bottom layer contains the community codes themselves.

3.2. N-body code: PH4

The proposed scenarios presented in Section 2 requires high numerical precision to accurately model close encounters and direct collisions. We therefore employ the `ph4` N-body code from the AMUSE framework (Zwart and McMillan, 2018). In this section, we include a description of the code, and the external routines implemented to handle collisions.

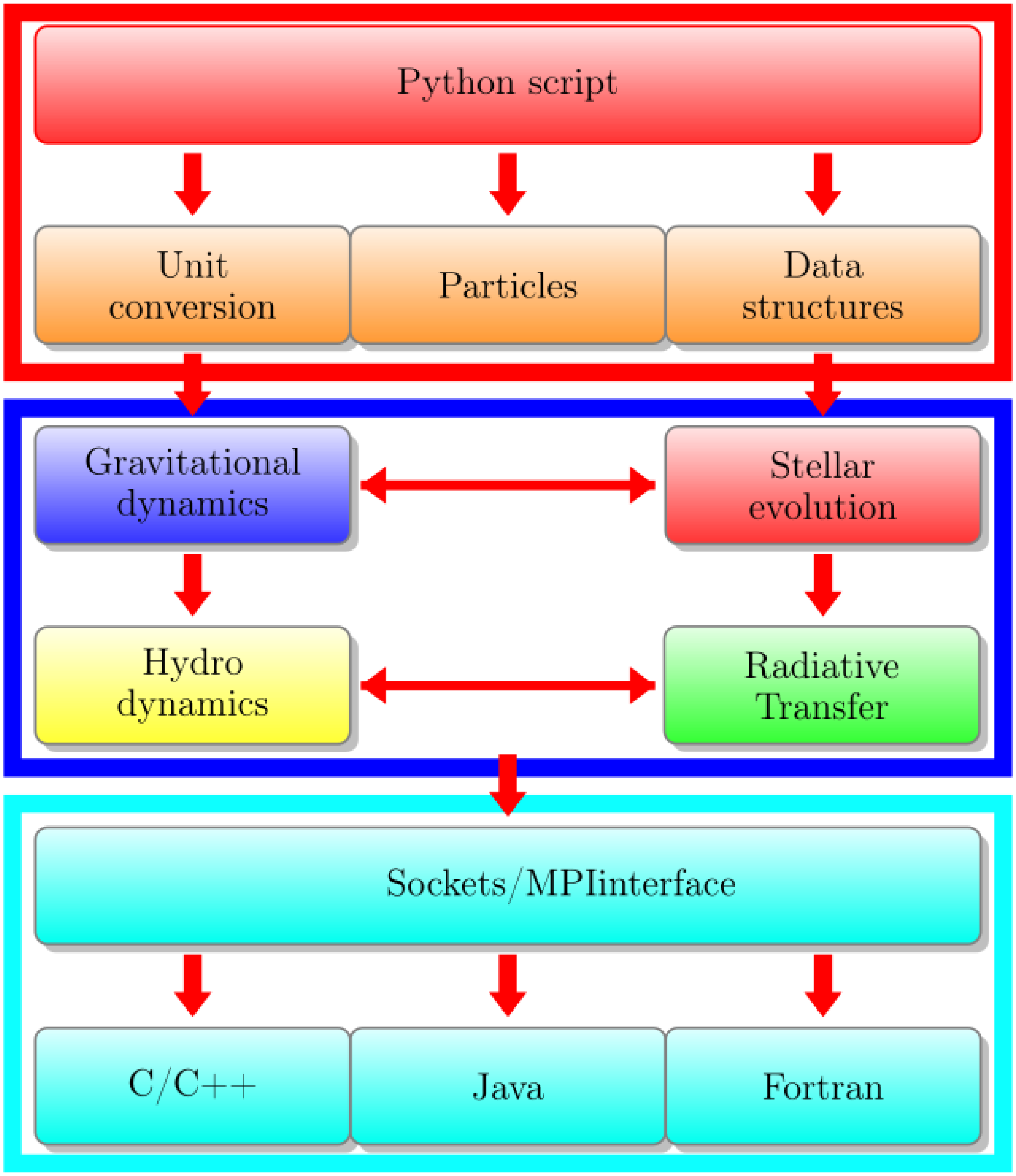

Figure 3.2: Schematic representation of the three-layer architecture of the AMUSE framework. The user interacts with the system via a Python script (top layer), which communicates through interfaces (middle layer) with the community codes (bottom layer). Credits: Zwart and McMillan (2018).

3.2.1. Hermite integrator

The `ph4` code uses a fourth-order Hermite integrator (Makino and Aarseth, 1992) to compute the positions and velocities of stars under gravity. This predictor-corrector method employs adaptive individual time steps, ensuring high accuracy in regions of strong gravitational interactions. Considering the particle i with its own time t_i, time step Δt_i, position $\mathbf{x}_i$, velocity $\mathbf{v}_i$, acceleration $\mathbf{a}_i$ and jerk $\dot{\mathbf{a}}_i$, all calculated at time t_i. The integration algorithm proceeds as follows:

1. Select the particle with the minimum $t_i + \Delta t_i$. Set the global time to $t = t_i + \Delta t_i$.
2. Calculate the predicted positions ($\mathbf{x}_p$) and velocities ($\mathbf{v}_p$) for all particles, using the current values for $\mathbf{x}$, $\mathbf{v}$, $\mathbf{a}$, $\dot{\mathbf{a}}$.
3. Calculate the acceleration ($\mathbf{a}_i$) and jerk ($\dot{\mathbf{a}}_i$) for particle i at time $t_i + \Delta t_i$ using the predicted positions and velocities.
4. Calculate the second and third time derivatives of acceleration ($\mathbf{a}_i^{(2)}$ and $\mathbf{a}_i^{(3)}$) using Hermite interpolation.
5. Apply corrections to the position and velocity of particle i.
6. Calculate and update the time step Δt_i.
7. Repeat the algorithm.

The interpolation uses the acceleration and jerk at the beginning and end of the time step, allowing the scheme to achieve fourth-order accuracy without explicitly computing higher-order derivatives. A detailed derivation of the algorithm and the full set of equations are provided in Appendix B. From Eqs. (B.2) and (B.3), the predicted positions and velocities are:

$$\mathbf{x}_{p,j} = \mathbf{x}_j + \mathbf{v}_j(t - t_j) + \frac{\mathbf{a}_{0,j}}{2}(t - t_j)^2 + \frac{\dot{\mathbf{a}}_{0,j}}{3!}(t - t_j)^3 \tag{3.1}$$

$$\mathbf{v}_{p,j} = \mathbf{v}_j + \mathbf{a}_{0,j}(t - t_j) + \frac{\dot{\mathbf{a}}_{0,j}}{2!}(t - t_j)^2 \tag{3.2}$$

Here, $\mathbf{x}_j$ and $\mathbf{v}_j$ are the position and velocity of particle j at time t_j. The current acceleration $\mathbf{a}_{0,j}$ and jerk $\dot{\mathbf{a}}_{0,j}$ are computed as:

$$\mathbf{a}_{0,j} = \sum_{k \neq j} G m_k \frac{\mathbf{r}_{0,jk}}{(\mathbf{r}_{0,jk}^2 + \varepsilon^2)^{3/2}} \tag{3.3}$$

$$\dot{\mathbf{a}}_{0,j} = \sum_{k \neq j} G m_k \left[\frac{\mathbf{v}_{0,jk}}{(\mathbf{r}_{0,jk}^2 + \varepsilon^2)^{3/2}} - \frac{3(\mathbf{v}_{0,jk} \cdot \mathbf{r}_{0,jk})\mathbf{r}_{0,jk}}{(\mathbf{r}_{0,jk}^2 + \varepsilon^2)^{5/2}} \right] \tag{3.4}$$

$$\mathbf{r}_{0,jk} = \mathbf{r}_k - \mathbf{r}_j \tag{3.5}$$

$$\mathbf{v}_{0,jk} = \mathbf{v}_k - \mathbf{v}_j \tag{3.6}$$

Using Hermite interpolation, the second and third time derivatives of the acceleration can be estimated. This estimation is subsequently used to correct the positions and velocities. The relevant expressions, derived in the appendix, are:

$$\mathbf{a}_{0,i}^{(2)} = \frac{-6(\mathbf{a}_{0,i} - \mathbf{a}_{1,i}) - \Delta t_i(4\dot{\mathbf{a}}_{0,i} + 2\dot{\mathbf{a}}_{1,i})}{\Delta t_i^{\,2}} \tag{3.7}$$

$$\mathbf{a}_{0,i}^{(3)} = \frac{12(\mathbf{a}_{0,i} - \mathbf{a}_{1,i}) + 6(\dot{\mathbf{a}}_{0,i} + \dot{\mathbf{a}}_{1,i})\Delta t_i}{\Delta t_i^{\,3}} \tag{3.8}$$

The final corrected position and velocity of particle i are then given by:

$$\mathbf{x}_i(t_i + \Delta t_i) = \mathbf{x}_{p,i} + \frac{\mathbf{a}_{0,i}^{(2)} \Delta t_i^{\,4}}{24} + \frac{\mathbf{a}_{0,i}^{(3)} \Delta t_i^{\,5}}{120} \tag{3.9}$$

$$\mathbf{v}_i(t_i + \Delta t_i) = \mathbf{v}_{p,i} + \frac{\mathbf{a}_{0,i}^{(2)} \Delta t_i^{\,3}}{6} + \frac{\mathbf{a}_{0,i}^{(3)} \Delta t_i^{\,4}}{24} \tag{3.10}$$

These corrections incorporate terms up to fourth order. The main advantage of this method is that it only requires the explicit calculation of the acceleration and jerk at each step, with higher-order derivatives being obtained through the interpolation scheme.

3.2.2. Block time steps

While the Hermite integrator computes individual time steps for each particle, evolving particles one by one is computationally expensive. `ph4` therefore groups particles into block time steps, which are powers of two of the N-body time unit. This allows particles with the same time step to be integrated simultaneously, improving efficiency while maintaining synchronization. The choice of time step directly affects the accuracy of the integration, the conservation of energy, and the total computational cost.

In the block scheme, time steps are set in two steps. First, a dynamic time step is calculated. Then, the time step is assigned using a block scheme. For the dynamical time step, the standard formula (B.35) is used:

$$\Delta t_i = \sqrt{\eta \frac{|\mathbf{a}_{1,i}||\mathbf{a}^{(2)}_{1,i}| + |\dot{\mathbf{a}}_{1,i}|^2}{|\dot{\mathbf{a}}_{1,i}||\mathbf{a}^{(3)}_{1,i}| + |\mathbf{a}^{(2)}_{1,i}|^2}} \tag{3.11}$$

As can be noted, $\mathbf{a}_{1,i}$ and $\dot{\mathbf{a}}_{1,i}$ are calculated in each step, while $\mathbf{a}^{(3)}_{1,i}$ is constant. Only $\mathbf{a}^{(2)}_{1,i}$ needs to be estimated:

$$\mathbf{a}^{(2)}_{1,i} = \mathbf{a}^{(2)}_{0,i} + \Delta t_i \mathbf{a}^{(3)}_{1,i} \tag{3.12}$$

When the simulation begins, the initial time steps are calculated using (B.37):

$$\Delta t = \eta_s \frac{|\mathbf{a}|}{|\dot{\mathbf{a}}|} \tag{3.13}$$

To avoid the computational cost of evolving particles individually, `ph4` assigns particles to discrete block time steps that are powers of two of the N-body time unit. Given the initial time step $\Delta t_{0,j}$ (calculated from Equation (3.13)), the N-body time τ, and an integer n is adjusted to the condition:

$$0.5 < \frac{\tau 2^n}{\Delta t_{0,j}} < 1 \tag{3.14}$$

Then, the initial time step of particles is then set to $\tau 2^n$, with n being able to be positive or negative. As the model evolves, the new time step is computed using (3.11). The time step is halved if the calculated time step is smaller than the current time step, and doubled if it is larger than twice the current time step. The advantage of this time step criterion is that the time step in the code is equal to or lower than that required by the dynamics. Moreover, since particles share the same time steps, they can be integrated simultaneously. Additionally, all particles can be synchronized, as their times are multiples of powers of two of the same time unit.

3.2.3. Collisions and mergers

Stellar collisions are detected through a two-stage filter. First, `ph4` checks whether two particles are closer than the sum of their radii: $|\Delta\mathbf{x}| < r_i + r_j$. A second filter then verifies close encounters using the criterion $\varepsilon|\Delta\mathbf{x}||\Delta\mathbf{v}| > \Delta\mathbf{x}\cdot\Delta\mathbf{v}$, where $\varepsilon = 0.001$. This avoids false detections caused by hyperbolic encounters with predominantly tangential relative velocities. Once a collision is detected, the flag `collision_detection` in the code is activated, and a custom handling routine is executed. In our implementation, both particles involved in the collision are identified as particles i and j, for the most and less massive between them, respectively. For particle i their properties are modified, setting its mass to the sum of the masses, its position to the center of mass, its velocity to the center-of-mass velocity, and its radius to conserve mass density:

$$m_{i,\text{new}} = m_i + m_j \tag{3.15}$$

$$\mathbf{x}_{i,\text{new}} = \frac{m_i\mathbf{x}_i + m_j\mathbf{x}_j}{m_i + m_j} \tag{3.16}$$

$$\mathbf{v}_{i,\text{new}} = \frac{m_i\mathbf{v}_i + m_j\mathbf{v}_j}{m_i + m_j} \tag{3.17}$$

$$r_{i,\text{new}} = r_i\left(\frac{m_i + m_j}{m_i}\right)^{1/3}. \tag{3.18}$$

$$\tag{3.19}$$

Finally, particle j is removed from the simulation.

3.2.4. Escaping stars

After each time step, potential escapers are identified through a multi-stage detection algorithm. To maintain mass and energy conservation, escaped stars are not removed from the simulation. Instead we add an extra attribute, `escaped` (all stars begin with `particles.escaped = False`), to track bound and unbound stars for subsequent analysis.

To detect escaping stars, the first step is to define an appropriate length scale. In the initialization, the escape scale is defined as the minimum between ten times the virial radius and the Lagrangian radius containing the total mass:

$$r_{0,\text{esc}} = \min(10\times r_{\text{vir}}, r_{100}) \tag{3.20}$$

where r_{100} represents the Lagrangian radius containing the total stellar mass. In subsequent iterations, the escape distance is updated as:

$$r_{\text{esc}} = \max(r_{\text{esc}}, 10\times r_{b,\text{vir}}) \tag{3.21}$$

where $r_{b,\mathrm{vir}}$ denotes the virial radius of bound stars. The escape detection algorithm, executed at the end of each time step, proceeds as follows:

1. Compute the core center and identify stars beyond the escape radius.
2. Select gravitationally unbound stars.
3. Identify stars with outward radial motion.
4. Verify minimal gravitational influence on nearest neighbor.

The core center location is determined using a density-weighted scheme following Sweatman (1993). For each star, the distance to its 7th nearest neighbor ($r_{i,7}$) is computed, and the density center is calculated as:

$$\mathbf{x}_c = \frac{\sum_{i=1}^{N} \mathbf{x}_i / r_{i,7}^3}{\sum_{i=1}^{N} 1 / r_{i,7}^3} \tag{3.22}$$

This formulation resembles a center of mass calculation but uses local stellar density instead of mass. Stars satisfying the condition

$$|\mathbf{x}_i - \mathbf{x}_c| > r_{\mathrm{esc}} \tag{3.23}$$

are identified as potential escapers. Gravitational unbinding is evaluated using the total energy $E_i = E_{p,i} + E_{k,i}$, where $E_{p,i}$ is the gravitational potential energy (including contributions from both stars and gas) and $E_{k,i}$ is the kinetic energy. Stars with $E_i \leq 0$ are considered bound, while those with $E_i > 0$ are classified as unbound. Among unbound stars, outward-moving candidates are identified through the condition:

$$\Delta\mathbf{x}_i \cdot \Delta\mathbf{v}_i > 0 \tag{3.24}$$

where $\Delta\mathbf{x}_i = \mathbf{x}_i - \mathbf{x}_c$ and $\Delta\mathbf{v}_i = \mathbf{v}_i - \mathbf{v}_c$ represent position and velocity relative to the cluster center. Similar to (3.22), the velocity of the core center is computed as

$$\mathbf{v}_c = \frac{\sum_{i=1}^{N} \mathbf{v}_i m_i / r_{i,7}^3}{\sum_{i=1}^{N} m_i / r_{i,7}^3} \tag{3.25}$$

The final validation step quantifies the gravitational influence on neighboring stars. For each candidate escaper, the relative potential energy ($e_{p,i}$) and kinetic energy ($e_{k,i}$) with respect to its nearest neighbor are computed. The energy ratio is defined as:

$$X_E := \frac{|e_{p,i}|}{e_{k,i}} \tag{3.26}$$

Stars satisfying $X_E < 0.01$ are confirmed as escapers, ensuring minimal gravitational perturbation to neighboring stars. These stars are tagged as escapers (`particles.escaped = True`). Tagged escapers are excluded from subsequent dynamical calculations (e.g., core radius, density profiles), but they are not removed from the simulation, ensuring mass and energy conservation.

3.3. SPH code: Fi

The gas dynamics are modelled using the `Fi` code (Gerritsen et al., 1997), an MPI-parallel smoothed particle hydrodynamics (SPH) implementation within the AMUSE framework. We focus on the aspects most relevant to our simulations; a comprehensive description of the code can be found in (Pelupessy, 2005).

3.3.1. Barnes-Hut tree

A key feature of `Fi` is its ability to compute the self-gravity of the gaseous component. Particle-based codes offer a range of methods for gravitational force calculation. The most accurate approach is the direct summation of all pairwise interactions, which scales as $\mathcal{O}(N^2)$ and is computationally expensive for large N. While such precision is often necessary for collisional stellar dynamics (e.g., in our N-body component), the gas system in our simulations is treated as a collisionless fluid for the purpose of gravity calculation, allowing for faster, approximate methods, sacrificing accuracy for computational efficiency.

`Fi` employs the Barnes-Hut (BH) tree algorithm (Barnes and Hut, 1986). This method uses a hierarchical oct-tree spatial decomposition and a geometric opening criterion to approximate gravitational forces.

The tree is constructed by enclosing all particles within a root cell. If this cell contains more than one particle, it is subdivided into eight daughter cells of equal volume. This subdivision process repeats recursively for each cell containing multiple particles, until every cell holds at most one particle (a leaf node). Figure 3.3 illustrates this structure in two dimensions.

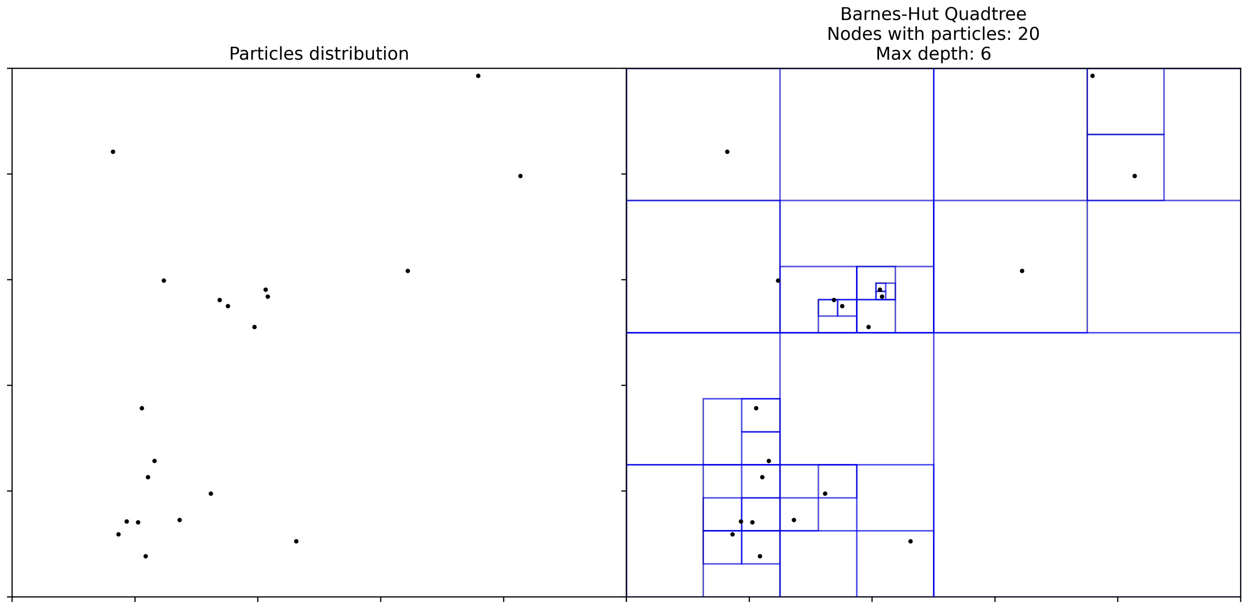

Figure 3.3: Representation of an oct tree

Once the tree is built, the gravitational force on a given particle is calculated by going through the tree from the root. For each node encountered (linked to a cell), a decision is made: either open the node to examine its children, or accept the node as a single interacting entity. The classic Barnes-Hut opening criterion is:

$$l/d < \theta \tag{3.27}$$

where l is the side length of the cubic cell, d is the distance from the target particle to the node's center of mass, and θ is an accuracy parameter. If this condition is met, the entire mass of the node is treated as a single point mass at its center of mass, providing an acceptable approximation when the node is sufficiently distant (i.e., appears small in angular size). The recursion stops upon reaching a leaf node, which by definition represents a single particle and is interacted with directly. This approximation reduces the computational cost of the force calculation from $\mathcal{O}(N^2)$ to $\mathcal{O}(N \log N)$ and adapts efficiently to clustered mass distributions, making it suitable for collisionless systems (Barnes and Hut, 1986).

The standard geometric criterion (Eq. 3.27), however, can fail in some circumstances (Salmon and Warren, 1994). Therefore, `Fi` implements a modified criterion following (Springel and Hernquist, 2002):

$$M_n l^4 < \alpha |\mathbf{a}| d^6 \tag{3.28}$$

where M_n is the total mass contained in the node, $\mathbf{a}$ is an estimate of the gravitational acceleration (typically from the previous timestep), and α is a parameter to control accuracy.

3.3.2. Smoothed Particle Hydrodynamics

The gas dynamics are solved using the Smoothed Particle Hydrodynamics (SPH) method (Gingold and Monaghan, 1977; Lucy, 1977). In SPH, the hydrodynamic equations are solved from a Lagrangian perspective, representing the fluid with a discrete set of particles rather than a continuous medium. For an inviscid fluid described by density ρ, pressure p, and specific internal energy u, the Lagrangian takes the form:

$$L = \int \left[\frac{1}{2}v^2 - u(\rho)\right] \rho d^3\mathbf{x}, \tag{3.29}$$

where the specific internal energy satisfies $\partial u/\partial\rho = p/\rho^2$, and $\mathbf{r} = (x_1, x_2, x_3)$ is the position vector. From this Lagrangian, the Euler equations of motion can be derived via a variational principle (Rasio, 2000):

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \tag{3.30}$$

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} = -\frac{\nabla p}{\rho} \tag{3.31}$$

$$\frac{\partial u}{\partial t} + (\mathbf{v} \cdot \nabla)u = -\frac{p}{\rho}(\nabla \cdot \mathbf{v}) \tag{3.32}$$

Equations (3.30), (3.31), and (3.32) are the continuity, momentum, and energy equations, respectively.

To obtain a discrete representation, the continuous Lagrangian is replaced by a particle-based analogue:

$$L_{\mathrm{SPH}} = \sum_{i=1}^{N} m_i \left[\frac{1}{2}\mathbf{v}_i^2 - u_i\right], \tag{3.33}$$

where m_i, $\mathbf{v}_i$, and ρ_i are the mass, velocity, and density of each particle (representing a fluid element), and $u_i = u(\rho_i)$ is the specific internal energy. To use Eq. (3.33) as the system's Lagrangian, a suitable expression for the density in terms of the particle coordinates is required.

A key ingredient of the method is the density estimate. In SPH, the density at any point $\mathbf{r}$ is estimated by interpolating over neighboring particles:

$$\rho(\mathbf{r}) = \sum_j m_j W(\mathbf{r} - \mathbf{r}_j; h), \tag{3.34}$$

where $W(\mathbf{r} - \mathbf{r}_j; h)$ is a smoothing kernel of characteristic width h (the smoothing length). The kernel must satisfy two conditions: its integral over all space must be unity, and it should approach a Dirac delta function in the limit $h \to 0$. The first condition ensures that integrating Eq. (3.34) over space recovers the total mass of the system.

In the case of Fi, the cubic spline kernel is adopted (Monaghan, 1992; Pelupessy, 2005):

$$W(r,h) = \frac{1}{\pi h^3} \begin{cases} 1 - \frac{3}{2}\left(\frac{r}{h}\right)^2 + \frac{3}{4}\left(\frac{r}{h}\right)^3, & 0 \leq \frac{r}{h} \leq 1, \\ \frac{1}{4}\left(2 - \frac{r}{h}\right)^3, & 1 \leq \frac{r}{h} \leq 2, \\ 0, & \text{otherwise.} \end{cases} \tag{3.35}$$

With the density estimation and kernel defined, we now turn to the treatment of the smoothing length h. In modern SPH implementations, it is common to use an adaptive smoothing length h_i for each particle, adjusted to keep the number of neighboring particles approximately constant. However, the use of adaptive smoothing lengths introduces ∇h terms in the equations, which can lead to violations of energy and entropy conservation if not properly accounted for (Nelson and Papaloizou, 1994). A rigorous way to incorporate variable smoothing lengths is to treat h_i as additional variables in the Lagrangian formulation, subject to constraints (Springel and Hernquist, 2002). In Fi, the following constraint is imposed:

$$\frac{4\pi}{3} h_i^3 (\rho_i + \tilde{\rho}) = M_{\mathrm{SPH}}, \tag{3.36}$$

where $M_{\mathrm{SPH}} = \bar{m} N_{\mathrm{nb}}$, with $\bar{m}$ the mean particle mass and N_{nb} the target number of neighbors. The term $\tilde{\rho}$ is an extra density introduced to prevent h_i from diverging as $\rho_i \to 0$ (Pelupessy, 2005). This constraint can be incorporated into the variational principle via Lagrangian multipliers, yielding correction factors in the equations of motion (see Appendix C).

With the notation $W_{ij}(h_i) \equiv W(|\mathbf{r}_i - \mathbf{r}_j|, h_i)$ and the symmetrized kernel $\bar{W}_{ij} = [W_{ij}(h_i) + W_{ij}(h_j)]/2$, the equation of motion for particle i becomes:

$$\frac{d\mathbf{v}_i}{dt} = -\sum_{j=1}^{N} m_j \left[f_i \frac{P_i}{\rho_i^2} \nabla_i W_{ij}(h_i) + f_j \frac{P_j}{\rho_j^2} \nabla_i W_{ij}(h_j) + \Pi_{ij} \nabla_i \bar{W}_{ij} \right], \tag{3.37}$$

where the correction factor f_i accounts for the variability of the smoothing length:

$$f_i = \left(1 + \frac{h_i}{3(\rho_i + \tilde{\rho})} \frac{\partial \rho_i}{\partial h_i} \right)^{-1}. \tag{3.38}$$

The first two terms come from a conservative derivation of SPH equations.To handle discontinuities such as shocks, an artificial viscosity (AV) term Π_{ij} is included. In SPH, entropy generation is confined to shocks through this viscous term, which also prevents unphysical particle interpenetration.

`Fi` implements the standard Monaghan artificial viscosity (Monaghan, 1992; Pelupessy, 2005):

$$\Pi_{ij} = \frac{-\alpha \mu_{ij} \bar{c}_{ij} + \beta \mu_{ij}^2}{\bar{\rho}_{ij}} \tag{3.39}$$

with

$$\mu_{ij} = \begin{cases} \frac{\mathbf{v}_{ij} \cdot \mathbf{r}_{ij}}{\bar{h}_{ij}(\mathbf{r}_{ij}^2/\bar{h}_{ij}^2 + \eta^2)}, & \text{for } \mathbf{v}_{ij} \cdot \mathbf{r}_{ij} < 0, \\ 0, & \text{for } \mathbf{v}_{ij} \cdot \mathbf{r}_{ij} \geq 0 \end{cases} \tag{3.40}$$

where $\mathbf{v}_{ij} = \mathbf{v}_i - \mathbf{v}_j$ and $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$. The dimensionless parameters α and β control the strength of the viscosity (default values are $\alpha = 0.5$, $\beta = 1.0$), and $\eta \ll 1$ prevents numerical divergences. Quantities overlaid with a bar are symmetrized averages, e.g., $\bar{c}_{ij} = (c_i + c_j)/2$.

The AV is only activated when particles are approaching each other ($\mathbf{v}_{ij} \cdot \mathbf{r}_{ij} < 0$), ensuring that $\mu_{ij} \leq 0$ and therefore $\Pi_{ij} \geq 0$. As the term $-\nabla_i W_{ij}(h_i)$ is proportional to $(\mathbf{r}_i - \mathbf{r}_j) = \mathbf{r}_{ij}$, the AV acts to decrease kinetic energy, with the dissipated energy converted into thermal energy, thereby increasing the internal energy (or, equivalently, the entropic function, depending on the thermodynamic formulation chosen) of the particles.

Finally, the thermodynamic evolution of the gas must be specified. `Fi` offers two formulations: evolving either the specific internal energy u_i or the entropic function A_i (related to the specific entropy). For an ideal gas with adiabatic index γ, these quantities are related through

$$P_i = A_i \rho_i^\gamma, \tag{3.41}$$

$$u_i = \frac{A_i}{\gamma - 1}\rho_i^{\gamma-1} = \frac{P_i}{(\gamma-1)\rho_i}, \tag{3.42}$$

where A_i is the entropic function, which remains constant for adiabatic processes in the absence of shocks. The evolution equation for the entropic function is

$$\frac{dA_i}{dt} = -\frac{\gamma-1}{\rho_i^\gamma}(\Gamma - \Lambda) + \frac{1}{2}\frac{\gamma-1}{\rho_i^{\gamma-1}}\sum_{j=1}^{N} m_j \Pi_{ij} \mathbf{v}_{ij} \cdot \nabla_i \bar{W}_{ij} \tag{3.43}$$

where Γ and Λ represent external heating and cooling rates (both set to zero in our simulations). Alternatively, the internal energy equation can be integrated:

$$\frac{du_i}{dt} = \sum_{j=1}^{N} m_j \left[\frac{f_i}{2}\frac{p_i}{\rho_i^2}\nabla_i W_{ij}(h_i) + \frac{f_j}{2}\frac{p_j}{\rho_j^2}\nabla_i W_{ij}(h_j) + \Pi_{ij}\nabla_i \bar{W}_{ij}\right] \mathbf{v}_{ij} \tag{3.44}$$

For our simulations, we adopt the energy equation (Eq. (3.44)), as it provides better numerical stability in the coupled stellar-gas framework (see Appendix C).

3.4. Coupling Strategy: The BRIDGE Method

The N-body and SPH codes operate as independent modules and must be coupled for mixed star-gas simulations. A key advantage of using SPH for the gas component is that both systems are represented by particles, which simplifies the implementation of gravitational interactions. In this section, we describe the coupling scheme employed in this work, based on the `BRIDGE` method (Fujii et al., 2007) and following the implementation of Reinoso Reinoso (2024).

3.4.1. Symplectic Integrator

The `BRIDGE` method (Fujii et al., 2007) employs a symplectic integrator, specifically the second-order leapfrog scheme. The evolution of a dynamical system can be described by Hamilton's equations:

$$\frac{df}{dt} = \{f, H\}, \tag{3.45}$$

where f, H denotes the Poisson bracket:

$$\{f, H\} = \sum_i \left(\frac{\partial f}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial H}{\partial q_i} \right) = D_H f(t) \tag{3.46}$$

with (q_i, p_i) being the canonical coordinates. The formal solution of this equation is given by the exponential operator:

$$f(t) = e^{tD_H} f(0) \tag{3.47}$$

For a gravitational Hamiltonian of the form

$$H(\mathbf{x}, \mathbf{p}) = \frac{\mathbf{p}^2}{2m} + \Phi(\mathbf{x}) \equiv H_{\text{kin}}(\mathbf{p}) + H_{\text{pot}}(\mathbf{x}), \tag{3.48}$$

the Liouville operator D_H can be decomposed as:

$$\begin{aligned} D_H &= \frac{\partial H}{\partial \mathbf{p}} \cdot \frac{\partial}{\partial \mathbf{x}} - \frac{\partial H}{\partial \mathbf{x}} \cdot \frac{\partial}{\partial \mathbf{p}} && (3.49) \\ &= \frac{\mathbf{p}}{m} \cdot \frac{\partial}{\partial \mathbf{x}} + (-\nabla \Phi(\mathbf{x})) \cdot \frac{\partial}{\partial \mathbf{p}} && (3.50) \\ &\equiv D_{\text{drift}} + D_{\text{kick}}. && (3.51) \end{aligned}$$

The solution after a time step Δt is then:

$$f(t + \delta t) = e^{\delta t D_H} f(t) \tag{3.52}$$

When applied to the phase-space coordinates $(\mathbf{x}, \mathbf{v})$, the exponential operator $e^{\delta t D_H}$ has no closed-form expression due to the coupling between position and velocity (Zwart and McMillan, 2018). However, the individual operators D_{drift} and D_{kick} have simple actions:

$$\begin{aligned} D_{\text{drift}}(\mathbf{x}, \mathbf{v}) &= (\mathbf{v}, \mathbf{0}), && (3.53) \\ D_{\text{kick}}(\mathbf{x}, \mathbf{v}) &= (\mathbf{0}, -\nabla \Phi(\mathbf{x})). && (3.54) \end{aligned}$$

The drift operator advances positions at constant velocity, while the kick operator updates velocities at constant position. Importantly, these operators do not commute: $[D_{\text{drift}}, D_{\text{kick}}] \neq 0$. However, the exponential operator can be approximated using an operator splitting technique (Fujii et al., 2007):

$$e^{\Delta t D_H} = e^{\delta t (D_{\text{kick}} + D_{\text{drift}})} = \prod_{i=1}^{k} e^{a_i \delta t D_{\text{kick}}} e^{b_i \delta t D_{\text{drift}}} + \mathscr{O}(\Delta t^{k+1}). \tag{3.55}$$

For $k = 2$ with coefficients $a_1 = a_2 = 1/2$, $b_1 = 1$, $b_2 = 0$, we obtain the second-order accurate approximation:

$$e^{\delta t (D_{\text{kick}} + D_{\text{drift}})} \approx e^{(\delta t/2) D_{\text{kick}}} e^{\delta t D_{\text{drift}}} e^{(\delta t/2) D_{\text{kick}}} + \mathscr{O}(\delta t^3). \tag{3.56}$$

This leads to the kick-drift-kick leapfrog integrator, which proceeds as follows for a time step Δt:

1. Kick: Advance velocities by $\Delta t/2$ using accelerations computed at the current positions:

$$\mathbf{v}_0 \to \mathbf{v}_{1/2} = \mathbf{v}_0 + \frac{\Delta t}{2} \mathbf{a}_0$$

2. Drift: Advance positions by Δt using the new half-step velocities:

$$\mathbf{x}_0 \to \mathbf{x}_1 = \mathbf{x}_0 + \Delta t \mathbf{v}_{1/2}$$

3. Kick: Advance velocities by another $\Delta t/2$ using accelerations computed at the updated positions:

$$\mathbf{v}_{1/2} \to \mathbf{v}_1 = \mathbf{v}_{1/2} + \frac{\Delta t}{2} \mathbf{a}_1$$

where $\mathbf{a}_0 = \mathbf{a}(\mathbf{x}_0)$ and $\mathbf{a}_1 = \mathbf{a}(\mathbf{x}_1)$ are the accelerations due to the gravitational potential. The subscripts 0, 1/2, and 1 denote quantities evaluated at times t, $t + \Delta t/2$, and $t + \Delta t$, respectively.

3.4.2. BRIDGE

AMUSE includes an implementation of the original BRIDGE scheme. An example coupling an N-body code with an SPH code can be found in the AMUSE distribution at:

```
amuse/examples/simple/gas_in_cluster.py
```

This example models a star cluster with an embedded gas component, coupling GADGET-2 (SPH) and HERMITE (N-body). In this implementation, each code performs its own drift step internally. The kick step, however, is handled externally as follows:

- Accelerations $\mathbf{a}_{\mathrm{NB},i}$ for each N-body particle due to all SPH particles are computed using the SPH code's **`get_gravity_at_point`**(*eps*, x, y, z) method, which returns the gravitational acceleration at a given point (x, y, z) with softening length *eps*. The velocities of N-body particles are then kicked using these accelerations.
- Similarly, accelerations $\mathbf{a}_{\mathrm{SPH},i}$ for each SPH particle due to all N-body particles are computed using the **`get_gravity_at_point`**(*eps*, x, y, z) method of a tree code (BHTree in the example), with the softening length set to the SPH smoothing length h_i. The velocities of SPH particles are then kicked accordingly.

As identified by Reinoso Reinoso (2024), this implementation has two main issues, both related to the kick steps. First, SPH and N-body particles use different gravitational softening lengths (h and ε, respectively), leading to asymmetric forces:

$$F_{\mathrm{NB}} = \frac{G m_{\mathrm{NB}} m_{\mathrm{SPH}}}{(r^2+\varepsilon^2)^{3/2}} r \neq F_{\mathrm{SPH}} = \frac{G m_{\mathrm{NB}} m_{\mathrm{SPH}}}{(r^2+h^2)^{3/2}} r, \tag{3.57}$$

where $r = |\mathbf{r}_{\mathrm{NB}} - \mathbf{r}_{\mathrm{SPH}}|$. This asymmetry can be resolved by introducing an effective softening length $\bar{\varepsilon}$ for the mutual gravitational interaction during the kick steps.

Second, the force calculation may use different softening kernels: SPH codes often employ a cubic spline softening, while N-body codes typically use a Plummer (or similar) softening. The difference between these softening formulations is illustrated in Figure 3.4.

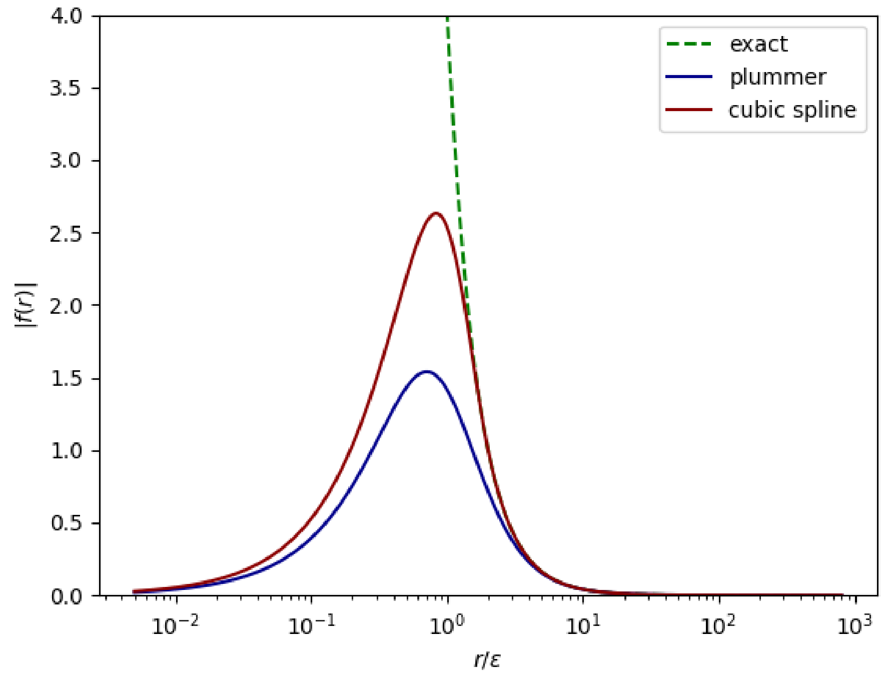

Figure 3.4: Comparison of force calculation normalized (masses and gravitational constant set to 1). In dashed green, the exact gravitational force. In blue, the force with plummer softening, and in red, the force calculated with cubic spline softening.

To handle these issues, Reinoso Reinoso (2024) developed a modified BRIDGE implementation that uses the FASTKICK code to compute gravitational interactions between SPH and N-body particles. FASTKICK computes forces using a Plummer softening with a constant softening length $\bar{\varepsilon}$, ensuring symmetry and consistency. The code provides a Python interface, making it compatible with AMUSE simulation scripts, with the computationally intensive force calculations performed in compiled languages (C/Fortran), and a GPU-accelerated version (C++/CUDA) is also available.

The modified BRIDGE algorithm proceeds as follows for each global time step Δt:

1. At time t, compute accelerations $\mathbf{a}_{\mathrm{NB},i}$ for all N-body particles due to SPH particles, and $\mathbf{a}_{\mathrm{SPH},i}$ for all SPH particles due to N-body particles, using FASTKICK with softening length $\bar{\varepsilon}$. Kick velocities by $\Delta t/2$.
2. Evolve each code independently for a full time step Δt (drift), updating positions via their own internal integrators.
3. Recompute accelerations using FASTKICK with the updated positions, and apply a second velocity kick of $\Delta t/2$.

3.4.3. Adaptive timestep

The evolution of the coupled system is controlled by the BRIDGE time step. At the end of each BRIDGE step, the time step is adjusted to accurately resolve the physics of the gas and the star-gas interaction. For the gas, the minimum free-fall time is calculated:

$$t_{\rm ff} = \sqrt{\frac{3\pi}{32G\rho_{\rm max}}} \, , \tag{3.58}$$

where $\rho_{\rm max}$ is the maximum density among all SPH particles. For the gas-stars interaction, Courant-type time steps that depend on the acceleration between gas and star particles are calculated:

$$\Delta t_{\rm coup,gas} = \min_i \left(f_{\rm coup} \sqrt{\frac{2h_i}{\mathbf{a}_{{\rm SPH},i}}} \right) \tag{3.59}$$

$$\Delta t_{\rm coup,stars} = \min_i \left(f_{\rm coup} \sqrt{\frac{\varepsilon_{\rm coup}}{\mathbf{a}_{\rm NB,i}}} \right) \, , \tag{3.60}$$

where h_i is the softening length of SPH particles, $\mathbf{a}_{{\rm SPH},i}$ and $\mathbf{a}_{\rm NB,i}$ are the forces calculated by FASTKICK in the kick step, and $\varepsilon_{\rm coup}$ is the softening length used by FASTKICK in the calculation of gravitational interactions. For $f_{\rm coup} = 1$, Equations (3.59) and (3.60) set the minimum resolution for gas and stars interactions to be solved in the BRIDGE steps. Since $f_{\rm coup}$ is a security parameter, it is recommended to use values < 1.

Taking into account the minimum gas free-fall time (Equation (3.60)) and the Courant-type time steps (Equations (3.59) and (3.60)), the relevant time scale yields

$$\tau_{\rm step} = \min(t_{\rm ff}, \Delta t_{\rm coup,gas}, \Delta t_{\rm coup,stars}) \, . \tag{3.61}$$

Finally, the time step of BRIDGE is adjusted by the condition

$$0.5\tau_{\rm step} \leq \frac{\tau}{2^n} \leq \tau_{\rm step} \, , \tag{3.62}$$

with τ the crossing time of the stellar cluster and n an integer. The time step of the BRIDGE $\Delta t_{\rm BRIDGE} = \tau/2^n$ is adjusted at the end of each BRIDGE step, and is also used in SPH code. Note that the adjusted time step is a power of two of the initial N-body time unit, consistent with the block time steps used for the N-body particles. Consequently, the three interacting codes are in synchrony. Figure 3.5 shows a schematic of the coupling strategy.

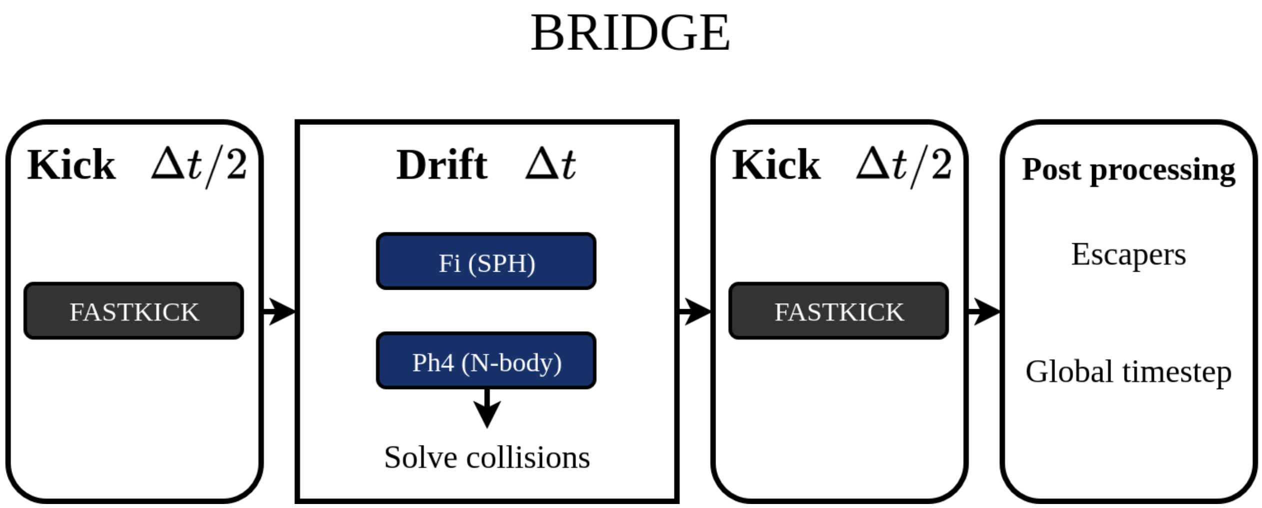

Figure 3.5: Scheme of the BRIDGE coupling strategy. In the kick steps, FASTKICK computes the mutual gravitational forces between stars and gas, and the velocities of both components are updated. In the drift step, each code evolves its particles independently: Ph4 handles stellar dynamics (including collision detection and mergers), while Fi evolves the gas. After the kick-drift-kick cycle, the global BRIDGE time step is updated based on the gas free-fall time and the coupling timescales ($t_{\rm ff}$, $\Delta t_{\rm coup,gas}$, $\Delta t_{\rm coup,stars}$). This updated timestep is used for both the next BRIDGE step and the Fi integration. Additionally, escaped N-body particles are identified and tagged in the post-processing phase.

Chapter 4

Initial Conditions and Setup

Chapter 3 described the numerical methods used in this work, including the SPH code for the gas component, the N-body code for the stellar component, and the coupling scheme. This chapter provides a complete specification of the simulations, detailing the initial conditions for both the stellar and gaseous systems, as well as the numerical parameters used in the code initialization.

4.1. Plummer distribution

The scenario presented in Chapter 2 and proposed by Escala (2021) was previously explored by Vergara et al. (2023) through simulations sampling different collisional regimes. These models represent stellar systems characterized by their initial mass and size scale. The stellar clusters are initialized using a Plummer distribution (Plummer, 1911). The Plummer model is characterized by the potential (Binney and Tremaine, 2008):

$$\Phi_p(r) = -\frac{GM}{\sqrt{r^2+b^2}}, \quad \text{with } r = |\mathbf{r}| \tag{4.1}$$

where b is the Plummer scale. From the Poisson equation, the associated density is

$$\rho_p(r) = \frac{3M}{4\pi b^3}\left(1+\frac{r^2}{b^2}\right)^{-5/2} \tag{4.2}$$

The potential energy of a Plummer model is:

$$U_p = -\frac{3\pi GM^2}{32b} \tag{4.3}$$

Now we need a description for velocity. The velocity dispersion σ for an isotropic spherical stellar system can be obtained through the Jeans equation (Binney and Tremaine, 2008):

$$\frac{d(\rho\sigma^2)}{dr} = -\rho\frac{d\Phi}{dr} \tag{4.4}$$

Solving for the Plummer model we have the dispersion velocity:

$$\sigma_p(r)^2 = \frac{GM}{6\sqrt{r^2+b^2}} \tag{4.5}$$

We can relate the Plummer scale parameter b with a physical scale. First, we use the definition of virial radius as the characteristic system radius R via the potential energy U:

$$-\frac{GM^2}{2R} = U \tag{4.6}$$

$$\frac{1}{R} = -\frac{2}{GM^2}U\ , \tag{4.7}$$

where U is the total gravitational potential energy. In the case of a system of N particles, this radius is defined as:

$$R = \left(\frac{1}{M^2}\sum_{i,j\neq i}^{N}\frac{m_i m_j}{|\mathbf{r}_i - \mathbf{r_j}|}\right)^{-1} \tag{4.8}$$

In the case of a Plummer model, replacing Eq. 4.3 into Eq. 4.7, this radius can be expressed in function of the Plummer scale parameter b:

$$\frac{1}{R} = \left(-\frac{2}{GM^2}\right)\left(-\frac{3\pi GM^2}{32b}\right) \tag{4.9}$$

$$R = \frac{16b}{3\pi} \tag{4.10}$$

Once the physical scale of the system is defined via b (or equivalently R), we need to generate the actual particle distribution. This is typically done in a dimensionless system of units, which we now describe.

4.1.1. N-body units

Most gravitational N-body codes employ a dimensionless system of units, known as N-body units or Hénon units (see Heggie and Mathieu (1986)). In this system, physical quantities are scaled to be of order unity, which brings two main benefits: (i) it eliminates the need to multiply by G in every force calculation, and (ii) it improves numerical precision by keeping variables close to unity, thereby minimizing round-off errors that would otherwise arise when working with astrophysical scales spanning many orders of magnitude. Additionally, N-body units facilitate comparison between results obtained with different codes.

N-body units are defined by setting the gravitational constant, total mass, and total energy to convenient values:

$$G = 1 \tag{4.11}$$

$$M = 1 \tag{4.12}$$

$$E = -\frac{1}{4} \tag{4.13}$$

With this, the unit of mass, length and time can been defined. The mass unit is simply the total mass:

$$u_m = M \tag{4.14}$$

Assuming virial equilibrium ($2T + U = 0$), the kinetic and potential energies become:

$$T = \frac{1}{4} = -E \tag{4.15}$$

$$U = -\frac{1}{2} = 2E \tag{4.16}$$

From the equation of virial radius (Eq. 4.7), we can replace the potential energy from Eq. 4.16:

$$R = -\frac{GM^2}{4E} \tag{4.17}$$

Using $G = 1$, $M = 1$ and $E = -1/4$ we can check that $R = 1$, so the length unit u_l is:

$$u_l = R = -\frac{GM^2}{4E} \tag{4.18}$$

Finally, we derive the time unit. A characteristic velocity follows from the kinetic energy:

$$T = \frac{M\sigma^2}{2} \tag{4.19}$$

$$\sigma^2 = \frac{2T}{M} = \frac{-2E}{M} \tag{4.20}$$

The crossing time is defined as:

$$t_{cross} = \frac{2R}{\sigma} = \frac{GM^{5/2}}{(-2E)^{3/2}} \tag{4.21}$$

With $G = 1$, $M = 1$, and $E = -1/4$, we obtain $t_{\mathrm{cross}} = 2^{3/2}$. The standard N-body time unit is then defined as:

$$u_t = t_{cross}2^{-2/3} = \frac{GM^{5/2}}{(-4E)^{3/2}} \tag{4.22}$$

which indeed yields $u_t = 1$ for $G = 1$, $M = 1$ and $E = -1/4$. Note that defining the unit system requires only a scale radius and the total mass of the system. AMUSE provides its own physical units system, which can be converted to N-body units using a converter. In AMUSE, N-body units are handled through the **`nbody_system`** class, which requires at least two physical quantities, typically the total mass and the virial radius. The following example illustrates how to initialize a converter:

```python
from amuse.units import units, nbody_system

R_vir = 1 | units.pc          # Define a scale radius
M_tot = 1e4 | units.MSun  # Define the total mass

```

```
converter = nbody_system.nbody_to_si(R_vir, M_tot)
```

Code 4.1: How to define a converter in AMUSE

The converter serves a bidirectional role: it translates physical quantities (e.g., positions, velocities, masses) into dimensionless N-body units for the numerical calculations, and converts the simulation output back to dimensional physical units for subsequent analysis.

4.2. Stars and Gas Initial Conditions

For the results presented in Chapter 5, we adopt the same initial conditions as the models in Vergara et al. (2023). For each model, we perform four simulations with different gas fractions, using the same stellar initial conditions (same as in Vergara et al. (2023)) for all gas fractions. We use the notation of family model to reference the four variations of a specific model. For example, the model family A is composed of the four models A_0, A_5, A_{10} and A_{20}, where the capital letter indicate that they share the same initial conditions for stellar cluster, and the suffix indicate the percentage of mass of the gas component added. Table 4.1 summarizes the initial conditions of stellar components of each model.

Table 4.1: Initial conditions for stars systems.

Model ID[1]	R_v [pc]	M [$M_\odot$]	N	m_s [$M_\odot$]	r_s [$R_\odot$]	σ [km/s]
A	0.005	5×10^4	10^3	50	11.7	146.66
B	0.005	2.5×10^4	5×10^2	50	11.7	103.71
C	0.005	10^4	10^3	10	4.7	65.59
D	0.005	5×10^3	5×10^2	10	4.7	46.38
F	0.005	10^3	10^3	1	1.0	20.74
H	0.01	5×10^4	5×10^3	10	4.7	103.71
I	0.01	10^4	10^3	10	4.7	46.38
J	0.01	5×10^3	5×10^3	1	1.0	32.79
K	0.01	10^3	10^3	1	1.0	14.66
L	0.1	5×10^5	10^4	50	11.7	103.71
M	0.1	10^5	10^4	10	4.7	46.38
N	0.1	10^4	10^4	1	1.0	14.67

[1] The "Model ID" column identifies the family of models. All members of a given family (e.g., A_0, A_5, A_{10}, and A_{20}) share the same stellar initial conditions, which are listed here with capital letters with no suffix (A, B, C, etc.).

In the table, R_v, M, N, and σ denote the virial radius, total mass, number of stars, and velocity dispersion of the stellar system, respectively. The initial mass and radius of each star are given by m_s and r_s. Each variant of a family model have the same initial conditions for the stellar system. For the gas-free models, the stellar clusters are initialized in virial equilibrium ($2K+U=0$). For models with gas, the stellar component retains the same initial velocities and is therefore not in virial equilibrium with the total mass (stars and gas). The total system is subvirial, with $2K < |U_{\text{total}}|$, and will evolve dynamically toward a new equilibrium. The gas is initialized in hydrostatic equilibrium, supported by pressure in the gravitational potential of the stars. As the gas has zero initial velocity, it is far from virial equilibrium and will adjust on a dynamical timescale. The initial conditions of gas components are presented in Table 4.2, where $q = M_{\text{SPH}}/M$, M_{SPH} is the total mass of SPH particles, N_{SPH} and m_{SPH} are the total number and mass per SPH particle, and c_s is the sound speed of the gas component, calculated as the mass-weighted average within the virial radius.

4.2.1. Particles generation

Both the stellar and gaseous systems are generated using the AMUSE implementation of the Plummer model. To generate a stellar system with the initial conditions of model A for example, we first initialize a converter:

```
from amuse.units import units, nbody_system

M_stars = 5e4 | units.MSun
R_vir = 0.005 | units.pc
converter = nbody_system.nbody_to_si(R_vir, M_stars)
```

Code 4.2: Converter example for model A

After importing the **plummer** module, the stellar particles are generated by calling the function **new_plummer_model** with the converter and number of particles:

```
from amuse.ic import plummer

N = 1000
stars = plummer.new_plummer_model(N, converter, radius_cutoff=5., do_scale=True)
stars.radius = 11.7 | units.RSun
```

Code 4.3: Stars particles in model A

The **radius_cutoff** parameter truncates the distribution at $5b$, where b is the Plummer scale parameter (here corresponding to a virial radius $R_v = 0.005$ pc). Setting **do_scale=True** ensures that the generated particles are scaled to N-body units, satisfying the virial conditions described in Section 4.1.1 (total mass $M=1$, total energy $E=-1/4$, kinetic energy $T=1/4$, and potential energy $U=-1/2$).

For the gaseous component, we use **gasplummer** function from module **amuse.ic** to generate

Table 4.2: Initial conditions for gas systems

Model ID	q	$M_{\rm SPH}$ $[M_\odot]$	$N_{\rm SPH}$	$m_{\rm SPH}$ $[M_\odot]$	c_s $[km/s]$
A_5	0.05	2.5×10^3	5×10^3	0.5	95.97
A_{10}	0.1	5×10^3	10^4	0.5	96.67
A_{20}	0.2	10^4	2×10^4	0.5	96.90
B_5	0.05	1.25×10^3	2.5×10^3	0.5	66.35
B_{10}	0.1	2.5×10^3	5×10^3	0.5	67.17
B_{20}	0.2	5×10^3	10^4	0.5	68.11
C_5	0.05	5×10^2	5×10^3	0.1	43.10
C_{10}	0.1	10^3	10^4	0.1	43.24
C_{20}	0.2	2×10^3	2×10^4	0.1	43.35
D_5	0.05	2.5×10^2	2.5×10^3	0.1	29.77
D_{10}	0.1	5×10^2	5×10^3	0.1	30.51
D_{20}	0.2	10^3	10^4	0.1	30.46
F_5	0.05	5×10^1	5×10^3	0.01	13.63
F_{10}	0.1	10^2	10^4	0.01	13.64
F_{20}	0.2	2×10^2	2×10^4	0.01	13.72
H_5	0.05	2.5×10^3	2.5×10^4	0.1	68.50
H_{10}	0.1	5×10^3	5×10^4	0.1	68.57
H_{20}	0.2	10^4	10^5	0.1	69.04
I_5	0.05	5×10^2	5×10^3	0.1	30.25
I_{10}	0.1	10^3	10^4	0.1	30.24
I_{20}	0.2	2×10^3	2×10^4	0.1	30.72
J_5	0.05	2.5×10^2	2.5×10^4	0.01	21.64
J_{10}	0.1	5×10^2	5×10^4	0.01	21.85
J_{20}	0.2	10^3	10^5	0.01	21.78
K_5	0.05	5×10^1	5×10^3	0.01	9.65
K_{10}	0.1	10^2	10^4	0.01	9.62
K_{20}	0.2	2×10^2	2×10^4	0.01	9.63
L_5	0.05	2.5×10^4	5×10^4	0.5	68.84
L_{10}	0.1	5×10^4	10^5	0.5	68.85
L_{20}	0.2	10^5	2×10^5	0.5	69.15
M_5	0.05	5×10^3	5×10^4	0.1	30.76
M_{10}	0.1	10^4	10^5	0.1	30.84
M_{20}	0.2	2×10^4	2×10^5	0.1	30.90
N_5	0.05	5×10^2	5×10^4	0.01	9.76
N_{10}	0.1	10^3	10^5	0.01	9.71
N_{20}	0.2	2×10^3	2×10^5	0.01	9.78

the SPH particles. The number of gas particles is chosen such that the mass ratio $m_{\text{SPH}}/m_{\text{NB}} = 0.01$, where m_{NB} is the mass of each N-body particle. For example, for the variation with 20%, A_{20}, the gas particles are generated as follows:

```
from amuse.ic import gasplummer

f_gas = 0.2
N_gas = N*100
N_gas = int(N_gas*f_gas)

gas = gasplummer.new_plummer_gas_model(N_gas,
                                        convert_nbody=converter,
                                        do_scale=True,
                                        mass_cutoff=0.95535)
gas.mass = (M_stars*f_gas / N_gas)
```

Code 4.4: Gas particles in model A

While the stellar component is initialized in virial equilibrium with respect to its own mass, the gas particles are initially at rest. To prevent immediate collapse, their internal energy is set to achieve hydrostatic equilibrium in the gravitational potential of the stellar cluster. For a fluid in hydrostatic equilibrium,

$$\nabla p = -\rho \nabla \Phi \, , \tag{4.23}$$

where p, ρ, Φ stand for the pressure, density and gravitational potential. Assuming a polytropic equation of state (Eqs. 3.41 and 3.42), this condition gives a relation for the specific internal energy u:

$$u = -\frac{\Phi}{\gamma} \, , \tag{4.24}$$

where γ is the polytropic index.

The initialization described in Listing 4.4 raises a potential issue. The gas particles are generated using a converter scaled to the total stellar mass ($M = 5 \times 10^4 \, \text{M}_\odot$), and their internal energy is set accordingly to achieve hydrostatic equilibrium in the combined potential. However, after generation, we explicitly reduce the gas mass to $f_{\text{gas}} M$ (in this case, 20% of the stellar mass). In isolation, this reduction would lead to an imbalance: the gravitational force from the gas diminishes, while the internal pressure remains unchanged, causing the gaseous component to expand. However, because the gas is embedded in and gravitationally coupled to the stellar system, the dominant contribution to the potential comes from the stars. We therefore expect the gas to remain near dynamical equilibrium, with any minor adjustments occurring on a dynamical timescale as the simulation evolves.

4.3. Numerical Parameters

Having described the numerical methods in the previous chapter, we now detail the specific parameter configurations used for each code. As discussed in Section 3.1, the AMUSE framework provides a unified interface that connects Python scripts with compiled codes through class structures. Each code is accessible via the community module, e.g., **`amuse.community.ph4.interface`** for `Ph4`. The following example illustrates how to create an instance of `Ph4`:

```python
from amuse.community.ph4.interface import ph4

grav = ph4()
```

Code 4.5: Instance creation for `Ph4`

In listing 4.5, **`grav`** is an instance of the class **`amuse.community.ph4.interface.ph4`** (shortened **`ph4`**). The code classes have a variety of parameters and internal methods, whose principal functionality is to set up the parameters and particles before running the code. Most codes run in N-body units; however, they have the possibility to work with physical units. In the following, we show how to correctly initialize and run an instance of **`ph4`**, with the creation of particles using a Plummer distribution, and using a converter:

```python
from amuse.community.ph4.interface import ph4
from amuse.units import units, nbody_system
from amuse.ic.plummer import new_plummer_model

# Step 1, creation of a converter and particles
virial_radius = 1 | units.pc
total_mass = 1000 | units.MSun
N_stars = 1000

converter = nbody_system.nbody_to_si(total_mass, virial_radius)

bodies = new_plummer_model(N_stars,
                           convert_nbody=converter,
                           do_scale=True)
bodies.radius = 1 | units.RSun

# Step 2, creation of an instance of the code
```

```
grav = ph4(converter,
           mode = 'cpu',
           number_of_workers=1)

grav.initialize_code()
grav.parameters.epsilon_squared = (1 | units.RSun) ** 2

# Step 3, adding particles
grav.particles.add_particles(bodies)
grav.commit_particles()

# Step 4, enable collision detection
sc_col_det = grav.stopping_conditions.collision_detection
sc_col_det.enable()
```

Code 4.6: Example setup for Ph4 with particles

In listing 4.6, we have created a converter. The principal function of the converter is to translate units from the N-body system of units (see Section 4.1.1), to the international system of units. The converter needs the virial radius and total mass, then escalates quantities such that one N-body unit of length is equal to the virial radius. When the instance of **ph4** is created, we provided the converter, the **mode** (where options are **'gpu'** and **'cpu'**), and the **number_of_workers**, that controls the number of cpus used in parallel calculations. Since the stopping condition for detection of collisions is enabled, is necessary to provide a function to handle collisions. Then, we can evolve the instance of **grav**. The next listing shows a minimalist loop:

```
from amuse.community.ph4.interface import ph4
from amuse.units import units, nbody_system
from amuse.ic.plummer import new_plummer_model

t_end = 1000 | units.yr

# Principal loop
while grav.model_time < t_end:
  grav.evolve_model(t_end)

  if sc_col_det.is_set():
    solve_collision(grav, sc_col_det)
```

Code 4.7: Loop with evolution for Ph4

where **solve_collision** is a function that handles the collisions. Note that when a collision is detected, the code stop before reaching the final time. Then, if there is no function to correctly solve the collisions, the loop could be infinite. Since the structures provided by AMUSE are generic, the initialization of most codes is very similar.

4.3.1. Ph4

The principal parameters of **ph4** are summarized in Table 4.3.

Table 4.3: Key runtime parameters for the PH4 N-body code. Parameters shown with their default and the values used in this work.

Parameter	Description	Default	Used
epsilon_squared	Gravitational softening length squared	0.0	$(0.05 R_\odot)^2$
timestep_parameter	Accuracy parameter for Hermite integrator	0.14	0.01
force_sync	Force synchronization of particles	0 (False)	1 (True)
manage_encounters	Close encounter management	4	4
initial_timestep_fac	Initial timestep factor	0.0625	0.0625
initial_timestep_limit	Initial timestep limit	0.03125	0.03125

The smaller timestep parameter (0.01) increases the accuracy of the Hermite integrator, which is necessary to resolve close encounters and collisions accurately. The gravitational softening is set to a small fraction ($(0.05 R_\odot)^2$) of the stellar radii to avoid singularities while minimally affecting the dynamics.

In our simulations we use a gravitational softening of 0.05 $R_\odot$. Since the smaller stars are solar-stars, the force calculation has small differences from real forces, while infinite forces are avoided. To use the GPU, it is necessary to set mode='gpu' during initialization. By default, **ph4** instance is created with mode='cpu'. All our simulations run with cpu, and with number_of_workers=4, to run parallel calculations.

4.3.2. Fi

To initialize Fi, we create an instance of **amuse.community.fi.interface.Fi**. However, SPH particles are stored as gas_particles. The following shows how to initialize Fi with SPH particles following a plummer distribution:

```
from amuse.community.fi.interface import Fi
from amuse.units import units, nbody_system
from amuse.ic.gasplummer import new_plummer_gas_model

# Creation of SPH particles
gas_pl = gasplummer.new_plummer_gas_model
SPH_bodies = gas_pl(N_gas,
```

```python
                    convert_nbody=converter,
                    do_scale=True)

# Creation of Fi instance
SPH = Fi(converter, mode='openmp', number_of_workers=8)
SPH.initialize_code()

# Adding gas particles
SPH.gas_particles.add_particles(SPH_bodies)
```

Code 4.8: Example setup for FI with gas particles

Now, we present the principal parameters used in Fi, summarized in Table 4.4. In our simulations, Fi runs with 8 parallel process, through **Fi(number_of_workers = 8)** during the initialization.

Table 4.4: Key runtime parameters for the FI SPH code. Parameters shown with their default and the values used in this work.

Parameter	Description	Default	Used
integrate_entropy_flag	Integrate entropy (True) or internal energy (False)	True	False
radiation_flag	Include radiative cooling/heating	False	False
star_formation_flag	Include star formation	False	False
use_hydro_flag	Enable SPH hydrodynamics	True	True
eps_is_h_flag	Gravitational softening = SPH smoothing length	True	True
gamma	Adiabatic index (polytropic)	1.6666667	1.6666667
n_smooth	Target number of SPH neighbours	64	64
alpha	Artificial viscosity parameter (Monaghan)	0.5	0.5
beta	Artificial viscosity parameter (Monaghan)	1.0	1.0
courant	Courant condition parameter	0.3	0.3
acc_timestep_flag	Use acceleration-based timestep criterion	True	True
timestep	Initial system timestep	1.0	1.0

The only parameter modified from its default is integrate_entropy_flag, which is set to False to evolve the specific internal energy instead of the entropic function. This choice provides better numerical stability for energy conservation (see Section 3.3). All other parameters are left at their default values.

4.3.3. Bridge

Finally, we present the parameters used in our BRIDGE implementation. Although the implementation[1] runs in Python and is independent of AMUSE, it is designed to work with AMUSE data structures. The principal parameters used to initialize BRIDGE are summarized in Table 4.5.

Table 4.5: Key runtime parameters for the BRIDGE coupling scheme.

Parameter	Description	Value
`mode`	Use GPU-accelerated kicks (1) or CPU (0)	1 (GPU enabled)
`f_coup`	Coupling safety factor for timestep calculation	0.5
`eps_nb`	Softening length for gravitational coupling	5 AU
`dt_bridge`	Maximum BRIDGE timestep (N-body units)[1]	1.0
`do_accretion`	Use accretion implementation	False

[1] The BRIDGE timestep is adaptive and determined by the gas free-fall time and the coupling timescales ($t_{\rm ff}$, $\Delta t_{\rm coup,gas}$, $\Delta t_{\rm coup,stars}$). The value listed here is the maximum allowed timestep in N-body units, which is adjusted downward as needed. GPU acceleration is enabled via `mode=1` for the FASTKICK force calculations. The N-body and SPH components run on CPU (PH4 with 4 workers, FI with 8 workers). The parameter `do_accretion` is set to `False` in our simulations, as we do not include gas accretion.

4.3.4. Code validation

To verify the accuracy and stability of our numerical setup, we present the results of a test simulation. Figure 4.1 shows the relative energy error $\Delta E/E$ (top panel) and the virial ratio $2K/|U|$ (bottom panel) as a function of time over 50 crossing times. The system evolved is a star cluster of 5000 stars of 1 $M_\odot$ (total star mass $M = 5000\ M_\odot$), with virial radius $R_{\rm v} = 1.0$ pc, embedded in a gas cloud of mass 5000 $M_\odot$, with 5000 SPH particles. The initial crossing time of the system is 0.1491 Myr.

The relative energy error remains below $\sim 10^{-4}$ throughout the simulation, confirming that the integration scheme conserves energy to high accuracy. The virial ratio fluctuates around unity, indicating that the system remains in virial equilibrium. These results validate the numerical setup described in this chapter.

[1] This implementation was developed by B. Reinoso, a co-author of the paper presented in Chapter 5.

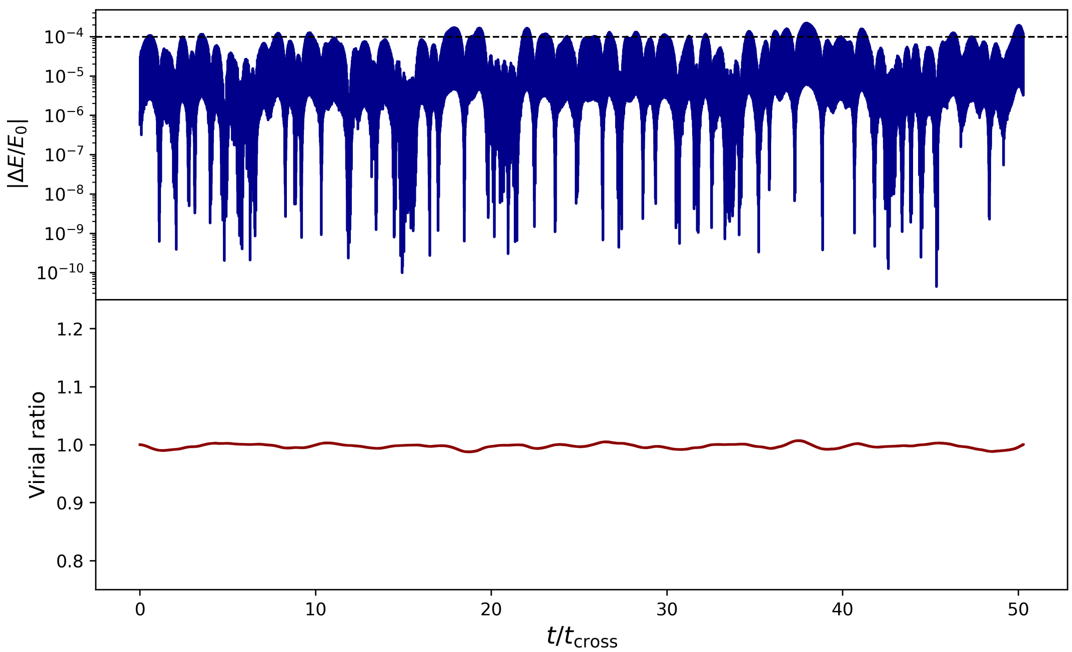

Figure 4.1: Energy conservation and virial equilibrium for a test simulation of a gas-embedded star cluster. Top panel: relative energy error $\Delta E/E$. Bottom panel: virial ratio $2K/|U|$. The system remains in virial equilibrium with an energy error of $\lesssim 10^{-4}$ over 50 crossing times.

Chapter 5

Results

5.1. Statement about my contribution

This section presents the research article entitled "The role of a gaseous component on the collisional runaway path to supermassive black hole seeds". My contributions include the derivation of the generalized critical mass, which incorporates gravitational focusing and an external potential, as well as the definition of the relaxation mass scale. I performed all simulations, carried out the data analysis, and produced all tables and figures shown in the article. All authors contributed to the discussion and to the revision of the manuscript. In particular, Bastian Reinoso developed the `BRIDGE` implementation used for the coupling strategy.

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The role of a gaseous component on the collisional runaway path to supermassive black hole seeds

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ABSTRACT

Context. The origin of supermassive black holes (SMBHs) remains unclear, despite observational evidence for their rapid formation in the early Universe, as evidenced by active galactic nuclei (AGN) at high redshift.
Aims. In this paper, we investigate stellar dynamics in dense, gas-embedded clusters. We explore how gas influences the runaway collision process and the subsequent formation of SMBH seeds, in a global instability scenario.
Methods. We perform a series of coupled N-body and SPH simulations using the AMUSE framework, with the stellar and gaseous components interacting gravitationally via the BRIDGE method. We use 12 initial condition models with equal-mass stars following Plummer distributions, with total masses ranging from 10^3 to 5×10^5 $M_\odot$, virial radii from 0.005 to 0.1 pc, and stellar masses of 50, 10 and 1 $M_\odot$. For each model, we perform four simulations with gas fractions ranging from 0% to 20%, resulting in a total of 48 simulations. The gas is sampled with the same spatial distribution as the stars, with a mass ratio per particle of $m_{SPH}/m = 0.01$, where m is the stellar mass.
Results. We find that in most of our simulations, higher gas fractions lead to the formation of more massive black holes, as the core collapses earlier, triggering collisions through gas-star interactions. Family N models, which lie in the stable regime, form only a negligible BH even at high gas fractions. Our results show increases in BH mass of $\sim 8\%$ for low gas fractions, to $\sim 44\%$ at 20% gas fraction. The core collapse time is also shortened, with the strongest reduction at 20% gas fraction (reduction of $\sim 27\%$).
Conclusions. The transonic regime of our initial conditions is where dynamical friction is most efficient in principle. Although our simulations do not fully resolve this effect, and in the absence of other dissipation mechanisms (e.g., supernovae, stellar feedback, cooling), gas-star interactions are the principal mechanism driving the enhancement in BH formation efficiency. This implies that the presence of gas during runaway collisions can lead to the formation of more massive SMBH seeds.

Key words. Galaxies: star clusters: general – Gas dynamics – Hydrodynamics – Methods: numerical – Stars: black holes – Stars: kinematics and dynamics

1. Introduction

The origin of supermassive black holes (SMBHs) is currently a topic of active research, and the formation mechanism remains unclear (Woods et al. 2019). There is evidence of SMBHs with masses in the range $\sim 10^6$–10^{10} $M_\odot$ (Natarajan & Treister 2009; Volonteri 2010; Woods et al. 2019; Latif & Ferrara 2016; King 2015). However, the origin of SMBHs becomes more enigmatic with the observations of quasars at $z \gtrsim 6$ (Willott et al. 2007; Jiang et al. 2015), and the more recent detection of 31 quasars at $6.6 < z < 7.8$ from the Euclid Wide Survey (Yang et al. 2026). In particular, Mortlock et al. (2011) reported a quasar at $z \sim 7.085$ (0.77 Gyr after the Big Bang) hosting a black hole of 2×10^9 $M_\odot$.

These observations imply that the seeds of the first SMBHs must have formed and grown on very short timescales (Latif & Ferrara 2016). Several formation channels have been proposed to produce SMBH seeds, each with distinct initial masses and associated challenges. The remnants of Population III stars are expected to form light seeds of $\sim 10^2$ $M_\odot$ (Fryer et al. 2001). These seeds can grow into intermediate-mass black holes (IMBHs) in dense early globular clusters through gas accretion (Vesperini et al. 2010); collisions between Pop III stars can also contribute to building more massive seeds (Vergara et al. 2021; Reinoso et al. 2018, 2020), and both mechanisms may occur simultaneously (Boekholt et al. 2018). However, their growth to SMBH masses requires sustained Eddington accretion or super-Eddington bursts, which are difficult to achieve (Regan & Volonteri 2024; Smith et al. 2018). The direct collapse of primordial gas clouds can form heavier seeds of $\sim 10^5$ $M_\odot$ (Bromm & Loeb 2003; Lodato & Natarajan 2006; Latif & Schleicher 2015; van Dokkum et al. 2025); but this pathway requires specific conditions to suppress fragmentation and maintain effective cooling (Shang et al. 2010; Suazo et al. 2019). Moreover, direct collapse black holes (DCBHs) typically form in metal-free environments, and subsequent accretion can be severely suppressed by supernovae (SNe) feedback and by the large relative velocity that the BH acquires due to its formation at $\sim$ 1 kpc from the galactic centre (Chon et al. 2021). Runaway collisions in dense stellar clusters provide another pathway, efficiently forming IMBHs with masses of $\sim 10^3$–10^4 $M_\odot$, which can serve as seeds for SMBHs (Sakurai et al. 2017; Chung et al. 2026). However, current simulations of the collisional runaway typically produce IMBHs rather than SMBHs (Portegies Zwart & McMillan

2002; Vergara et al. 2023; Vergara et al. 2026; Reinoso et al. 2018), partly due to the computational cost of simulating the large number of stars required to model more massive clusters. Once formed, these IMBHs could grow through subsequent gas accretion and mergers with other BHs (Inayoshi et al. 2022; Kritos et al. 2023; Inayoshi et al. 2016; Chung et al. 2026; Pelupessy et al. 2007).

Galactic centers are one of the most likely places where SMBHs can form, since the process of mass segregation and dynamical friction (DF) can funnel material to the center (Shlosman et al. 1990; Escala 2006; Naiman et al. 2015). Besides SMBHs, nuclear stellar clusters (NSCs) can also be found in galactic centers (Neumayer et al. 2020; Schiavi et al. 2021), with both referred to under the nomenclature of central massive objects (CMOs; Ferrarese et al. (2006)). There are correlations between properties of the host galaxy and the CMO it contains, including the M–σ relation between BH mass and bulge velocity dispersion (Gültekin et al. 2009; Tremaine et al. 2002), the $M_{\rm BH}$–$M_{\rm bulge}$ relation (Häring & Rix 2004; Marconi & Hunt 2003), as well as correlations between the host galaxy mass and the mass of the SMBH or NSC it harbors (Ferrarese et al. 2006). These correlations suggest that there could be a link in the origin of SMBHs and NSCs (Escala 2021). NSCs dominate in galaxies with masses $\lesssim 10^{10}\ M_\odot$, while SMBHs dominate in those with masses $\gtrsim 10^{12}\ M_\odot$; in the intermediate-mass regime, both types coexist (Georgiev et al. 2016). Runaway collisions are the primary candidate for forming a massive BH in the center (Escala 2021; Chung et al. 2026; Das et al. 2021). Mass segregation and the Spitzer instability drive the migration of the most massive stars toward the center. This increase in the core density triggers a runaway collision process, forming a massive BH as a consequence (Gürkan et al. 2004; Rasio et al. 2003).

Observations with the James Webb Space Telescope (JWST; Gardner et al. (2023)) have discovered a considerable population of SMBHs at $5 \lesssim z \lesssim 11$ (Schneider et al. 2023; Natarajan et al. 2024; Eilers et al. 2023; Harikane 2025), as well as young massive clusters (YMCs) that could eventually form IMBHs through runaway collisions (Schleicher et al. 2026). The presence of such SMBHs challenges the light seed scenario, since Pop III remnants would require sustained super-Eddington accretion to grow to these masses (Kiyuna 2026). In contrast, heavy seeds can reach SMBH masses through Eddington-limited accretion (Jeon et al. 2025; Silk et al. 2024). Recent observations of extremely red, point-like objects, known as little red dots (LRDs), may help elucidate the origins of early SMBHs, as they could represent an intermediate population between seeds and the observed SMBHs (Vaida & Farber 2026). Observations of LRDs span a redshift range of $3 \lesssim z \lesssim 10$. LRDs were initially identified as extremely red galaxies (Endsley et al. 2023; Labbé et al. 2023). There is no consensus on their nature. Evidence supporting an AGN interpretation includes the detection of high-ionization coronal lines, such as [FeX], and broad Hα emission (Kocevski et al. 2023; Furtak et al. 2024). Although many LRDs are radio and X-ray quiet, this could be expected from super-Eddington accretion models, where thick disks suppress high-energy emission (Secunda et al. 2026). From a stellar interpretation, LRDs are thought to be dusty, intensely star-forming galaxies (Escala et al. 2025; Kritos & Silk 2026; Rantala 2026). If this is the case, these objects would be among the densest stellar systems ever observed, with core densities as high as $\sim 10^8\ M_\odot$ pc^{-3}, largely exceeding the threshold for runaway collisions, and may include tidal disruption events (TDEs) in some models (Bellovary 2025).

In this paper, we investigate the role of gas in the formation of SMBH seeds. We test the scenario of global instability in NSCs proposed by Escala (2021) through coupled N-body/SPH simulations, by selecting a set of initial conditions from Vergara et al. (2023) and adding a gas component. We first describe the scenario in Section 2 and then present the numerical methods in Section 3. We present the main results in Section 4. We discuss the results and their implications for SMBH seed formation, along with caveats, in Section 5. Finally, we summarize the main results and outline considerations for future work in Section 6.

2. The collisional runaway scenario for SMBH seeds

A promising channel for the formation of massive black hole seeds involves runaway stellar collisions in dense stellar clusters Escala (2021). This scenario is motivated by the observation that NSCs and SMBHs are found in galactic centers (Georgiev et al. 2016), suggesting a potential evolutionary link. While stellar collisions are negligible in typical galactic fields (Binney & Tremaine 2008), dense star clusters could have reached densities high enough for the collision timescale ($t_{\rm coll}$) to become shorter than a characteristic timescale $t_{\rm H}$ (e.g., the age of the system or the Hubble time), allowing collisions to become a dominant dynamical process. Escala (2021) established a critical density criterion, derived from $t_{\rm coll} < t_{\rm H}$, to define this unstable, collision-dominated regime. This work showed that observed NSCs generally avoid this parameter space, whereas systems with massive black holes (MBHs) are consistent with having undergone such a phase.

2.1. Timescale criteria and the instability regime

The viability of the collisional runaway scenario depends on the interplay between three fundamental timescales: the collision time ($t_{\rm coll}$), the relaxation time ($t_{\rm relax}$), and a characteristic timescale ($t_{\rm H}$). Following the scenario proposed by Escala (2021), the collisional runaway predicts two global collisional instabilities, both defined by the condition that the collision time is comparable to or shorter than the age of the system ($t_{\rm coll} \le t_{\rm H}$). In the first regime, defined by systems where $t_{\rm coll} < t_{\rm relax}$ ($\le t_{\rm H}$), collisions dominate before relaxation can act, leading to runaway collisions that form a massive remnant without the system being able to expand (Escala et al. 2025). In the second regime, defined by systems where $t_{\rm relax} < t_{\rm coll}$ ($\le t_{\rm H}$), the system is initially in the collisional regime but expands before collisions are triggered, as two-body relaxation operates on shorter timescales. However, core collapse could be present in this scenario (Portegies Zwart & McMillan 2002), enhancing collisions within the core. We now derive the relations that quantify these regimes, following the approach of Vergara et al. (2023).

Consider a stellar system of total mass M and characteristic radius R, composed of N stars of mass m and radius $r_\star$. We assume the system is virialized, with a velocity dispersion $\sigma = (GM/R)^{1/2}$. The collision time can be estimated as $t_{\rm coll} = (n\Sigma\sigma)^{-1}$, where $n = 3M/(4\pi R^3 m)$ is the number density and Σ is the effective cross-section (Binney & Tremaine 2008). Identifying terms in the collisional rate $1/t_{\rm coll} = 16\sqrt{\pi} n\sigma r_\star^2(1+\Theta)$, the cross-section can be defined as $\Sigma_0 = 16\sqrt{\pi} r_\star^2(1+\Theta)$, where $\Theta = Gm/(2r_\star\sigma^2)$ is the Safronov number. Substituting these expressions yields $t_{\rm coll} = (4\pi m R^{7/2})/(3\Sigma_0 G^{1/2} M^{3/2})$.

For a typical velocity dispersion of $\sim$ 100 kms^{-1}, the Safronov number becomes $9.54(m/M_\odot)(R_\odot/r_\star)$ (Binney & Tremaine 2008). Consequently, the effective cross-section is determined primarily by the stellar radii, corresponding to the limit where the Safronov number is small. Then, the condition for collisions to be significant over the age of the system, $t_{\rm coll} \leq t_{\rm H}$, leads to a mass-radius relation that serves as the first boundary in an M–R diagram:

$$M \geq \left(\frac{4\pi m}{3\Sigma_0 t_H G^{1/2}}\right)^{2/3} R^{7/3}. \tag{1}$$

If, instead, we include the virialized velocity dispersion in the Safronov number, the dependence of Σ_0 on σ translates into a dependence on the total mass M through $\sigma = \sqrt{GM/R}$. This distinguishes our derivation from that of Vergara et al. (2023), where the cross-section is dominated by stellar radii and the Safronov number is neglected. Moreover, in the absence of an external potential, M denotes the stellar mass. When an external potential of mass $M_{\rm ext}$ is present, the total mass is $M_{\rm tot} = M + M_{\rm ext}$, and the velocity dispersion increases to $\sigma = \sqrt{GM/R}(1+q)$ (Reinoso et al. 2020), where $q = M_{\rm ext}/M$. This makes Eq. (1) an implicit relation for M. Solving this relation for M (see Appendix A) yields:

$$M_{\rm crit} = \left(\frac{\beta R^{7/2}}{2} + \sqrt{\frac{\beta^2 R^7}{4} + \frac{\alpha^3 R^3}{27}}\right)^{2/3} + \left(\frac{\beta R^{7/2}}{2} - \sqrt{\frac{\beta^2 R^7}{4} + \frac{\alpha^3 R^3}{27}}\right)^{2/3} - \frac{2\alpha R}{3}, \tag{2}$$

where

$$\alpha = \frac{9.54 m R_\odot v_{100}^2}{r_\star G M_\odot (1+q)^2},\ \beta = \frac{\sqrt{\pi} m}{12 G^{1/2} r_\star^2 (1+q) t_H}. \tag{3}$$

Here, $v_{100} = 100$ km s^{-1}. The external potential is assumed to have the same virial radius and mass distribution as the stellar system. Eq. (2) is the marginal condition for $t_{\rm coll} = t_H$. Systems that exceed this mass ($M > M_{\rm crit}$) have $t_{\rm coll} < t_{\rm H}$.

The relaxation time, which characterizes the timescale for stellar orbits to be significantly altered by cumulative weak encounters, is given by Binney & Tremaine (2008): $t_{\rm relax} = t_{\rm cross} \frac{0.1N}{\ln N}$, where $t_{\rm cross} = R/\sigma$, and $\sigma = \sqrt{GM/R}$ for virialized system. Replacing the number of stars $N = M/m$, the relaxation time can be written as $t_{\rm relax} = (0.1 R^{3/2} M^{1/2})/(m G^{1/2} \ln(M/m))$.

The condition $t_{\rm relax} \leq t_{\rm H}$ defines a second relation:

$$R \leq \left(\frac{t_{\rm H} m \ln(M/m)}{0.1}\right)^{2/3} \left(\frac{G}{M}\right)^{1/3}. \tag{4}$$

Solving $t_{\rm relax} = t_{\rm H}$ for M including the effects of an external potential ($M = M_{\rm star}(1+q)$) yields the mass scale (Appendix B):

$$M_{\rm relax} = \frac{m}{(1+q)^2} \exp\left(-2W_{-1}\left(-\frac{1}{2\lambda}\right)\right), \tag{5}$$

where W_{-1} is the lower branch of the Lambert W function evaluated at $z = -1/(2\lambda)$, and $\lambda = t_{\rm H}\sqrt{mG}/(0.1 R^{3/2}(1+q)^2)$. We use the lower branch of the Lambert function since it is physically more relevant (see Appendix B.1). The relation in Eq. (5) defines the characteristic mass scale above which a system of radius R can evolve over a time $t_{\rm H}$ without two-body relaxation processes significantly altering its initial conditions.

The timescales $t_{\rm coll}$ and $t_{\rm relax}$ define three distinct dynamical regimes for a stellar system of mass M, radius R and external potential of mass ratio q. The collisional regime (CR) occurs when $t_{\rm coll} \leq t_{\rm H} \leq t_{\rm relax}$ (i.e., $M \geq M_{\rm crit}$ and $M \geq M_{\rm relax}$), where collisions dominate before two-body relaxation can expand the cluster, potentially leading to a global collapse and the formation of a massive BH. The stable regime (SR) is defined by $t_{\rm relax} \leq t_{\rm H} \leq t_{\rm coll}$ ($M \leq M_{\rm crit}$ and $M \leq M_{\rm relax}$), where relaxation expands the cluster before collisions become relevant. Finally, we define an intermediate regime (IR) by $t_{\rm relax} \leq t_{\rm coll} \leq t_{\rm H}$ ($M_{\rm crit} \leq M \leq M_{\rm relax}$), where two-body relaxation begins to expand the cluster before global collisions become frequent; however, the core may still undergo collisional collapse while the outer envelope expands. The described regimes characterize the stability of the system against a global collisional collapse. In the CR, we expect that most of the mass ends up in a central BH, resulting in a BH-dominated system. In the SR, the gravitational potential remains dominated by the stellar cluster. The IR represents an intermediate case, where a broad range of BH masses may form embedded in a surrounding NSC.

3. Numerical methods and initial conditions

We model the coupled evolution of stars and gas using the Astrophysical Multipurpose Software Environment, AMUSE[1] (Portegies Zwart et al. 2009, 2013; Pelupessy et al. 2013; Portegies Zwart & McMillan 2018). The stellar component is evolved with a pure N-body code, while the gas is modeled with the smoothed particle hydrodynamics (SPH) method (Gingold & Monaghan 1977). The two components interact gravitationally via the BRIDGE method (Fujii et al. 2007). We provide more details below.

3.1. Stellar dynamics and collision treatment

The stellar dynamics are computed with the pure N-body code PH4 (McMillan et al. 2012), which implements a 4th-order Hermite integrator (Makino & Aarseth 1992), with adaptive block time steps. Stars are treated as point masses with an associated physical radius.

A collision between two stars is detected when the distance between their centers is less than the sum of their radii, $|\mathbf{x}_i - \mathbf{x}_j| < R_i + R_j$. When a collision is detected, the integration is interrupted and the two stars are replaced by a new single star. The merger product inherits the total mass, the center-of-mass position, and center-of-mass velocity of the progenitors. The new radius is calculated under the assumption of constant density: $R_{\rm coll} = R_1((m_1+m_2)/m_1)^{1/3}$. This simple treatment conserves linear momentum and mass.

3.2. Gas dynamics

The gas is evolved using the SPH code FI (Gerritsen & Icke 1997), which solves the hydrodynamic equations in Lagrangian form. The code utilizes a cubic spline kernel for interpolation (Monaghan 1992), with an adaptive smoothing length to keep a total of 64 neighbors. Self-gravity of the gas is computed using a Barnes-Hut tree code (Barnes & Hut 1986), with a softening length that is equal to the smoothing length. The gas is modeled as an ideal, adiabatic fluid without radiative cooling or external heating.

Although fully conservative equations for entropy and energy are implemented in some SPH codes (Springel & Hernquist 2002), we set up the FI so that it evolves the internal energy, which consequently becomes the relevant thermodynamic quantity. For an ideal adiabatic gas with adiabatic index γ, the equa-

[1] https://github.com/amusecode/amuse

tion of state is $P = A\rho^{\gamma}$, where P is the pressure and ρ the density. In SPH, the equation of state is written in terms of the specific internal energy u:

$$P = (\gamma - 1)\rho u \, . \tag{6}$$

The sound speed, which is the relevant dynamical quantity in the absence of radiative cooling and heating, is given by (Horedt 2004):

$$c_s = \sqrt{\frac{\partial P}{\partial \rho}} = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\gamma(\gamma - 1)u} \, . \tag{7}$$

3.3. Gravitational coupling

The gravitational interaction between the stellar (PH4) and gaseous (FI) systems is handled by the BRIDGE coupling scheme (Fujii et al. 2007). This method is necessary to efficiently integrate systems with very different intrinsic time scales. BRIDGE uses a symplectic, kick-drift-kick (Verlet) integrator to couple the subsystems. In this scheme, the gravitational force from one subsystem on the other is applied as impulses (kicks) for a half timestep, updating the velocity. Then, each subsystem is evolved independently for a full timestep (drift). Finally, a kick is applied for a half timestep. The gravitational force between gas and stars is calculated with a Plummer softening kernel and with a softening length given by

$$\epsilon_{g,s} = 0.5(h + \epsilon_{\rm coup}) \, , \tag{8}$$

where h is the smoothing length of the gas particle and $\epsilon_{\rm coup}$ = 5 AU is the coupling softening length. The initial BRIDGE timestep is chosen to be the crossing time of the stellar component, and we implement an adaptive timestep $\Delta t_{\rm BRIDGE}$. After every BRIDGE timestep, $\Delta t_{\rm BRIDGE}$ is set to be the minimum between the free-fall time in the densest SPH particle and a Courant-type timestep that depends on the acceleration between gas and star particles. These timesteps are calculated as

$$t_{\rm ff} = \sqrt{\frac{3\pi}{32 G \rho_{\rm max}}}, \tag{9}$$

$$\Delta t_{\rm coup,gas} = \min\left(f_{\rm coup} \sqrt{\frac{2h}{\rm a_{g,s}}} \right) \tag{10}$$

$$\Delta t_{\rm coup,stars} = \min\left(f_{\rm coup} \sqrt{\frac{\epsilon_{\rm coup}}{\rm a_{s,g}}} \right) \tag{11}$$

$$\tau = \min(t_{\rm ff}, \Delta t_{\rm coup,gas}, \Delta t_{\rm coup,stars}) \, , \tag{12}$$

where $\rho_{\rm max}$ is the maximum density among SPH particles. h and $\mathbf{a}_{g,s}$ are the softening length and the highest acceleration among all SPH particles due to the stars. $\epsilon_{\rm coup}$ is set to be 5 AU, and $\mathbf{a}_{g,s}$ is the highest acceleration among all stars due to the gas. $f_{\rm coup}$ = 0.5 is a free parameter that controls the time-step accuracy.

Finally, the BRIDGE timestep is set to fulfill the condition

$$0.5\tau \leq \Delta t_{\rm BRIDGE} \leq \tau \, . \tag{13}$$

This adaptive timestep ensures stability against strong gas-stars gravitational interactions.

3.4. Initial conditions

In order to test the scenario proposed by Escala (2021) we simulated the same initial conditions as Vergara et al. (2023), keeping the same model IDs to facilitate comparison. For each model, we perform four simulations with different gas fractions, using the same stellar initial conditions for all gas fractions. Table D.2 summarizes the initial conditions for the stellar component, where ID is the model identifier, R_v is the initial virial radius, M is the total mass, N is the number of stars, m is the initial star mass, r is the initial star radius and σ is the velocity dispersion. Each model is a Plummer distribution (Plummer 1911) with equal-mass stars. The initial virial radii are $R_v = 0.005, 0.01, 0.1$ pc, and the number of stars $N = 500, 1000, 5000, 10000$. The stellar masses and radii are the same as in Vergara et al. (2023): for $m = 1, 10$ and 50 $M_\odot$, we adopt $r = 1, 4.7$ and 11.7 $R_\odot$, respectively, following the mass–radius relations of Bond et al. (1984) and Demircan & Kahraman (1991). Due to computational limitations, we simulate clusters that are less massive but more compact than real NSCs. This allows us to explore the instability on shorter timescales (t_H = 5 Myr) while maintaining the same physical regime.

Fig. 1 shows the initial conditions of models in Table D.2 (see Appendix D.1), in a mass-radius space. The solid lines represent the relaxation mass (Eq. (5)), and the dashed lines are for the critical mass (Eq. (2)), calculated for the final simulation time (t_H = 5 Myr) for q = 0; the colored areas indicate the range of gas fractions, from q = 0 to q = 0.2. Black dot-dashed line stands for the condition $t_{\rm coll} = t_{\rm relax}$, for q = 0. Colors indicate the initial star mass, with yellow for m = 1 $M_\odot$, red for m = 10 $M_\odot$ and blue for m = 50 $M_\odot$. Each model is represented by a different symbol, with the same color convention. We note that family models A, B, C, D, H and I lie in the IR, while family models F, J, K, L, M and N are in the SR, with no models in the CR.

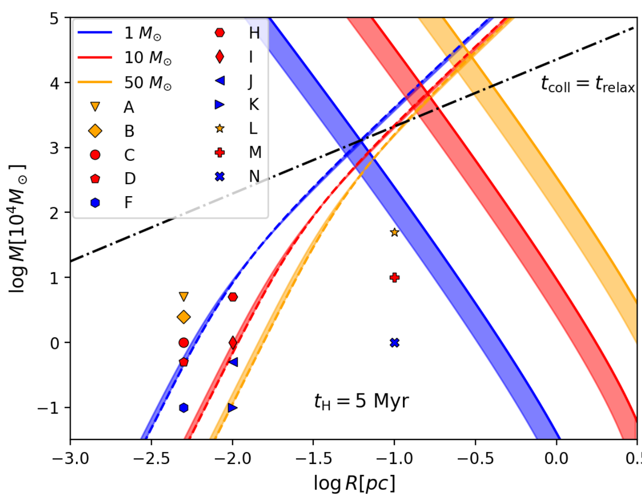

Fig. 1: SMBH formation scenario represented in a mass-radius space. For all symbols and lines, the colors represent different stellar masses, with yellow for m = 1 $M_\odot$, red for m = 10 $M_\odot$ and blue for m = 50 $M_\odot$. Solid and dashed lines are for the relaxation and critical mass, with t_H = 5 Myr and q = 0, while colored areas indicate the range of gas fractions from q = 0 to q = 0.2. Black dot-dashed line represents the condition $t_{\rm coll} = t_{\rm relax}$ for q = 0. The symbols represent the initial conditions of Table D.2.

For each star cluster model, we first generate a Plummer distribution of SPH particles with the same virial radius and total mass as the stellar system (so that the gas would be in virial equilibrium if considered alone). We then scale the gas mass to 5%, 10% and 20% of the stellar mass by adjusting the particle masses while keeping the same spatial distribution. For all models, the SPH particle mass is $m_{\rm SPH} = m/100$, where m is the stellar mass. The number of SPH particles for a given gas fraction f is then $N_{\rm SPH} = f \times N \times 100$, with N the number of stars. Since the added gas mass is small compared to the stellar mass, we assume that the overall virial equilibrium of the stellar component is not significantly disturbed.

To characterize the gas further, we consider the initial sound speed profiles (Eq. (7)) for the gas variants of the A family (A_5, A_{10} and A_{20}) and the N family (N_5, N_{10} and N_{20}), which are the most and least collisional models families in our sample (in their respective regimes), respectively. For the A gas variants, the stellar velocity dispersion decreases from 186.80 km s^{-1} in the central region (r_5) to 101.60 km s^{-1} in the outer regions (r_{95}), while the gas sound speed falls from 129.89 km s^{-1} at the center to 65.51 km s^{-1} in the outermost layers (see Appendix E).

The fact that $\sigma/c_{\rm s} > 1$ throughout the stellar cluster has important consequences. The transonic motion of stars through the gas produces an efficient DF, as described by Ostriker (1999), which leads to stars losing orbital energy and sinking toward the center, enhancing the occurrence of collisions in the central region. Although our simulations do not fully resolve DF (see Appendix D.3), the dissipation of kinetic energy due to gas-star interactions is still present. For the N gas variants (N_5, N_{10} and N_{20}), the velocities are an order of magnitude lower ($\sigma \sim 15$ km s^{-1}, $c_{\rm s} \sim 10$ km s^{-1}), but the condition $\sigma/c_{\rm s} > 1$ still holds.

4. Results

Here we present the main results from our simulations. We use the black hole efficiency defined in Escala (2021):

$$\epsilon_{\rm BH} = (1 + M_{\rm NSC}/M_{\rm BH})^{-1}\,, \tag{14}$$

where $M_{\rm BH}$ corresponds to the most massive object, and $M_{\rm NSC}$ is the mass of gravitationally bound stars. Considering $M_{\rm ESC}$ as the escaped stars mass, the total initial mass of the stellar system is $M_{\rm s} = M_{\rm ESC} + M_{\rm NSC} + M_{\rm BH}$. The central massive object is defined as $M_{\rm CMO} = M_{\rm NSC} + M_{\rm BH}$, i.e., the total bound mass of the stellar system. In terms of the efficiency, the BH candidate mass is then $M_{\rm BH} = \epsilon_{\rm BH} M_{\rm CMO}$.

4.1. Dynamical evolution

In this section we present the time evolution of the mass components of the system. We select two model families from different regimes, A and N. For each family, we consider two variants: a stars-only simulation (denoted A_0 and N_0), and simulations with gas masses of 20% of the initial stellar mass (denoted A_{20}, N_{20}). The A variants have an initial stellar mass of $M = 5 \times 10^4\ M_\odot$, virial radius $R_{\rm vir} = 0.005$ pc, and is composed of 10^3 stars. The N variants have an initial mass of $M = 10^4\ M_\odot$, virial radius $R_{\rm vir} = 0.1$ pc and 10^4 stars. As shown in Fig. 1, the A family is in the IR, where the collision time is shorter than the simulation time. In contrast, the N family is in the SR where collisions are not globally relevant.

Each row in Fig. 2 corresponds to A_0 and A_{20}, from top to bottom. The left column shows the evolution over the full 5 Myr, while the right column zooms in to 40 relaxation timescales. The dashed blue line marks the initial collision time, the dot-dashed green line indicates the initial relaxation time, and the dashed black line shows the estimated core collapse time (see Appendix C). We observe that the inner 10% and 30% Lagrangian radii rapidly contract towards the center, on scales of the collision time. After 1 Myr, A_0 and A_{20} show that the Lagrangian radii remain stable, with A_0 showing a slight increase in the 90% Lagrangian radii compared with A_{20}. In the second column we can observe slight differences in the core collapse (slightly shorter for A_{20}), and no considerable changes during 1 $t_{\rm relax}$.

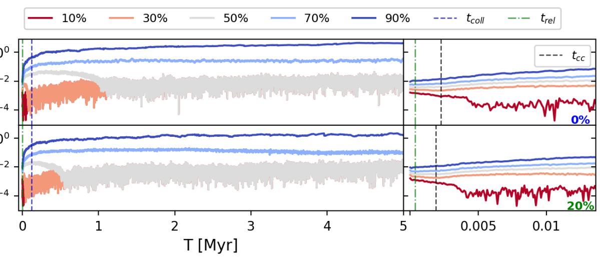

Fig. 2: Lagrangian radii for stars, from 10% (red) to 90% (blue). The top panel corresponds to A_0, and bottom panel to A_{20}. The left column shows the full simulation time (5 Myr), while the right column shows the evolution up to 40 relaxation times. Blue dashed and green dot-dashed lines are the initial collision and relaxation timescales, respectively. Black dot-dashed lines are the core collapse.

The time evolution of the mass components and $\epsilon_{\rm BH}$ of stars for A_0 and A_{20} is displayed in Fig. 3. The top panel shows the escaped mass normalized by the initial stellar mass $M_{\rm s}$. We find that approximately 40% of the initial stellar mass escapes during the first 1 Myr of evolution for both variants, and the total escaped mass reaches the range of 50-60% by 5 Myr. The A_0 variant shows only 57.3% of escaped mass vs 50.2% for A_{20}. The second panel shows collisions normalized by the initial number of stars, with solid lines for total collisions and dashed lines for collisions involving the most massive object. We can see that most collisions happen during the first Myr, augmenting as the gas mass increases. During the first Myr, A_0 shows 27.2% of collisions, reaching 27.5% at 5 Myr. A_{20} shows 32.5% of collisions at 1 Myr, and 33.5% of collisions at 5 Myr. The influence of gas is stronger during the early evolution of the cluster, increasing the number of collisions by 19.49% relative to A_0. The gas also increases the number of collisions at late times. The third panel shows the evolution of the black hole efficiency (Eq. (14)), which is slightly higher for A_{20}. As described in Vergara et al. (2023), the increase in efficiency in the late stage of the system is dominated by the escaped mass, as collisions are less common. The bottom panel shows the mass evolution of the black hole candidate, exhibiting a similar behavior to the collision panel, with a rapid increase at early times, and a negligible increase in late evolution. The final mass of the most massive object is 13 300 $M_\odot$ for A_0, and 16 400 $M_\odot$ for A_{20}, representing 62.30% and a 65.86% of the CMO mass, respectively. This represents, respectively, 26.6 and 32.8% of the initial stellar mass, which means that most of the collisions end in a single object. From the right column, the black dash-dotted line is the mean core collapse time, where we note that it coincides with the beginning of the onset of runaway collisions, as expected. By the collision

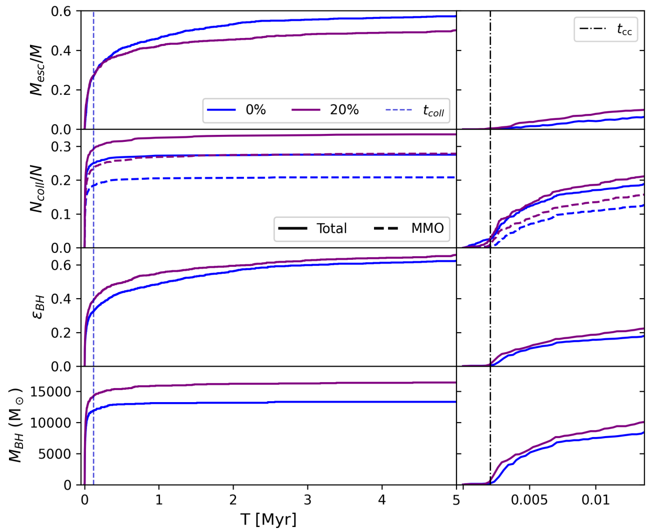

Fig. 3: Mass components and $\epsilon_{\rm BH}$ evolution over time of stars. Blue curves are for A_0, and purple for A_{20}. Top panel is for escaped mass normalized by the total stellar mass. Second panel shows collisions normalized by the initial number of stars, where solid lines are for the total and dashed lines for collisions involving the most massive object. Third panel shows $\epsilon_{\rm BH}$ (Eq. 14), and bottom panel shows the mass of the most massive object. Left column is for 5 Myr, with the vertical blue dashed line representing $t_{\rm coll}$. Right column is a zoom to 40 relaxation times (~ 0.014 Myr), where the dot-dashed black line is the mean core collapse time.

time (vertical dashed blue line), an important part of collisions have occurred, while $\epsilon_{\rm BH}$ has reached approximately half of its final value.

Fig. 4 shows the Lagrangian radii of N_0 and N_{20}. For N_0, the 90% and 70% radii expand slowly, while the 30% and 10% radii contract slightly. N_{20} exhibits qualitatively the same behavior, with no significant differences compared with N_0.

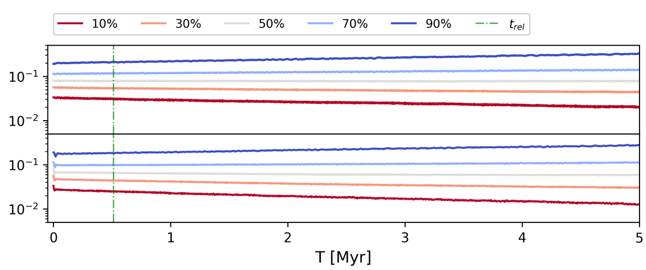

Fig. 4: Lagrangian radii for stars, from 10% (red) to 90% (blue). The top panel corresponds to N_0, and bottom panel to N_{20}. Green and black dot-dashed line are the initial relaxation timescales and core collapse, respectively.

The mass and $\epsilon_{\rm BH}$ evolution of N_0 and N_{20} is presented in Fig. 5. The top panel shows that the escaped mass is lower than 1%, which means that most of the initial stellar mass remains gravitationally bound. From the second panel, collisions are near zero, which may be related to a long collisional timescale. As there are no significant collisions and escaped mass, the efficiency is near zero, as can be observed in the third panel. Consequently, as can be observed from the bottom panel, the most massive object shows only 4 collisions.

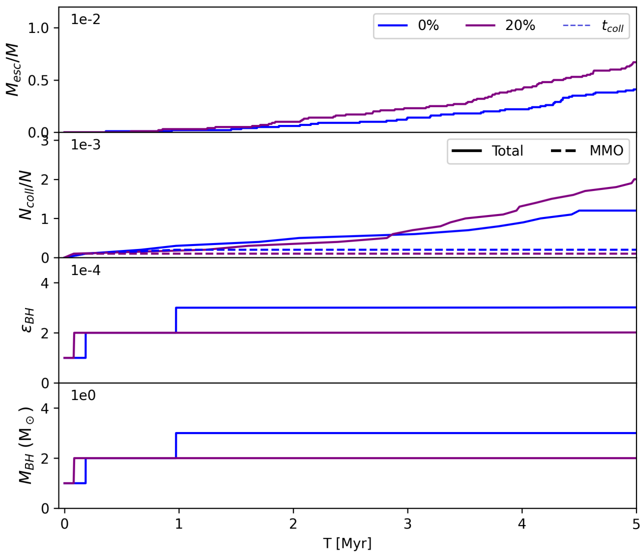

Fig. 5: Mass components evolution over time of stars for four variants of model N. Equivalent to Fig. 3 with no zoom.

Models of family N lie in the regime where $t_{\rm coll} > t_{\rm H}$, i.e., collisions are not globally relevant. The results shown in Figs. 4 and 5 are therefore expected: no significant mass loss, negligible collisions, near-zero BH efficiency, and no formation of a massive object.

4.2. *Effect of gas on core collapse timescale*

For each model, we estimated the core collapse time $t_{\rm cc}$ using the method described in Appendix C. Table 1 shows the ratio $x_{\rm cc} = t_{\rm cc,gas}/t_{\rm cc,0}$ averaged over models, where the columns, from left to right are for models in the IR, SR and all, respectively. Each row corresponds to a gas fraction. In the case of models in the IR, for a gas fraction of 5%, the mean ratio is 0.8765 ± 0.1818; for 10%, 0.8345 ± 0.1345; and for 20%, 0.7410 ± 0.1094. All ratios are below 1, indicating a reduced $t_{\rm cc}$ in the presence of gas. The core collapse time tends to be shorter as the gas fraction increases, while the dispersion decreases. The decrease in dispersion with higher gas fraction suggests that the effect becomes more predictable when more gas is present.

For models in the SR, the mean ratio is 1.0111 ± 0.1570 for 5% gas fraction, 0.8571 ± 0.1068 for 10% gas fraction, and 0.7082 ± 0.0533 for 20% gas fraction. The reduction of $t_{\rm cc}$ is less pronounced for SR models at 5% and 10% gas fractions, and at 5% it is almost negligible. The dispersions are smaller for all gas fractions compared with models in the IR. For a gas fraction of 20%, the reduction on $t_{\rm cc}$ is slightly more pronounced in the SR models (0.7082 ± 0.0533) than in the IR models (0.7410 ± 0.1094). Despite these small differences between IR and SR models, the general tendency is to shorten the core collapse time as the gas mass is larger.

Fig. 6 shows the core collapse time as a function of the relaxation time and collision time, on left and right panels respectively. In addition to the described behavior, we observe correlations between the $t_{\rm cc}$ and both the relaxation and collision times:

Table 1: Mean ratio $x_{cc} = t_{cc,gas}/t_{cc,0}$ of core collapse times for gas-embedded models relative to their gas-free counterparts, and standard deviation for IR family models (A, B, C, D, H, I), SR family models (F, J, K, L, M), and the combined sample.

	IR		SR		Total	
X_{gas} (%)	$\bar{x}_{cc}$	σ	$\bar{x}_{cc}$	σ	$\bar{x}_{cc}$	σ
5	0.8765	0.1818	1.0111	0.1570	0.9377	0.1836
10	0.8345	0.1345	0.8571	0.1068	0.8448	0.1232
20	0.7410	0.1094	0.7082	0.0533	0.7261	0.0899

t_{cc} decreases with shorter t_{relax} (left panel) and with shorter t_{coll} (right panel).

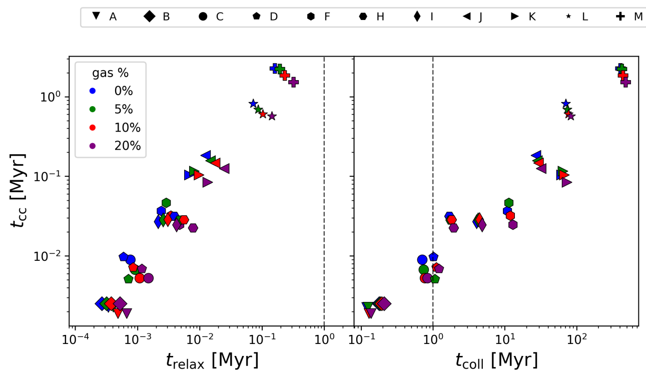

Fig. 6: Core collapse time as a function of t_{relax} (left panel) and t_{coll} (right panel).

4.3. Critical mass and BH formation efficiency

In this subsection, we analyze the black hole formation efficiency of our models. First, we compare our stars-only results with those of Vergara et al. (2023) to validate our setup. Then, we show how the presence of gas modifies the efficiency as a function of the critical mass, testing the scenario proposed by Escala (2021).

To make a first comparison, Table 2 presents results for the same models as Vergara et al. (2023). From left to right, the columns are: model identifier, escaped mass (in $M_\odot$), black hole mass (in $M_\odot$) and formation efficiency (ϵ_{BH}). For each quantity, two sub-columns show our stars-only results (left) and those of Vergara et al. (2023) (right). For the escaped mass, the differences are small: the ratio of our values to those of Vergara et al. (2023) ranges from less than 1% to approximately 10%, corresponding to less than 5% of the initial stellar mass. In the case of the BH mass, the largest differences occur for the most collisional models (A_0, B_0, C_0, in the IR), with relative differences of 34.11%, 36.36%, and 15.66%, respectively. These correspond to differences of 6.77%, 5.60%, and 2.60% of the initial stellar mass.

These differences probably arise from the different numerical codes used. Our simulations employ PH4 (via AMUSE), while Vergara et al. (2023) use NBODY6++GPU (Wang et al. 2015; Aarseth & Aarseth 2003). Both use a fourth-order Hermite integrator with hierarchical block time-steps. However, NBODY6++GPU additionally implements the Kustaanheimo–Stiefel algorithm (Kustaanheimo et al. 1965) and chain regularization (Mikkola & Aarseth 1993) as special treatments for close encounters, binaries, and multiple systems.

The impact on BH mass translates into efficiency differences. Again, the most collisional models (A_0, B_0, C_0) show a considerable enhancement in BH formation efficiency, with differences of 15.97, 8.87, and 8.17 percentage points, respectively. For the remaining models, the differences are below 4%.

Significant differences are observed for N_0. In our simulations, our values are lower by 82.48% and 95.77% for escaped mass and BH mass, respectively. This discrepancy arises because the relaxation time of model N is relatively long (~ 0.5 Myr) and our simulation time is limited to 5 Myr. In contrast, models L_0 and M_0, despite also having large relaxation times, reach core collapse within 5 Myr (at ~ 0.9 and ~ 2.3 Myr, respectively). Since the simulations of Vergara et al. (2023) extend to 10 Myr, model N_0 has evolved considerably further in their case. Assuming core collapse occurs on a timescale of $\sim 20 t_{relax}$ (Quinlan 1996), and given that $t_{relax} \sim 0.5$ Myr for N_0, the core collapse is expected at ~ 10 Myr.

Next, we show results for all models, including their gas variants. Fig. 7 shows, ϵ_{BH} (top panel), and normalized escaped mass (bottom panel). In the left column, quantities are plotted as a function of M/M_{relax}, while in the right column they are plotted as a function of M/M_{crit}. From the bottom panels, we observe a tendency for the escaped mass to grow as M/M_{relax} decreases and M/M_{crit} increases, with no clear distinction between gas variants. There are two processes that produce escaping stars: diffusive or small-angle relaxation evaporates stars slowly from the cluster; on the other hand, two-body encounters produce ejections of high-energy stars, more likely in dense cores (Khalisi et al. 2007). Family models in the IR that are more collisional (A, B, and C) have shorter core collapse and collision times, showing an important fraction of escapers (probably due to ejections). In the other extreme, the N family models, where core collapse is not reached and collisions are not globally relevant, exhibit a small fraction of escapers.

In the top-right panel, we observe a clear tendency for efficiency to grow as M/M_{crit} increases. All models follow the same overall trend, but with a systematic vertical shift: the stars-only variants (blue) lie systematically below the fitted curves, while the 20% gas variants lie above. This is confirmed by the fits. Near the first regime ($M/M_{crit} < 10^{-4}$) and the third regime ($M/M_{crit} > 10^5$), where the efficiency approaches 0 and 1 respectively, these differences become smaller, as all curves converge. To model this effect, we performed a unified fit where the shape of the curve is determined by a common set of parameters (x_0, k, a), while the gas fraction q introduces a horizontal scaling of the mass ratio. We define an effective mass ratio $x_{eff} = (M/M_{crit})(1 + \alpha_q q)$, where α_q is a free parameter that depends on q. Similar to Vergara et al. (2026), the efficiency is then modeled by the form:

$$\epsilon_{BH} = f(x_{eff}) = (1 + \exp\left[-k\left(\log_{10}(x_{eff}) - \log_{10}(|x_0|)\right)\right])^{a} \quad . \qquad (15)$$

The parameters x_0, k and a were obtained by fitting the stars-only variants ($q = 0$). For each gas fraction, we then fitted a

Table 2: Comparison of stars-only (sub zero variants) simulations with Vergara et al. (2023). Our values are from single realizations; reference values include standard deviations from three realizations.

Model ID	$M_{\rm esc}$ [$M_\odot$]		$M_{\rm BH}$ [$M_\odot$]		$\epsilon_{\rm BH}$ [%]	
	This work	Vergara+23	This work	Vergara+23	This work	Vergara+23
A_0	28 650	28 533 ± 147	13 300	9917 ± 143	62.30	46.33 ± 2.5
B_0	13 300	14 300 ± 687	5250	3850 ± 324	44.87	36.00 ± 3.7
C_0	5620	5370 ± 42	1920	1660 ± 14	43.84	35.67 ± 0.5
D_0	2540	2767 ± 111	630	610 ± 43	25.61	27.33 ± 3.1
F_0	411	372 ± 5	88	85 ± 12	14.94	13.67 ± 2.1
H_0	21 750	23 137 ± 1966	12 100	11 397 ± 143	42.83	42.67 ± 2.1
I_0	4500	4253 ± 183	980	1110 ± 83	17.82	19.33 ± 1.7
J_0	1469	1459 ± 22	435	545 ± 42	12.32	15.67 ± 1.2
K_0	344	327 ± 8	59	59 ± 2	8.99	8.67 ± 0.5
L_0	69 300	75 517 ± 1812	64 000	67 450 ± 2585	14.86	15.67 ± 0.5
M_0	7170	7850 ± 283	5510	6987 ± 246	5.94	7.67 ± 0.5
N_0	41	234 ± 44	3	71 ± 5	0.03	0.73 ± 0.0

This work: simulations run up to 5 Myr. Vergara+23: simulations of Vergara et al. (2023), run up to 10 Myr. The stellar initial conditions for each of our models correspond to the same letter in Vergara+23; e.g., our model A_0 shares the same stellar initial conditions as model A in Vergara+23.

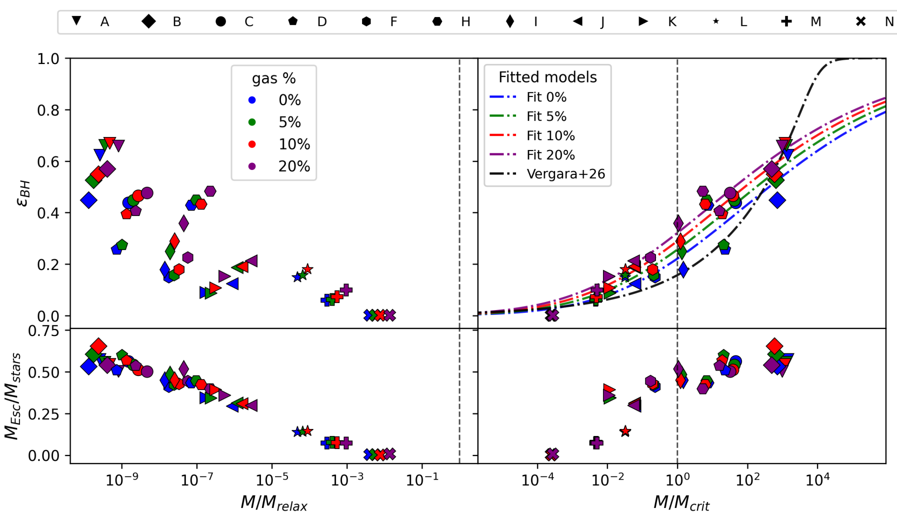

Fig. 7: In the y-axis, top panel shows the BH efficiency ($\epsilon_{\rm BH}$), and bottom panel the normalized escaped mass ($M_{\rm ESC}/M_{\rm stars}$). The left column shows quantities as a function of $M/M_{\rm relax}$, while the right column shows quantities as a function of $M/M_{\rm crit}$. Symbols correspond to the different models (see legend above the figure), and colors indicate the gas fraction: blue (0%), green (5%), red (10%), purple (20%). Dashed black lines mark mass ratios equal to unity. The dash-dotted curves in the top-right panel are fits of an asymmetric sigmoid function (Eq. (15)) for each gas fraction, sharing the same shape parameters x_0, k, a (fitted to the stars-only data) while α_q is adjusted for each gas fraction. The black dash-dotted curve is the model from Vergara et al. (2026)

single parameter α_q. The resulting parameters are $x_0 = 0.039$, $k = 0.349$, $a = -3.163$ and $\alpha_q = 26.7,\ 35.6,\ 47.8$ for $q = 0.05,\ 0.1,\ 0.2$, respectively. The fitted function shows three regimes. For $M/M_{\rm crit} < 10^{-4}$, the efficiency is effectively 0. Between 10^{-4} and 10^5, the efficiency increases approximately linearly with $\log_{10}(M/M_{\rm crit})$, reaching 70-80%. For $M/M_{\rm crit} > 10^5$, it asymptotically approaches unity. The main differences appear in the intermediate regime, where the efficiency increases with gas fraction. Compared with the fitted model of Vergara et al. (2026), our models exhibit a smoother transition between the regimes described above.

Table 3 lists the mean values for the normalized escaped mass and efficiencies, where each gas variant is normalized by its corresponding stars-only counterpart. The results are consistent with Fig. 7. For the escaped mass, the means are within one standard deviation of unity for all gas fractions, indicating no significant difference. In particular, for the 20% gas case, the mean ratio is closest to 1 and the dispersion is the smallest among all gas fractions. The black hole mass ratios show a clearer trend. The mean ratios are 1.0840, 1.2252 and 1.4414 for 5%, 10%, and 20% gas, corresponding to increases of 8.40%, 22.52%, and 44.14% relative to the stars-only cases. Since the variance in escaped mass is negligible compared to the increases in BH mass, the efficiency ratios follow a similar trend, with mean values of 1.1256, 1.2701, and 1.4510 (increases of 12.56%, 27.01%, and 45.10%) for 5%, 10%, and 20% gas, respectively.

Table 3: Ratios of gas runs to stars-only runs. Values are means ± standard deviation over all models.

Gas fraction	$\varepsilon_{\mathrm{BH,gas}}/\varepsilon_{\mathrm{BH,0}}$	$M_{\mathrm{BH}}/M_{\mathrm{BH,0}}$	$M_{\mathrm{esc}}/M_{\mathrm{esc,0}}$
5%	1.1256 ± 0.1627	1.0840 ± 0.1589	1.0471 ± 0.0633
10%	1.2701 ± 0.2079	1.2251 ± 0.2057	1.0442 ± 0.0905
20%	1.4510 ± 0.3139	1.4414 ± 0.2132	1.0042 ± 0.0869

5. Discussion

In this work, we have performed simulations for a set of initial conditions that include coupled star-gas variants. We studied the effect of gas on SMBH seed formation in nuclear stellar clusters. Using the Astrophysical Multipurpose Software Environment (AMUSE), we coupled an N-body integrator (PH4) with an SPH code (FI) via the BRIDGE scheme. In this implementation, the interaction between gas and stars is purely gravitational. Using the initial conditions from Vergara et al. (2023) with different gas fractions, we tested the scenario proposed by Escala (2021). Although our stars-only simulations show some differences with respect to Vergara et al. (2023), the simulations with gas show an enhancement in black hole mass and efficiency, as well as an acceleration of core collapse.

5.1. The role of gas on SMBH formation

Stellar collisions in dense star clusters have been studied extensively in N-body simulations. Portegies Zwart et al. (1999) simulated young star clusters and showed that collisions are more frequent than expected from simple timescale estimates, and that they rejuvenate stars, extending their lifetimes before supernova explosions. As stars interact, massive stars tend to reach the core due to mass segregation over the half-mass relaxation timescale (Spitzer 1987). Mass segregation also implies that massive objects, such as binaries or massive stars, can return to the core after being ejected by strong encounters, as a result of DF (Spitzer 1987; Hut et al. 1992). The products of stellar collisions lose kinetic energy, causing them to sink toward the center. DF and mass segregation explain why most collisions involve a single object, and therefore make this an effective mechanism for BH formation. Numerical simulations have shown that energy equipartition is not reached under realistic initial conditions (Trenti & van der Marel 2013). This implies that massive stars remain dynamically decoupled from the lighter population, continuing to segregate to the core and enhancing the collision rate, which favors runaway growth and the formation of a massive central object. Comparing our stars-only results with those of Vergara et al. (2023), we find differences in BH mass ranging from 0% to 36% (or 0% to 6% of the initial star cluster mass), with the largest discrepancies occurring in the most collisional models (A, B, and C).

For family models A, B, C, and H, which lie in the IR, the BH masses are 13 300, 5 250, 1 920, and 12 100 $M_\odot$, respectively, placing them in the IMBH mass range ($\sim 10^2$–10^5 $M_\odot$). Although models L and M are in the SR, their BH masses are 64 000 and 5 510 $M_\odot$, respectively. This is because models L and M are the most massive clusters; despite being in the SR, their large total mass still allows the formation of a massive BH. In contrast, they show lower efficiencies, meaning that the dynamics are controlled by the star cluster. If we compare models with the same initial mass, we observe that larger M/M_{crit} corresponds to higher BH mass and efficiency. For example, family models C, F, and N, all with an initial mass of 10^4 $M_\odot$, form BHs with masses of 1 920, 88, and 3 $M_\odot$, and efficiencies of 43.84%, 13.67%, and 0.03%, respectively. This suggests that more massive clusters (eventually in the CR) could form even more massive BHs, with $\epsilon_{\mathrm{BH}} \sim 1$.

The gas can affect the cluster dynamics through two distinct mechanisms. When considered as a static background potential, it increases the stellar velocities, causing a delay in core collapse. As shown by Reinoso et al. (2020), this delay also affects the runaway growth, although for sufficiently long times the BH formation is enhanced, because the cluster maintains its compactness. On the other hand, a realistic gas can exert DF on stars through gas-star interactions (Ostriker 1999; Reinoso et al. 2020; Leigh et al. 2014). In our simulations, the stellar velocity dispersion is transonic ($\sigma \gtrsim c_s$), as shown in Figs. E.3 and E.4. In this regime, DF is expected to be particularly efficient (Ostriker 1999). While our simulations do not fully resolve the DF (see Appendix D.3), the dissipation of kinetic energy due to gas-star interactions is still present. Consistent with this, our simulations show that the efficiencies are enhanced, with increases of 12.56 ± 16.27%, 27.01 ± 20.79%, and 45.10 ± 31.39% relative to the stars-only simulations for gas fractions of 5%, 10%, and 20%, respectively. For the BH mass, the corresponding increases are 8.40 ± 15.89%, 22.51 ± 20.57%, and 44.14 ± 21.32%. We also find shorter core collapse times. On average, the core collapse time is reduced to 93.77 ± 18.36%, 84.48 ± 12.32% and 72.61 ± 8.99% relative to the stars-only variants for gas fractions of 5%, 10% and 20%, respectively. These results show that IMBHs can form in less massive clusters if a gaseous component is present. Although we use a modified critical mass relation compared with Vergara et al. (2023), the fitted function (Eq. (15)) describes the trend of BH efficiency ϵ_{BH} as a function of M/M_{crit}.

5.2. Implications for SMBH formation

We have shown that in our simplified models, the formation of an IMBH is possible, with gas playing an important role by enhancing the number of collisions and, consequently, the final mass of the BH. However, an important question is how massive a BH can become in more realistic NSCs. The sizes and masses of NSCs have been studied extensively (Neumayer et al. 2020), with masses ranging from 10^5 to 10^8 $M_\odot$. Using the scaling relations from Georgiev et al. (2016), a nuclear star cluster of mass $\sim 10^7$ $M_\odot$ has a minimum effective radius of $\sim$ 4 pc. We estimate the critical mass ratio M/M_{crit} for a cluster of equal-mass stars ($m = 1$ $M_\odot$, $r = 1$ $R_\odot$), finding $M/M_{\mathrm{crit}} = 4.3 \times 10^{-5}$ and 6.9×10^{-5} for $t_{\mathrm{H}} = 5$ Myr and 10 Myr, respectively. Using Eq. (15), the predicted BH efficiencies are ~ 0.014 and ~ 0.017. Since $M/M_{\mathrm{crit}} \sim 10^{-5}$ and $M/M_{\mathrm{relax}} > 1$ (see Fig. 7), we can assume no mass loss, and the upper limits on the BH mass are $\sim 1.5 \times 10^5$ $M_\odot$ and $\sim 1.7 \times 10^5$ $M_\odot$, respectively. When including a gas fraction of 20% and following the same calculation, the BH efficiencies increase to ~ 0.034 and ~ 0.039 for $t_{\mathrm{H}} = 5$ Myr and 10 Myr, respectively, yielding upper limits of $\sim 3.4 \times 10^5$ $M_\odot$ and $\sim 3.9 \times 10^5$ $M_\odot$. While the collisional and relaxation timescales are global quantities, the core of a nuclear star cluster could reach densities high enough to collapse into a massive BH, even as the outer stars expand due to relaxation processes (Escala 2021). This suggests that NSCs should have been more compact at their formation, consistent with the observed coexistence of NSCs and SMBHs in intermediate-mass galaxies (Georgiev et al. 2016; Kritos et al. 2023; Escala 2021), and with the absence of SMBHs in low-mass NSCs. Simulations of the most compact clusters, with radius R$\leq$ 0.5 pc and mass 10^7 $M_\odot$, show that a million solar mass black hole can form within 100 Myr (Kritos et al. 2023). For such a system (M= 10^7 $M_\odot$, R= 0.5 pc) composed of stars only, the efficiencies are 0.068 and 0.077, for $t_{\rm H}$ = 5 Myr and 10 Myr, respectively. The escaped mass fraction is negligible, and the BH mass is estimated to be $6.7{\times}10^5$ $M_\odot$ and $8.9{\times}10^5$ $M_\odot$. For a time of 100 Myr, the mass ratios are $M/M_{\rm crit}$ = 0.042 and $M/M_{\rm relax}$ = 1.080. The corresponding efficiency is 0.114, and the escaped mass fraction is $\sim$ 0.1 (estimated from Fig. 7; see Section 5.1). In consequence, the estimated BH mass is 1.024×10^6 $M_\odot$. Including a 20% of gas fraction, the estimated efficiencies are 0.126 and 0.140 for $t_{\rm H}$ = 5 Myr and 10 Myr, respectively. Despite the simplifications, our estimates show that in more massive NSCs it is possible to form a SMBH of $\sim 10^6$ $M_\odot$, with gas playing an important role (increasing the BH mass by approximately 50%). However, for realistic NSCs the predicted efficiencies are low ($\leq$ 5%), indicating that the BH would represent only a small fraction of the total NSC mass. For more compact systems ($R \lesssim$ 0.5 pc), a SMBH can form on a timescale of 100 Myr without gas, and within 10 Myr with a 20% of gas fraction.

Several mechanisms have been proposed in the literature for the growth of IMBHs to masses in the SMBH range. Pelupessy et al. (2007) simulated halos containing IMBHs formed from Population III remnants and found that in collapsing halos with $M_{\rm halo} \lesssim 10^{10}$ $M_\odot$, accretion at the Eddington rate is not achieved due to stellar feedback. Even in the most massive halos, the growth of SMBH seeds is driven principally by BH–BH mergers. They propose that the BH mass scales with the host halo mass as $M_{\rm BH} = 5 \times 10^6$ $M_\odot(M_{\rm halo}/10^{11}$ $M_\odot)^{0.78}$. In contrast, Shi et al. (2024) describe a scenario in which subclusters and IMBHs form within a gas cloud complex of mass 10^8 $M_\odot$. The hierarchical merging of subclusters occurs on a timescale of $\lesssim$ 1 Myr, and the BHs in the subclusters migrate to the center within the same period. The formation of a disk ensures conditions for large accretion rates ($\sim$ 1–10 $M_\odot{\rm yr}^{-1}$), allowing an IMBH to grow to an SMBH of $\gtrsim 2 \times 10^6$ $M_\odot$ within 1 Myr. Chung et al. (2026) showed through simulations that an IMBH can form in dense stellar clusters, with masses ranging from $\sim$ 300 to $\sim$ 5000 $M_\odot$ via runaway collisions. The resulting BHs can grow to $\sim 6 \times 10^3$ $M_\odot$ on a timescale of $\sim$ 10 Myr through Eddington-limited gas accretion and TDEs. If a steady gas supply is assumed, the BHs can reach masses of $\sim 6 \times 10^5$ $M_\odot$, resulting in promising SMBH seeds. These mechanisms are consistent with the formation of IMBHs in dense NSCs (such as those produced in our simulations) and suggest that gas accretion and IMBH mergers provide a viable pathway for seeds to grow to the SMBH regime.

5.3. Caveats

In this section, we summarize the simplifications made in our simulations, which should be taken into account for improvements in future work. As we described in Section 3.2, the gas is evolved using an SPH method, with self-gravity and hydrodynamics, but without radiative cooling or heating. Radiative cooling allows the gas to lose thermal energy, which can help it remain gravitationally bound for longer periods. As shown by Silich & Tenorio-Tagle (2018), in sufficiently massive and compact clusters, radiative cooling can prevent the development of a global star cluster wind, thus suppressing gas expulsion. Moreover, recent work by Schleicher et al. (2026) suggests that in the most compact and massive young clusters detected by JWST ($M \gtrsim 6 \times 10^6 M_\odot$), supernova feedback may be insufficient to expel the gas, allowing it to be retained and potentially enhance both DF and Bondi accretion onto the central object. While our simulations do not include these dissipative processes, the findings of Schleicher et al. (2026) suggest that the gas-driven enhancement of runaway collisions we observe may be even stronger in more realistic scenarios. In addition, gas accretion could contribute to enhancing BH formation, since it increases the stellar density (Bonnell & Bate 2002; Reinoso et al. 2018). Consequently, the fraction of collisions can increase to 10%, compared with gas-free models. On the other hand, stellar winds and supernovae (SNe) contribute to the chemical enrichment of the remnant gas after star formation episodes. Although the gas is heated, it does not necessarily evaporate from the cluster. As shown by Silich & Tenorio-Tagle (2016), depending on the gas density, the remnant gas can be expelled from the cluster or remain, in which case it can contribute to the runaway collision process. Moreover, in the transonic regime where the stellar velocity dispersion is comparable to the gas sound speed, DF is particularly efficient (Ostriker 1999), contributing to mass segregation. To assess the realism of our simplifications, it is important to consider the timescales of the relevant processes. Runaway collisions begin near core collapse ($t_{\rm cc}$), and a massive object is formed at approximately the global collisional timescale $t_{\rm coll}$. Any process that evaporates the gas on a timescale $t \lesssim t_{\rm cc}$ has an important impact on the runaway collision process, while for timescales $t \gtrsim t_{\rm coll}$, the runaway is not significantly perturbed. In the intermediate regime, $t_{\rm cc} \leq t \leq t_{\rm coll}$, the influence may be relevant.

Some studies have investigated the timescales for gas expulsion in embedded clusters. Dinnbier & Walch (2020) studied gas expulsion from embedded star clusters, where the stellar mass is one third of the total mass (i.e., a gas fraction of q = 2). Their results show that for clusters with total mass $M_{\rm cluster} \lesssim 3 \times 10^3$ $M_\odot$, the gas expulsion timescale is $t_{\rm ge} \gtrsim 0.9$ Myr primarily through ionizing radiation and stellar winds. This mass range corresponds to models D, F, J, and K, for which $t_{\rm coll}$ (stars-only) is 1.018, 10.829, 27.032, and 61.53 Myr, respectively. We note that model D, which is in the IR, would evolve with a gas component in a more realistic scenario, at least during the runaway collision process. For more massive clusters, only 10% to 30% of the total mass could become unbound, and the majority could be ionized on a timescale of $\sim$ 1 Myr. In contrast, Calura et al. (2015) ran simulations of globular clusters with a total mass of $\sim 10^7$ $M_\odot$. Their hydrodynamical simulations included stellar feedback (winds and supernovae), radiative cooling and self-gravity. They found that after 3 Myr, stellar feedback removes 40% of the initial gas, and after 14 Myr, 99% is removed. Similarly, Pelupessy & Portegies Zwart (2012) showed that for relatively small embedded clusters, the gas could be expelled by different mechanisms. While high feedback coupling could remove gas on a timescale of $\sim$ 5 Myr, low feedback coupling is not sufficient to expel the gas via stellar winds, and supernovae become the dominant mechanism (their first supernova events occur at $\sim$ 10 Myr). More recently, Chung et al. (2026) showed that in hydrodynamical simulations of dense star cluster formation that include cooling, heating, and stellar feedback, the gas can be retained in sufficiently compact and massive clusters, affecting the IMBH formation process via runaway collisions. Taken together, these studies suggest that gas can persist during the core collapse phase in many realistic scenarios, particularly in low-mass clusters or with low stellar feedback. However, the survival of the gas depends on the details of gravity and stellar feedback, which we have not modeled. Therefore, although it seems plausible that gas could persist during an important part of the runaway collisions and could effectively enhance the formation of an IMBH, more realistic simulations that include stellar feedback, supernovae, and radiative cooling are necessary.

Another simplification concerns the initial conditions. Stars are initialized using a Plummer distribution, with equal-mass stars in virial equilibrium. The gas is initialized using a Plummer distribution with the same spatial distribution as the stars. Although the Plummer distribution is consistent with observations (Plummer 1911) and simulations (Chung et al. 2026), it is necessary to include an IMF for the stellar mass distribution. On the other hand, cosmological simulations could provide more realistic initial conditions, including stellar and gas distributions and the remnant gas fraction. Our most massive model (L family models) has a total stellar mass of $5\times10^5\ M_\odot$ and is therefore not able to form an SMBH. Modeling more massive systems would require a code more efficient than PH4 in AMUSE, and capable of detecting stellar collisions. Furthermore, as shown in Appendix D.3, DF is not fully resolved in our simulations, and the number of SPH particles within the gravitational radius is negligible. We therefore caution that gas DF is not fully captured. To achieve realistic DF, it would be necessary to increase the resolution of the SPH particles or to include a sub-grid model for unresolved gas DF, which could enhance the formation of an SMBH. As mentioned in Pelupessy & Portegies Zwart (2012) and Whitehead et al. (2013), small differences in initial conditions are amplified due to the chaotic nature of N-body systems. Therefore, future work should perform multiple runs with different random realizations for each set of initial conditions. Although the general results will remain, this would allow us to obtain statistics for the trends shown in Section 4.

6. Conclusions

In this work, we have studied the role of gas in the collisional runaway scenario for SMBH seed formation in nuclear stellar clusters. By performing a series of simulations with coupled N-body and SPH codes via the BRIDGE scheme, we have shown that the presence of a gaseous component, even at small fractions (5%–20% of the stellar mass), significantly affects the formation of a massive central object.

Our main findings are:

1. The gas shortens core collapse, particularly for gas fractions of 10% and 20%, with the strongest effect at 20% (reduction of 27.39% ± 8.99% relative to gas-free simulations). The effect is less robust at 5% gas fraction (6.23% ± 18.36%), suggesting that a minimum gas fraction is required for a significant effect.
2. The presence of gas enhances black hole formation. In our simulations, the BH mass increases by 8.40% ± 15.89%, 22.51% ± 20.57%, and 44.14% ± 21.32% for gas fractions of 5%, 10%, and 20%, respectively, and the BH formation efficiency shows a similar trend. The enhancement is most pronounced in the models with the highest collision rates within the IR (A, B, C, H), where the density is highest and the gas can most effectively contribute to the formation of an IMBH.
3. The fitted model for BH formation efficiency shows that the effect of gas is most important in the intermediate range $10^{-4} \leq M/M_{\rm crit} \leq 10^5$ where the efficiency increases with gas fraction. Since gas dynamical friction is not fully resolved in our simulations, this enhancement may be a conservative estimate, and higher resolution simulations could reveal an even stronger effect.

While our simulations demonstrate the potential of gas to enhance SMBH seed formation, several simplifications remain to be addressed. Our gas model is adiabatic and does not include radiative cooling, heating, stellar feedback, or supernovae. Our findings address the question of how gas affects the global instability that leads to runaway collisions and, eventually, to the formation of a SMBH. The next natural question is whether gas can remain during relevant timescales in stellar clusters. Future work should include these processes to determine whether the gas persists long enough during the core collapse phase to produce the effects observed here. Additionally, more realistic initial conditions (e.g., IMFs, cosmological initial conditions) and larger N-body systems (using more efficient codes, such as NBODY6++GPU) would allow us to confirm the robustness of our results and extend them to more massive clusters. The IMBHs formed in our simulations could serve as seeds that subsequently grow to SMBH masses via gas accretion and mergers, as suggested by recent studies (Inayoshi et al. 2022; Kritos et al. 2023; Chung et al. 2026).

We conclude that gas is a key ingredient in the collisional runaway scenario for SMBH seed formation, and its inclusion in future simulations is essential for a complete understanding of the formation of massive black holes in nuclear stellar clusters.

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Appendix A: Critical mass

In this section, we present a solution of the implicit equations for the critical mass presented in Vergara et al. (2023), where virialized system is assumed. Also, we consider an external potential (i.e., gas). We begin with the derivation of the collisional timescale. For a system with N identical stars of mass m and radius $r_\star$, velocity dispersion σ, total mass M and characteristic radius R, the collision rate is given by (Binney & Tremaine 2008):

$$\frac{1}{t_{\rm coll}} = 16\sqrt{\pi} n \sigma r_\star^2 (1+\Theta)\,, \tag{A.1}$$

where Θ is the Safronov number:

$$\Theta \equiv \frac{Gm}{2\sigma^2 r_\star} = 9.54 \frac{m}{M_\odot}\frac{R_\odot}{r_\star}\left(\frac{100\ \mathrm{km\ s^{-1}}}{\sigma}\right)^2\,, \tag{A.2}$$

and $n = 3M/(4\pi R^3 m)$ is the number density. Defining the cross-section $\Sigma_0 = 16\sqrt{\pi} r_\star^2(1+\Theta)$, the collision time can be rewritten as:

$$t_{\rm coll} = \frac{1}{n\sigma\Sigma_0}. \tag{A.3}$$

We consider a stellar system in virial equilibrium with characteristic radius R. In the absence of an external potential, the velocity dispersion satisfies $\sigma = \sqrt{GM/R}$. When the system is embedded in a gravitationally bound external potential of mass $M_{\rm ext}$ with the same distribution (same characteristic radius R), the modified dispersion velocity yields (Reinoso et al. 2020):

$$\sigma_{\rm ext} = \sqrt{\frac{GM}{R}(1+q)}\,, \tag{A.4}$$

where $q = M_{\rm ext}/M$ and $M + M_{\rm ext} = M(1+q)$ is the total mass. First, we replace the dispersion velocity (Eq. (A.4)) into the cross-section Σ_0:

$$\begin{aligned}
\Sigma_0 =& 16\sqrt{\pi} r_\star^2 (1+\Theta) \\
=& 16\sqrt{\pi} r_\star^2 \left(1 + 9.54\frac{m}{M_\odot}\frac{R_\odot}{r_\star}\left(\frac{v_{100}}{\sigma_{\rm ext}}\right)^2\right) \\
=& 16\sqrt{\pi} r_\star^2 + \frac{9.54\times 16\sqrt{\pi} r_\star m R_\odot v_{100}^2}{M_\odot}\frac{R}{GM(1+q)^2}\,.
\end{aligned} \tag{A.5}$$

From the relation $t_{\rm coll} = t_H$, we place all mass-dependent terms on left side:

$$\begin{aligned}
t_H =& t_{\rm coll} = \frac{1}{n\sigma_{\rm ext}\Sigma_0} \\
=& \frac{4\pi R^3 m}{3M}\frac{\sqrt{R}}{\sqrt{GM}(1+q)\Sigma_0} \\
\Sigma_0 M^{3/2} =& \frac{4\pi R^3 m}{3 t_H}\frac{\sqrt{R}}{\sqrt{G}(1+q)}\,.
\end{aligned} \tag{A.6}$$

Now, we replace Eq. (A.5) into Eq. (A.6):

$$\begin{aligned}
16\sqrt{\pi} r_\star^2 M^{3/2} + \frac{9.54\times 16\sqrt{\pi} r_\star m R_\odot v_{100}^2 R}{M_\odot G(1+q)^2} M^{1/2} =& \frac{4\pi m R^{7/2}}{3G^{1/2}(1+q)t_H} \\
M^{3/2} + \frac{9.54 m R_\odot v_{100}^2 R}{r_\star G M_\odot (1+q)^2} M^{1/2} =& \frac{\sqrt{\pi} m R^{7/2}}{12 G^{1/2} r_\star^2 (1+q) t_H}\,.
\end{aligned} \tag{A.7}$$

In order to simplify, we define the variables:

$$\alpha R = A = \frac{9.54 m R_\odot v_{100}^2}{r_\star G M_\odot (1+q)^2} R \tag{A.8}$$

$$\beta R^{7/2} = C = \frac{\sqrt{\pi} m}{12 G^{1/2} r_\star^2 (1+q) t_H} R^{7/2}\,. \tag{A.9}$$

Then, the equation for the mass is

$$M^{3/2} + A M^{1/2} - C = 0\,. \tag{A.10}$$

A.1. solving for M

The critical mass can be obtained solving Eq. (A.10). To simplify, we use the change of variable $u^2 = M$ to have a cubic polynomial. Therefore, Eq. (A.10) can be rewritten as:

$$u^3 + Au - C = 0\,. \tag{A.11}$$

This equation could have three roots, but we are only interested in real solutions. Given $p = A/3$ and $q = -C$, the discriminant of Eq. (A.11) is:

$$\Delta = 4p^3 + q^2 = \frac{4A^3}{27} + C^2\,. \tag{A.12}$$

As $A, C > 0$ is straight that the discriminant is positive, then there is only one real solution (Zwillinger 2002):

$$u_{\rm real} = \sqrt[3]{s_1} + \sqrt[3]{s_2} \tag{A.13}$$

$$s_1 = \frac{-q + \sqrt{q^2 + 4p^3}}{2} \tag{A.14}$$

$$s_2 = \frac{-q - \sqrt{q^2 + 4p^3}}{2}\,. \tag{A.15}$$

Replacing p and q by $A/3$ and $-C$:

$$\begin{aligned}
u_{\rm real} =& \sqrt[3]{\frac{-q + \sqrt{q^2+4p^3}}{2}} + \sqrt[3]{\frac{-q - \sqrt{q^2+4p^3}}{2}} \\
=& \sqrt[3]{\frac{C}{2} + \sqrt{\frac{C^2}{4} + \frac{A^3}{27}}} + \sqrt[3]{\frac{C}{2} - \sqrt{\frac{C^2}{4} + \frac{A^3}{27}}}\,.
\end{aligned} \tag{A.16}$$

Then, the critical mass is:

$$\begin{aligned}
M_{\rm crit} =& u_{\rm real}^2 \\
=& \left(\frac{C}{2} + \sqrt{\frac{C^2}{4} + \frac{A^3}{27}}\right)^{2/3} + \left(\frac{C}{2} - \sqrt{\frac{C^2}{4} + \frac{A^3}{27}}\right)^{2/3} + \\
& 2\sqrt[3]{\frac{C^2}{4} - \left(\frac{C^2}{4} + \frac{A^3}{27}\right)} \\
=& \left(\frac{C}{2} + \sqrt{\frac{C^2}{4} + \frac{A^3}{27}}\right)^{2/3} + \left(\frac{C}{2} - \sqrt{\frac{C^2}{4} + \frac{A^3}{27}}\right)^{2/3} - \\
& \frac{2A}{3}\,,
\end{aligned} \tag{A.17}$$

or in terms of α and β:

$$\begin{aligned}
M_{\rm crit} =& \left(\frac{\beta R^{7/2}}{2} + \sqrt{\frac{\beta^2 R^7}{4} + \frac{\alpha^3 R^3}{27}}\right)^{2/3} + \\
& \left(\frac{\beta R^{7/2}}{2} - \sqrt{\frac{\beta^2 R^7}{4} + \frac{\alpha^3 R^3}{27}}\right)^{2/3} - \frac{2\alpha R}{3}\,.
\end{aligned} \tag{A.18}$$

A.2. Analytical vs numerical solution

To test the analytical solution for the critical mass, we compare it with a numerical solution obtained by solving the implicit condition iteratively. Fig. A.1 shows the relative error between the numerical and analytical solutions as a function of the characteristic radius R. The relative difference between the numerical and

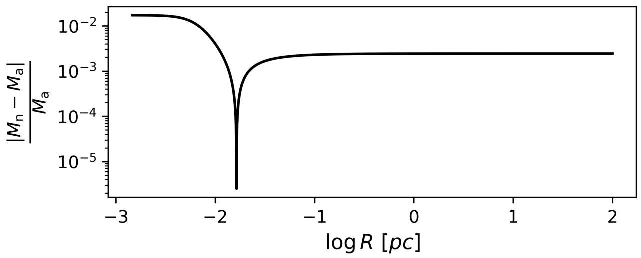

Fig. A.1: Relative error between the analytical solution (M_a) and the numerical solution (M_n) for the critical mass

analytical solutions is at most $\sim 1\%$ for small radii, decreasing to $\sim 0.1\%$ for larger radii. For all practical purposes, the analytical expression (Eq. (A.18)) is accurate and numerically stable for $M_{crit} \gtrsim 10\ M_\odot$.

Appendix B: Relaxation mass

In this section we derive a characteristic mass from the two-body relaxation timescale, in the presence of an external potential. Consider a system of equal N stars of mass m, with characteristic radius R and velocity dispersion σ. A star crossing the system suffers a mean-square velocity deflection given by (Binney & Tremaine 2008):

$$\Delta v^2 \simeq 8N \left(\frac{Gm}{Rv}\right)^2 \ln\left(\frac{b_{max}}{b_{min}}\right) , \tag{B.1}$$

where b_{max}, b_{min} are the maximum and minimum impact parameters. Usually, $b_{max} \simeq R$, and $b_{min} = 2Gm/v^2$ is the impact parameter for a 90° deflection. The number of crossings needed for the star to change its velocity significantly is therefore

$$n_{relax} = \frac{v^2}{\Delta v^2} = \left(\frac{Rv^2}{Gm}\right)^2 \frac{1}{8N \ln(b_{max}/b_{min})} . \tag{B.2}$$

As shown in Appendix A, the velocity dispersion in the presence of an external potential of mass $M_{ext} = qM$ (with the same spatial distribution) becomes $\sigma_{ext} = \sqrt{GNm/R}\,(1+q)$, where $M = Nm$ is the stellar mass. Setting $v = \sigma_{ext}$ we obtain $Rv^2/(Gm) = RGNm(1+q)^2/(RGm) = N(1+q)^2$ and

$$\begin{aligned}
\ln\left(\frac{b_{max}}{b_{min}}\right) &= \ln\left(\frac{R}{(2Gm/v^2)}\right) = \ln\left(\frac{Rv^2}{2Gm}\right) \\
&= \ln\left(\frac{N(1+q)^2}{2}\right) = \ln(N(1+q)^2) - \ln(2) \\
&\simeq \ln(N(1+q)^2) ,
\end{aligned} \tag{B.3}$$

where we have dropped the constant ln(2) for simplicity. Substituting into the expression for n_{relax} yields

$$n_{relax} = \frac{N(1+q)^4}{8\ln(N(1+q)^2)} \simeq \frac{0.1N(1+q)^4}{\ln(N(1+q)^2)} . \tag{B.4}$$

Since n_{relax} is the number of crossing times required for relaxation, the relaxation time is $t_{relax} = n_{relax} t_{cross}$, where $t_{cross} = R/\sigma_{ext}$. Thus,

$$t_{relax} = n_{relax} t_{cross} = \frac{0.1N(1+q)^4}{\ln(N(1+q)^2)} t_{cross} . \tag{B.5}$$

We now impose that the characteristic time t_H is shorter than the relaxation time ($t_H \leq t_{relax}$), which is the condition for the system to retain its initial properties (two-body relaxation is not yet effective). Similar to Vergara et al. (2023), we obtain a lower bound on the cluster radius:

$$\begin{aligned}
t_H &\leq \frac{0.1N(1+q)^4}{\ln(N(1+q)^2)} t_{cross} = \frac{0.1N(1+q)^4}{\ln(N(1+q)^2)} \frac{R}{\sigma_{ext}} \\
&= \frac{0.1M(1+q)^4}{m\ln(M(1+q)^2/m)} \frac{R^{3/2}}{\sqrt{GM}(1+q)} \\
&= \frac{0.1M^{1/2}(1+q)^3}{\ln(M(1+q)^2/m)} \frac{R^{3/2}}{G^{1/2}m} \\
R &\geq \left(\frac{t_H m}{0.1(1+q)^3} \ln\left(\frac{M(1+q)^2}{m}\right)\right)^{2/3} \left(\frac{G}{M}\right)^{1/3} .
\end{aligned} \tag{B.6}$$

B.1. Solving for the relaxation mass

Inequality (B.6) can be read as a relation $R(M)$. The relaxation mass is then defined as its inverse, $M_{relax}(R) = R^{-1}(M)$. Setting the equality gives

$$\begin{aligned}
R &= \left(\frac{t_H m \ln(M(1+q)^2/m)}{0.1(1+q)^3}\right)^{2/3} \left(\frac{G}{M}\right)^{1/3} \\
R^3 &= \left(\frac{t_H m \ln[M(1+q)^2/m]}{0.1(1+q)^3}\right)^2 \left(\frac{G}{M}\right) \\
M^{1/2}R^{3/2} &= \frac{m t_H \ln[M(1+q)^2/m]}{0.1(1+q)^3} G^{1/2} .
\end{aligned} \tag{B.7}$$

Considering the change of variable $u = M(1+q)^2/m$, Eq. (B.7) becomes

$$\begin{aligned}
\frac{(um)^{1/2}R^{3/2}}{1+q} &= \frac{m t_H \ln(u)}{0.1(1+q)^3} G^{1/2} \\
u^{1/2} &= \ln(u) \frac{t_H (Gm)^{1/2}}{0.1R^{3/2}(1+q)^2} .
\end{aligned} \tag{B.8}$$

Defining $\lambda = t_H(Gm)^{1/2}/(0.1R^{3/2}(1+q)^2)$, Eq. (B.8) yields:

$$u^{1/2} = \lambda \ln(u), \tag{B.9}$$

and introducing the change of variable $\omega = \ln(u)$

$$\begin{aligned}
e^{\omega/2} &= \lambda\omega \\
-1/(2\lambda) &= (-\omega/2)e^{-\omega/2} .
\end{aligned} \tag{B.10}$$

The inverse of the function $f(w) = we^w$ is the Lambert function W (i.e. $w = W(y)$ if $y = we^w$) (Weisstein 2002). Hence

$$\begin{aligned}
\frac{-\omega}{2} &= W\left(\frac{-1}{2\lambda}\right) \\
\omega &= -2W\left(\frac{-1}{2\lambda}\right) .
\end{aligned} \tag{B.11}$$

Returning to $u = e^{\omega}$ we obtain

$$u = \exp\left[-2W\left(\frac{-1}{2\lambda}\right)\right], \tag{B.12}$$

and finally the relaxation mass:

$$M_{\rm relax} = \frac{m}{(1+q)^2}\exp\left[-2W\left(\frac{-1}{2\lambda}\right)\right]. \tag{B.13}$$

Eq. (B.13) involves the Lambert function W evaluated at a negative argument $z = -1/(2\lambda)$ (since $\lambda > 0$). There are two real-valued branches, $W_0(z)$ and $W_{-1}(z)$ (Higham et al. 2015, Chapter III.17). The principal branch, $W_0(z)$, is defined for $z \in [-1/e, \infty)$. The lower branch $W_{-1}(z)$ is defined on $[-1/e, 0)$. Since $-1/(2\lambda) < 0$ and assuming $-1/e \leq -1/(2\lambda)$, both branches can be taken. From Fig. B.1, we see that for $z \in [-1/e, 0)$, the principal branch takes values in $[-1, 0]$, while $W_{-1}(z)$ takes values in $[-1, -\infty)$. Evaluating in Eq. (B.13), the

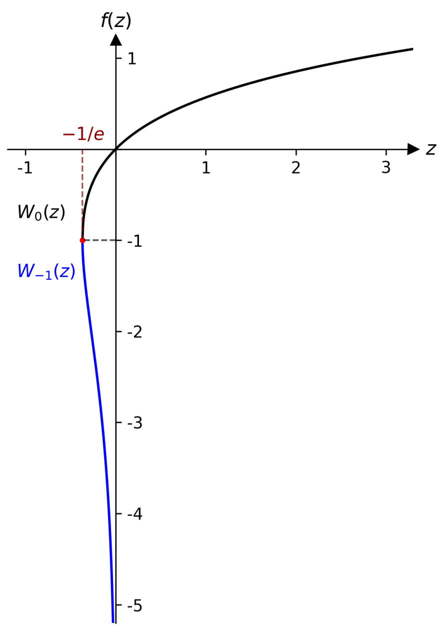

Fig. B.1: Lambert function. The black line shows the $W_0(z)$ branch, and the blue line shows the $W_{-1}(z)$ branch.

branch W_0 gives $me^2/(1+q) \geq M_{\rm relax} > 0$, while W_{-1} gives $me^2/(1+q) \leq M_{\rm relax} < \infty$. As we expect for the relaxation mass to be $M_{\rm relax} \gg m$, the physically relevant solution is the lower branch. Then, the relaxation mass can be written as:

$$M_{\rm relax} = \frac{m}{(1+q)^2}\ \exp\left[-2\,W_{-1}\left(-\frac{1}{2\lambda}\right)\right]. \tag{B.14}$$

A real solution exists only when $z \geq -1/e$, i.e. $\lambda \geq e/2$. For typical star clusters ($N \gg 1$, $q \lesssim 1$, and t_H large), this condition is amply satisfied, so the expression is well-defined and yields $M_{\rm relax} \gg m$.

Appendix C: Core Collapse Detection Method

In this section, we describe the implemented method for detecting the core collapse, using Lagrangian radii. The core collapse corresponds to the moment in which the inner Lagrangian radii reach their minimum (Di Cintio et al. 2022; Reinoso et al. 2020), followed by an expansion of the outer regions. This process is expected to occur on a timescale of $\sim 20t_{\rm relax}$ (Quinlan 1996), and can be observed from the evolution of Lagrangian radii. However, as the most massive BH reaches a mass comparable to that enclosed within a given Lagrangian radius, the evolution of that radius can be dominated by the motion of the BH rather than by the contraction of the stellar core. To avoid this, we use the smallest Lagrangian radius that is not dominated by the most massive BH. The algorithm proceeds as follows:

1. For a given Lagrangian radius $r_i(t)$, we select data within $t \leq 40t_{\rm relax}$.
2. We apply a Savitzky–Golay filter (implemented via the `savgol_filter` function from `scipy.signal`) with a quadratic polynomial to smooth the data using a smoothing window of size *sw*.
3. We define a sliding window of size $W = fN$, where f is a parameter in the range $[0, 1]$, and N is the number of data points.
4. For each window position, we perform two linear regressions: one on the first half of the data points in the window and another on the second half. We assign to the central time of the window the difference in slopes, computed as $\Delta m = m_{\rm right} - m_{\rm left}$.
5. After running over all the selected data, the core collapse time is identified as the time at which Δm reaches its maximum value. This corresponds to the point of maximum curvature change in the Lagrangian radius.

The window size is varied between 10% and 30% of the selected data. To preserve sufficient resolution, the smoothing window was set empirically to $ws = W/5$. The method was validated by visual inspection of the Lagrangian radii for each simulation, confirming that the identified $t_{\rm cc}$ consistently matches the inflection point in the $r_j(t)$ curves.

Appendix D: Code Setup and Performance

D.1. Code Parameters and Initial Conditions

Table D.1 summarizes the key parameters for the N-body code PH4 and the SPH code FI, from the AMUSE framework. Parameters not listed were left at their default values. Table D.2 show the initial conditions of each model.

D.2. Adaptive BRIDGE Timestep

The BRIDGE and FI codes use the same timestep, which is adjusted adaptively based on the local dynamical conditions of the gas and stars. Here, $t_{\rm nb}$ denotes the N-body time unit, and $t_{\rm relax}$ is the two-body relaxation time. The algorithm proceeds as follows:

1. The minimum free-fall time of the gas is computed as:

$$t_{\rm ff} = \sqrt{\frac{3\pi}{32G\rho_{\rm max}}}, \tag{D.1}$$

where $\rho_{\rm max}$ is the maximum density among SPH particles.

Table D.1: Key runtime parameters for PH4 and FI.

Code	Parameter	Value	Description
PH4	‘epsilon_squared‘	$(0.05\,R_\odot)^2$	Gravitational softening length squared.
	‘force_sync‘	‘True‘	Synchronizes forces across all particles.
	‘timestep_parameter‘	0.01	Controls the accuracy of the Hermite integrator.
FI	‘eps_is_h_flag‘	‘True‘	Sets gravitational softening equal to the SPH smoothing length.
	‘timestep‘	$1.0\ t_{\rm nb}$	Initial timestep (in N-body time units).
	‘radiation_flag‘	‘False‘	Disables radiative cooling and heating.
	‘use_hydro_flag‘	‘True‘	Enables SPH hydrodynamics.
	‘integrate_entropy_flag‘	‘False‘	Evolves specific internal energy (not entropy).
	‘isothermal_flag‘	‘False‘	Disables isothermal equation of state.
	‘star_formation_flag‘	‘False‘	Disables star formation.

2. A Courant-type timestep is computed from the accelerations between gas and star particles:

$$\Delta t_{\rm coup} = \min\left(f_{\rm coup}\sqrt{\frac{2h}{\mathbf{a}_{\rm g,s}}}, f_{\rm coup}\sqrt{\frac{\epsilon_{\rm nb}}{\mathbf{a}_{\rm s,g}}} \right) , \tag{D.2}$$

where h is the SPH smoothing length, $\epsilon_{\rm nb}$ is the coupling softening length, and $\mathbf{a}_{\rm g,s}$ and $\mathbf{a}_{\rm s,g}$ are the accelerations of gas and star particles due to the other component.
3. The reference timescale is then defined as:

$$\alpha = \min(t_{\rm ff}, \Delta t_{\rm coup}) . \tag{D.3}$$

4. The BRIDGE timestep is adjusted so that it satisfies:

$$\frac{\alpha}{2} \leq \Delta t_{\rm BRIDGE} \leq \alpha . \tag{D.4}$$

If $\Delta t_{\rm BRIDGE} < \frac{\alpha}{2}$, the timestep is multiplied by 2.

To maintain synchronization with the N-body integrator, the BRIDGE timestep is rounded to the nearest power of two of the N-body time ($t_{\rm nb}$):

$$\Delta t_{\rm BRIDGE} = 2^k t_{\rm nb} , \tag{D.5}$$

with k chosen such that the timestep lies within the range $[\alpha/2, \alpha]$. During the first 25 relaxation times ($t_{\rm relax}$), the maximum timestep is reduced to $2^{-4}\ t_{\rm nb}$ to improve temporal resolution during the core collapse phase. After this period, the timestep may grow up to the initial value of $1.0\ t_{\rm nb}$. For all simulations, $f_{\rm coup} = 0.5$ and $\epsilon_{\rm nb} = 5$ AU.

D.3. Dynamical friction

To verify that dynamical friction is well resolved, we check two conditions: (i) the gravitational radius should be larger than the coupling softening length ($\epsilon_{\rm nb}$ parameter); and (ii) the number of SPH particles within the gravitational radius for a typical star should be $\gtrsim 100$. The gravitational radius is given by

$$R_{\rm g} = GM/(c_{\rm s}^2 + v^2) , \tag{D.6}$$

where M is the mass of an individual star, $c_{\rm s}$ is the gas sound speed, and v is the velocity of the star. Among all models, the gravitational radius ranges between $\sim$ 0.1 AU and $\sim$ 10 AU, the first condition is only marginally satisfied for the most massive stars, and is not satisfied for lower-mass stars. Furthermore, the number of SPH particles within $R_{\rm g}$ is effectively zero for all models. We therefore conclude that dynamical friction from the gas is not fully resolved in our simulations. However, this implies that the gas-driven enhancement of core collapse and BH formation observed in our simulations may be even stronger if dynamical friction were fully resolved. Future work with higher resolution would be required to quantify this effect accurately.

D.4. Energy Conservation and Virialization

The validation of our results relies on the accuracy of energy conservation. Fig. D.1 shows, in the top panel, the error in total energy per timestep as a function of crossing time. The bottom panel shows the virial ratio, defined as the ratio of two times the total kinetic energy (including thermal energy of gas) to the absolute value of the gravitational potential energy, $2K/|U|$. The initial conditions for this test consist of a stellar system with virial radius $r_{\rm vir} = 1$ pc, composed of 5000 stars of 1 $M_\odot$ each, embedded in a gas cloud with the same virial radius, a total gas mass of 5000 $M_\odot$, and containing 5000 SPH particles. We observe that during the first 50 crossing times, the system remains virialized, with an energy error of $\lesssim 10^{-4}$.

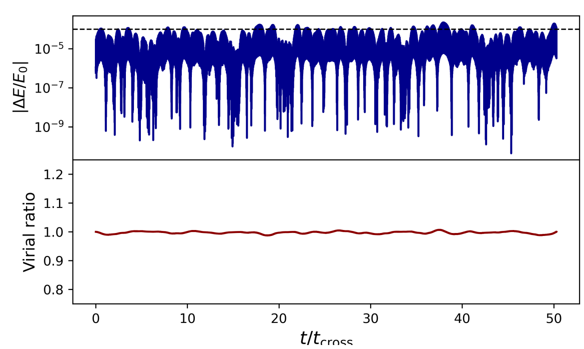

Fig. D.1: Numerical accuracy test for a system of 5000 stars of 1 $M_\odot$ embedded in a gas cloud with the same mass and 5000 SPH particles. Top panel: relative error in total energy per timestep as a function of crossing time. Bottom panel: virial ratio. The system remains virialized during the first 50 crossing times, with an energy error $\lesssim 10^{-4}$.

Appendix E: Extra plots

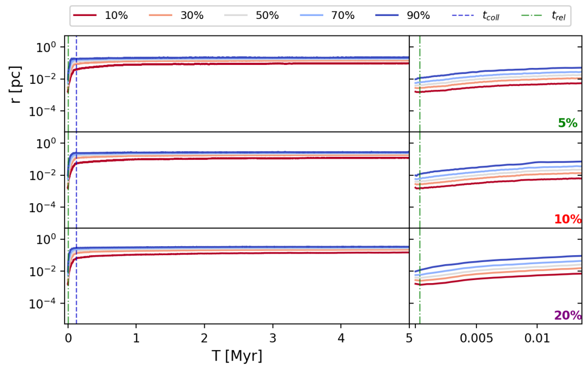

Fig. E.1: Lagrangian radii for gas of three variants of Model A. Top panel is for A_5, middle is for A_{10} and bottom panel is for A_{20}. Left column shows the full simulation time (5 Myr) while right column shows the evolution until 40 relaxation times.

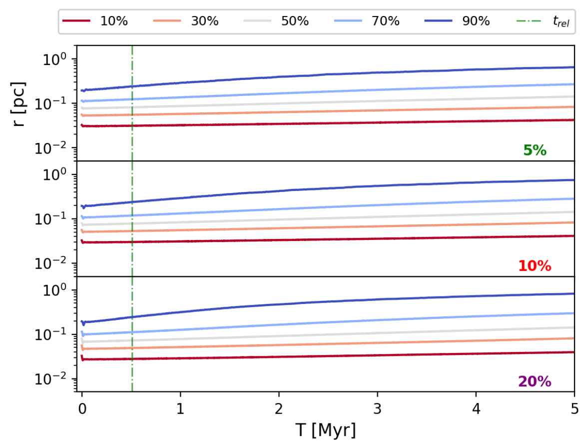

Fig. E.2: Lagrangian radii for gas of three variants of Model N. Top panel is for N_5, middle is for N_{10} and bottom panel is for N_{20}. Left column shows the full simulation time (5 Myr) while right column shows the evolution until 40 relaxation times.

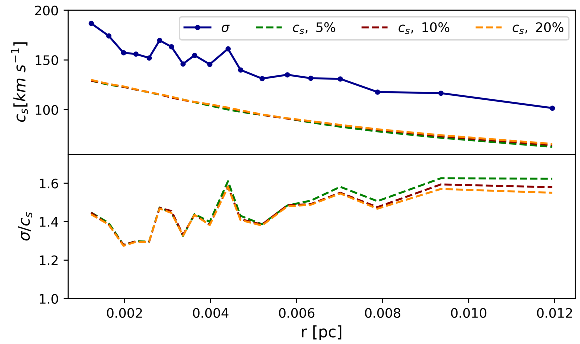

Fig. E.3: Sound speed of gas (c_s) and velocity dispersion (σ) for family A models.

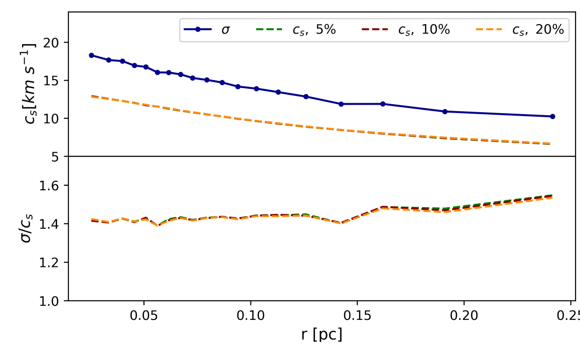

Fig. E.4: Sound speed of gas (c_s) and velocity dispersion (σ) for family N models.

Table D.2: Initial conditions for stellar clusters. R_v and M are the virial radius and total mass of the stellar cluster, r and m are the initial radius and mass of stars, N is the number of stars, $q_{\rm gas}$ is the gas fraction (gas mass divided by stellar mass), and σ is the velocity dispersion. For models with gas, $M_{\rm SPH}$ is the total gas mass, $N_{\rm SPH}$ is the number of SPH particles, and $m_{\rm SPH}$ is the mass per SPH particle, with $m_{\rm SPH} = m/100$. For $q_{\rm gas} = 0$, the gas-related columns are left blank.

ID	R_v[pc]	$M[M_\odot]$	N	$m[M_\odot]$	$r[R_\odot]$	$q_{\rm gas}$	$M_{SPH}[M_\odot]$	$N_{\rm SPH}$	$m_{\rm SPH}[M_\odot]$	$\sigma[\frac{\rm km}{\rm s}]$
A_0	0.005	5×10^4	10^3	50	11.7	0				146.66
A_5	0.005	5×10^4	10^3	50	11.7	0.05	2.5×10^3	5×10^3	0.5	146.66
A_{10}	0.005	5×10^4	10^3	50	11.7	0.1	5×10^3	10^4	0.5	146.66
A_{20}	0.005	5×10^4	10^3	50	11.7	0.2	10^4	2×10^4	0.5	146.66
B_0	0.005	2.5×10^4	5×10^2	50	11.7	0				103.71
B_5	0.005	2.5×10^4	5×10^2	50	11.7	0.05	1.25×10^3	2.5×10^3	0.5	103.71
B_{10}	0.005	2.5×10^4	5×10^2	50	11.7	0.1	2.5×10^3	5×10^3	0.5	103.71
B_{20}	0.005	2.5×10^4	5×10^2	50	11.7	0.2	5×10^3	10^4	0.5	103.71
C_0	0.005	10^4	10^3	10	4.7	0				65.59
C_5	0.005	10^4	10^3	10	4.7	0.05	5×10^2	5×10^3	0.1	65.59
C_{10}	0.005	10^4	10^3	10	4.7	0.1	10^3	10^4	0.1	65.59
C_{20}	0.005	10^4	10^3	10	4.7	0.2	2×10^3	2×10^4	0.1	65.59
D_0	0.005	5×10^3	5×10^2	10	4.7	0				46.38
D_5	0.005	5×10^3	5×10^2	10	4.7	0.05	2.5×10^2	2.5×10^3	0.1	46.38
D_{10}	0.005	5×10^3	5×10^2	10	4.7	0.1	5×10^2	5×10^3	0.1	46.38
D_{20}	0.005	5×10^3	5×10^2	10	4.7	0.2	10^3	10^4	0.1	46.38
F_0	0.005	10^3	10^3	1	1.0	0				20.74
F_5	0.005	10^3	10^3	1	1.0	0.05	5×10^1	5×10^3	0.01	20.74
F_{10}	0.005	10^3	10^3	1	1.0	0.1	10^2	10^4	0.01	20.74
F_{20}	0.005	10^3	10^3	1	1.0	0.2	2×10^2	2×10^4	0.01	20.74
H_0	0.01	5×10^4	5×10^3	10	4.7	0				103.71
H_5	0.01	5×10^4	5×10^3	10	4.7	0.05	2.5×10^3	2.5×10^4	0.1	103.71
H_{10}	0.01	5×10^4	5×10^3	10	4.7	0.1	5×10^3	5×10^4	0.1	103.71
H_{20}	0.01	5×10^4	5×10^3	10	4.7	0.2	10^4	10^5	0.1	103.71
I_0	0.01	10^4	10^3	10	4.7	0				46.38
I_5	0.01	10^4	10^3	10	4.7	0.05	5×10^2	5×10^3	0.1	46.38
I_{10}	0.01	10^4	10^3	10	4.7	0.1	10^3	10^4	0.1	46.38
I_{20}	0.01	10^4	10^3	10	4.7	0.2	2×10^3	2×10^4	0.1	46.38
J_0	0.01	5×10^3	5×10^3	1	1.0	0				32.79
J_5	0.01	5×10^3	5×10^3	1	1.0	0.05	2.5×10^2	2.5×10^4	0.01	32.79
J_{10}	0.01	5×10^3	5×10^3	1	1.0	0.1	5×10^2	5×10^4	0.01	32.79
J_{20}	0.01	5×10^3	5×10^3	1	1.0	0.2	10^3	10^5	0.01	32.79
K_0	0.01	10^3	10^3	1	1.0	0				14.66
K_5	0.01	10^3	10^3	1	1.0	0.05	5×10^1	5×10^3	0.01	14.66
K_{10}	0.01	10^3	10^3	1	1.0	0.1	10^2	10^4	0.01	14.66
K_{20}	0.01	10^3	10^3	1	1.0	0.2	2×10^2	2×10^4	0.01	14.66
L_0	0.1	5×10^5	10^4	50	11.7	0				103.71
L_5	0.1	5×10^5	10^4	50	11.7	0.05	2.5×10^4	5×10^4	0.5	103.71
L_{10}	0.1	5×10^5	10^4	50	11.7	0.1	5×10^4	10^5	0.5	103.71
L_{20}	0.1	5×10^5	10^4	50	11.7	0.2	10^5	2×10^5	0.5	103.71
M_0	0.1	10^5	10^4	10	4.7	0				46.38
M_5	0.1	10^5	10^4	10	4.7	0.05	5×10^3	5×10^4	0.1	46.38
M_{10}	0.1	10^5	10^4	10	4.7	0.1	10^4	10^5	0.1	46.38
M_{20}	0.1	10^5	10^4	10	4.7	0.2	2×10^4	2×10^5	0.1	46.38
N_0	0.1	10^4	10^4	1	1.0	0				14.67
N_5	0.1	10^4	10^4	1	1.0	0.05	5×10^2	5×10^4	0.01	14.67
N_{10}	0.1	10^4	10^4	1	1.0	0.1	10^3	10^5	0.01	14.67
N_{20}	0.1	10^4	10^4	1	1.0	0.2	2×10^3	2×10^5	0.01	14.67

Chapter 6

Conclusions

6.1. Interpretation of Gas-Induced Enhancements

6.2. Implications for SMBH Seed Formation

6.3. Limitations and Future Work

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Appendices

Appendix A. Derivation of Critical Masses for Collisions and Relaxation

In this section we derive two quantities relevant for the dynamical evolution of stellar clusters: the critical mass for collisions (following Escala (2021)) and the relaxation mass. For the critical mass, we first present the original derivation, then extend it to include an external potential (following Reinoso et al. (2020)), and finally generalize it by including gravitational focusing. For the relaxation mass, we derive an analogous mass scale from the two-body relaxation timescale.

A.1. Critical Mass in the Simplified Case

First, we show the derivation of the critical mass from Escala (2021). We begin with the collisional timescale (Equation (2.5)) from section 2.1, for a star cluster of equal stars with mass m and radius $r_\star$, total mass M and characteristic radius R:

$$t_{\rm coll} = \frac{1}{n\Sigma_0}\sqrt{\frac{R}{GM}}\,, \tag{A.1}$$

where $\Sigma_0 = 16\sqrt{\pi}r_\star^2(1+\Theta)$ is the effective cross-section, n is the number density, and

$$\Theta = 9.54\frac{m}{M_\odot}\frac{R_\odot}{r_\star}\left(\frac{100\ \text{km s}^{-1}}{\sigma}\right)^2, \tag{A.2}$$

is the Safronov number. This derivation relies on two assumptions. First, we assume the system is virialized, so the velocity dispersion σ can be expressed as a function of the total mass M and characteristic radius R. Second, we take $\sigma \sim 100$ km s^{-1}, then the Safronov number depends only on the stellar mass and radius. The implication of this assumption is that the effective cross-section depends only on the geometric cross-section of the stars, effectively neglecting gravitational focusing (i.e., the dependence on the velocity dispersion). Collisions become dynamically important when $t_{\rm coll} \leq t_{\rm H}$:

$$t_{\rm coll} = \frac{1}{n\Sigma_0}\sqrt{\frac{R}{GM}} \tag{A.3}$$

$$= \frac{4\pi R^3 m}{3M\Sigma_0}\sqrt{\frac{R}{GM}} \tag{A.4}$$

$$= \frac{4\pi R^{7/2} m}{3M^{3/2}\Sigma_0 G^{1/2}} \leq t_{\rm H} \ , \tag{A.5}$$

where $n = 3M/(4\pi R^3 m)$ is the number density for an homogeneous system. Solving for M:

$$M^{3/2} \geq \frac{4\pi R^{7/2} m}{3 t_{\rm H}\Sigma_0 G^{1/2}} \tag{A.6}$$

$$M \geq \left(\frac{4\pi m}{3\Sigma_0 G^{1/2} t_{\rm H}}\right)^{2/3} R^{7/3} \ . \tag{A.7}$$

The critical mass is then:

$$M_{\rm crit} = \left(\frac{4\pi m}{3\Sigma_0 G^{1/2} t_{\rm H}}\right)^{2/3} R^{7/3} \ . \tag{A.8}$$

This defines the mass limit for a virialized system of characteristic radius R, composed of equal-mass stars (mass m, radius $r_\star$), for which the collision time equals the characteristic time ($t_{\rm coll} = t_{\rm H}$). If a system exceeds this mass, its collision time is shorter than $t_{\rm H}$. This limit can be represented in a radius–mass diagram (Escala, 2021; Vergara et al., 2023).

A.2. Effect of an External Gravitational Potential

We now show how the critical mass changes when an external potential is considered. When no external potential is present, the virial theorem gives a relation for the time average kinetic and potential energies (K and W):

$$2K + W = 0 \ . \tag{A.9}$$

Although K and W are time-average quantities, in virial equilibrium they fluctuate about their mean values (for more details see Collins et al. (1978)). Then, we can express K and W in terms of characteristic quantities:

$$K = \frac{1}{2}\langle\sum_i m_i v_i^2\rangle = \frac{M}{2}\langle v^2\rangle \tag{A.10}$$

$$W = -\frac{1}{2}\langle\sum_{i,j\neq i}\frac{Gm_im_j}{|r_i - r_j|}\rangle = -\frac{GM^2}{2R_{\rm vir}}\ . \tag{A.11}$$

Note that m_i is the mass of ith star, and $M = \sum_i m_i$ is the total mass of the stellar system. For a stellar system, it is standard to replace $\langle v^2\rangle$ with the square of the velocity dispersion, σ^2. The quantity $R_{\rm vir}$ is known as the virial radius (for consistency, we use R instead of $R_{\rm vir}$ in the following). Replacing in Equation (A.9), we have:

$$\frac{M}{2}\sigma^2 = \frac{GM^2}{2R}\ . \tag{A.12}$$

The relation between σ and $R_{\rm vir}$ is then:

$$\sigma = \sqrt{\frac{GM}{R}}\ . \tag{A.13}$$

Following Reinoso et al. (2020), if we consider an external potential of mass $M_{\rm ext}$ with the same spatial distribution as the stellar system, then the total mass can be rewritten as $M + M_{\rm ext}$. Assuming the stars are in global virial equilibrium with this external potential yields:

$$\frac{M}{2}\sigma_{\rm ext}^2 = \frac{G(M+M_{\rm ext})^2}{2R}\ . \tag{A.14}$$

Defining $q = M_{\rm ext}/M$,

$$\sigma_{\rm ext} = \sqrt{\frac{GM}{R}}(1+q)\ . \tag{A.15}$$

Then, for an external potential, the collision time is:

$$
\begin{aligned}
t_{\rm coll} =& \frac{1}{n\Sigma_0}\sqrt{\frac{R}{GM}}\frac{1}{(1+q)} && \text{(A.16)}\\
=& \frac{4\pi R^3 m}{3M\Sigma_0}\sqrt{\frac{R}{GM}}\frac{1}{(1+q)} && \text{(A.17)}\\
=& \frac{4\pi R^{7/2} m}{3M^{3/2}\Sigma_0 G^{1/2}(1+q)} \ , && \text{(A.18)}
\end{aligned}
$$

and the critical mass can be written as

$$
M_{\rm crit,ext} = \left(\frac{4\pi m}{3\Sigma_0 G^{1/2} t_{\rm H}(1+q)}\right)^{2/3} R^{7/3} \ . \tag{A.19}
$$

A.3. Generalized Critical Mass Including Gravitational Focusing

In Section A.1, we assumed $\sigma \sim 100$ km s^{-1}. Consequently, the Safronov number depended only on the stellar mass and radius. To account for gravitational focusing, we now derive the critical mass using the virial velocity dispersion. In the following, we use M to denote the mass of stellar cluster. Then, the total mass of the system is $M(1+q)$ when considering an external potential (with $q = M_{\rm ext}/M$), and M when no external potential is present. First, we expand the effective cross section:

$$
\begin{aligned}
\Sigma_0 =& 16\sqrt{\pi} r_\star^2 (1+\Theta) && \text{(A.20)}\\
=& 16\sqrt{\pi} r_\star^2 \left(1 + 9.54 \frac{m}{M_\odot}\frac{R_\odot}{r_\star}\left(\frac{100 \text{ km s}^{-1}}{\sigma_{\rm ext}}\right)^2\right) && \text{(A.21)}\\
=& 16\sqrt{\pi} r_\star^2 + \frac{9.54 \times 16\sqrt{\pi}\ r_\star\ m R_\odot v_{100}^2}{M_\odot}\frac{R}{G\,M(1+q)^2} \ . && \text{(A.22)}
\end{aligned}
$$

We have introduced $\sigma_{\rm ext}$, and have defined $v_{100} = 100$ km s^{-1}. From $t_{\rm coll} = t_{\rm H}$ in Equation (A.18), we have

$$
\Sigma_0\, M^{3/2} = \frac{4\pi m}{3G^{1/2} t_H}\frac{R^{7/2}}{(1+q)} \ . \tag{A.23}
$$

Replacing Equation (A.22) into (A.23) yields,

$$16\sqrt{\pi}r_\star^2\, M^{3/2} + \frac{9.54\times 16\sqrt{\pi}\; r_\star\; mR_\odot v_{100}^2}{M_\odot}\frac{R\, M^{1/2}}{G(1+q)^2} = \frac{4\pi m}{3G^{1/2}t_H}\frac{R^{7/2}}{(1+q)} \tag{A.24}$$

$$M^{3/2} + \frac{9.54\; mR_\odot v_{100}^2 R}{r_\star G M_\odot (1+q)^2}\, M^{1/2} = \frac{\sqrt{\pi}m}{12G^{1/2}t_H r_\star^2}\frac{R^{7/2}}{(1+q)}\ . \tag{A.25}$$

Now, defining the coefficients

$$A = \frac{9.54\; mR_\odot v_{100}^2}{r_\star G M_\odot (1+q)^2} R \tag{A.26}$$

$$B = \frac{\sqrt{\pi}m}{12G^{1/2}t_H r_\star^2 (1+q)} R^{7/2}\ , \tag{A.27}$$

we can rewrite Equation (A.25):

$$M^{3/2} + AM^{1/2} = B\ . \tag{A.28}$$

To continue, we introduce the change of variable $u = M^{1/2}$:

$$u^3 + Au - B = 0\ . \tag{A.29}$$

Note that u must be positive since $M > 0$. As A and B are positive, the function $f(u) = u^3 + Au - B$ ranges from $-B$ to ∞. Then, $f(u)$ is strictly increasing, and there is exactly one real solution. Using Cardano's formula, the real solution for a depressed cubic equation is,

$$u_{\text{real}} = \sqrt[3]{\frac{B}{2} + \sqrt{\frac{B^2}{4} + \frac{A^3}{27}}} + \sqrt[3]{\frac{B}{2} - \sqrt{\frac{B^2}{4} + \frac{A^3}{27}}}\ . \tag{A.30}$$

Returning to $M = u^2$, we obtain

$$
\begin{aligned}
M &= u_{\rm real}^2 && \text{(A.31)}\\
&= \left(\frac{B}{2}+\sqrt{\frac{B^2}{4}+\frac{A^3}{27}}\right)^{2/3}+\left(\frac{B}{2}-\sqrt{\frac{B^2}{4}+\frac{A^3}{27}}\right)^{2/3} && \text{(A.32)}\\
&\quad +2\left(\frac{B^2}{4}-\left(\frac{B^2}{4}+\frac{A^3}{27}\right)\right)^{1/3}, && \text{(A.33)}\\
&= \left(\frac{B}{2}+\sqrt{\frac{B^2}{4}+\frac{A^3}{27}}\right)^{2/3}+\left(\frac{B}{2}-\sqrt{\frac{B^2}{4}+\frac{A^3}{27}}\right)^{2/3}-\frac{2A}{3}\,. && \text{(A.34)}
\end{aligned}
$$

Defining $\alpha = A/R$ and $\beta = B/R^{7/2}$, the critical mass is:

$$
M_{\rm crit} = \left(\frac{\beta R^{7/2}}{2}+\sqrt{\frac{\beta^2 R^7}{4}+\frac{\alpha^3 R^3}{27}}\right)^{2/3}+\left(\frac{\beta R^{7/2}}{2}-\sqrt{\frac{\beta^2 R^7}{4}+\frac{\alpha^3 R^3}{27}}\right)^{2/3}-\frac{2\alpha R}{3}\,. \tag{A.35}
$$

Equation (A.35) sets the upper mass limit for a stellar system of mass M, embedded in and in virial equilibrium with an external potential (with mass $M_{\rm ext} = qM$), such that its collision timescale satisfies $t_{\rm coll} = t_{\rm H}$. In consequence, if $M > M_{\rm crit}$, the collision time is shorter than $t_{\rm H}$.

A.4. Derivation of the Relaxation Mass

In this section, we derive a similar relation for the relaxation timescale $t_{\rm relax}$. In a stellar cluster, the energy of stars changes due to stellar encounters, which are typically divided into strong and weak encounters. If we consider a target star with velocity v, a strong encounter is considered, when after the interaction the change in velocity δv is comparable to v (i.e., $\delta v \sim v$). On the other hand, in a weak encounter, the change in velocity is small ($\delta v \ll v$). While strong encounters can produce notable effects (e.g., stellar collisions, binary formation) they are less frequent. Weak encounters, however, can accumulate over time and significantly alter the energy distribution and trajectories of stars. The relaxation timescale is the time required for a typical star's velocity to change significantly due to the cumulative effect of weak encounters. From Binney and Tremaine (2008), the deflection in the star velocity for crossing the system once is:

$$
\delta v^2 \simeq 8N\left(\frac{Gm}{Rv}\right)^2 \ln(\Lambda)\,, \tag{A.36}
$$

where N is the number of stars, m and v are the mass and velocity of the target star, and $\Lambda = R/b_{90}$ is the ratio of the maximum to minimum impact parameter for weak encounters. From Binney and Tremaine (2008), the minimum impact parameter is $b_{90} = 2Gm/v^2$, then $\Lambda = v^2R/(2Gm)$. The number of system crossings required to change v significantly is

$$n_{\rm cross} = \frac{v^2}{\delta v^2} = \frac{1}{8N\ln(\Lambda)}\left(\frac{Rv^2}{Gm}\right)^2 . \tag{A.37}$$

When an external potential is present, we can approximate $v^2 \sim \sigma_{\rm ext}^2$, and $n_{\rm cross}$ becomes

$$n_{\rm cross} = \frac{1}{8N\ln\left(\frac{GmN(1+q)^2}{R}\frac{R}{2Gm}\right)}\left(\frac{R}{Gm}\frac{GmN(1+q)^2}{R}\right)^2 \tag{A.38}$$

$$\simeq \frac{0.1N(1+q)^4}{\ln(N(1+q)^2)} . \tag{A.39}$$

In the last equation, we have used $M = Nm$, and we have neglected the $1/2$ inside $\ln(\Lambda)$. In the absence of an external potential (i.e., $q = 0$), Equation (A.39) reduces to the standard expression from Binney and Tremaine (2008). Since a typical star crosses the system in a crossing time $t_{\rm cross}$, the relaxation timescale is:

$$t_{\rm relax} = n_{\rm cross}t_{\rm cross} = \frac{0.1N(1+q)^4}{\ln(N(1+q)^2)}t_{\rm cross} . \tag{A.40}$$

With the crossing time estimated as $t_{\rm cross} = R/\sigma$, and using $\sigma_{\rm ext}$, Equation (A.40) becomes

$$t_{\rm relax} = \frac{0.1N(1+q)^3}{\ln(N(1+q)^2)}\frac{R^{3/2}}{\sqrt{GM}} . \tag{A.41}$$

Imposing $t_{\rm relax} < t_{\rm H}$ yields a relation between the radius R and mass M:

$$
\begin{aligned}
R^{3/2} \leq & \frac{t_{\mathrm{H}} \ln\left(N(1+q)^2\right)\sqrt{GM}}{0.1N(1+q)^3} && \text{(A.42)}\\
\leq & \frac{t_{\mathrm{H}} m \ln\left(M(1+q)^2/m\right)\sqrt{GM}}{0.1M(1+q)^3} && \text{(A.43)}\\
= & \frac{t_{\mathrm{H}} m}{0.1(1+q)^3} \ln\left(\frac{M(1+q)^2}{m}\right)\sqrt{\frac{G}{M}} && \text{(A.44)}\\
R \leq & \left(\frac{t_{\mathrm{H}} m}{0.1(1+q)^3} \ln\left(\frac{M(1+q)^2}{m}\right)\right)^{2/3} \left(\frac{G}{M}\right)^{1/3} . && \text{(A.45)}
\end{aligned}
$$

Equation (A.45) defines the condition for a system to significantly alter its initial conditions on a timescale $\sim t_{\mathrm{H}}$. If we denote the equality case by R_{relax}, systems with $R > R_{\mathrm{relax}}$ have a relaxation timescale longer than t_{H}. In a collisionless scenario, such systems are considered to be in virial equilibrium.

We now derive a mass scale from the condition in Equation (A.45). First, we rearrange the terms in (A.45) for the limit case:

$$
\begin{aligned}
R = & \left(\frac{t_{\mathrm{H}} m}{0.1(1+q)^3} \ln\left(\frac{M(1+q)^2}{m}\right)\right)^{2/3} \left(\frac{G}{M}\right)^{1/3} && \text{(A.46)}\\
R^3 M = & \left(\frac{t_{\mathrm{H}} m}{0.1(1+q)^3} \ln\left(\frac{M(1+q)^2}{m}\right)\right)^{2} G . && \text{(A.47)}
\end{aligned}
$$

We introduce the change of variable $u \equiv M(1+q)^2/m$,

$$
\begin{aligned}
\frac{R^3 u m}{(1+q)^2} = & \left(\frac{t_{\mathrm{H}} m}{0.1(1+q)^3} \ln(u)\right)^{2} G && \text{(A.48)}\\
\frac{R^{3/2} u^{1/2} m^{1/2}}{1+q} = & \frac{t_{\mathrm{H}} m}{0.1(1+q)^3} \ln(u) G^{1/2} && \text{(A.49)}\\
u^{1/2} = & \frac{t_{\mathrm{H}} \sqrt{Gm}}{0.1(1+q)^2 R^{3/2}} \ln(u) . && \text{(A.50)}
\end{aligned}
$$

Defining $\lambda = (t_{\mathrm{H}}\sqrt{Gm})/(0.1(1+q)^2 R^{3/2})$,

$$
u^{1/2} = \lambda \ln(u) , \tag{A.51}
$$

and introducing another change of variable, $\omega = \ln(u)$:

$$\begin{aligned} e^{\omega/2} =& \lambda \omega & \text{(A.52)} \\ \frac{1}{\lambda} =& \omega e^{-\omega/2} & \text{(A.53)} \\ \frac{-1}{2\lambda} =& \frac{-\omega}{2} e^{-\omega/2} \ . & \text{(A.54)} \end{aligned}$$

Equation (A.54) is solved by the Lambert W function (Weisstein, 2002). Indeed, for a function $f(w) = we^{w}$, its inverse is the Lambert $W(z)$ function, defined such that $W(f(w)) = w$. Hence,

$$\begin{aligned} \frac{-\omega}{2} =& W\left(\frac{-1}{2\lambda}\right) & \text{(A.55)} \\ \omega = -2W\left(\frac{-1}{2\lambda}\right) & \ . & \text{(A.56)} \end{aligned}$$

Returning to $u = e^{\omega}$,

$$u = \exp\left(-2W\left(\frac{-1}{2\lambda}\right)\right) \ . \tag{A.57}$$

Finally, we use $M = mu/(1+q)^2$:

$$M_{\text{relax}} = \frac{m}{(1+q)^2} \exp\left(-2W\left(\frac{-1}{2\lambda}\right)\right) \ . \tag{A.58}$$

Since $-1/2\lambda < 0$ ($\lambda > 0$), the argument of the Lambert W function is negative, and there are two real-valued branches (Higham et al., 2015, Chapter III.17). The principal branch, $W_0(z)$, is defined for $z \in [-1/e, \infty)$, and takes values $W_0(z) > -1$. In particular, for $z \in [-1/e, 0]$, it ranges from -1 to 0. The lower branch, $W_{-1}(z)$, is defined for $z \in [-1/e, 0]$, and ranges from -1 to $-\infty$, with $\lim_{z \to 0^-} W_{-1}(z) = -\infty$. Figure A.1 shows the principal and lower branches of the Lambert function.

If we consider the principal branch $W_0(z)$, we note that the exponential in Equation (A.58) takes values between e^0 and e^2. Consequently, the relaxation mass would be of the order of the individual stellar mass ($M_{\text{relax}} \sim m$). In contrast, for the lower branch $W_{-1}(z)$, the exponential ranges from e^2 to ∞, yielding a relaxation mass that can reach the mass scale of the system. We therefore adopt

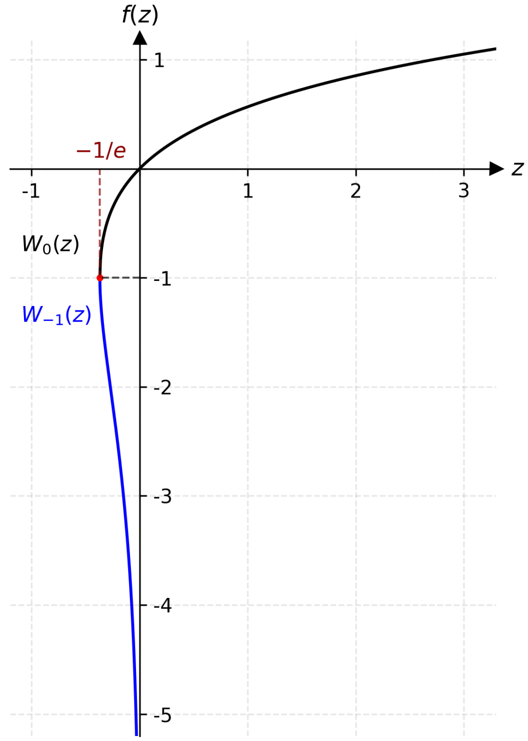

Figure A.1: Lambert function. The black line shows the $W_0(z)$ branch, and the blue line shows the $W_{-1}(z)$ branch.

the lower branch as the physically relevant solution:

$$M_{\text{relax}} = \frac{m}{(1+q)^2} \exp\left(-2W_{-1}\left(\frac{-1}{2\lambda}\right)\right) . \tag{A.59}$$

Finally, we must verify that the argument of the Lambert W function lies within the domain of the lower branch: $-1/e \leq -1/2\lambda < 0$. The right inequality is satisfied since $\lambda > 0$. The left part gives a condition over the characteristic timescale:

$$-1/e \leq -1/2\lambda \tag{A.60}$$

$$-2\lambda \leq -e \tag{A.61}$$

$$\lambda \geq e/2 \tag{A.62}$$

$$\frac{t_{\mathrm{H}}\sqrt{Gm}}{0.1(1+q)^2 R^{3/2}} \geq \frac{e}{2} \tag{A.63}$$

$$t_{\mathrm{H}} \geq \frac{0.1e(1+q)^2 R^{3/2}}{2\sqrt{Gm}} \tag{A.64}$$

Equation (A.64) implies that, for a system of characteristic radius R composed of equal-mass stars m, there exists a minimum possible relaxation time. From Equation (A.41), for fixed R and m, the number of stars that minimizes t_{relax} is $N_{\mathrm{opt}} = e^2/(1+q)^2$. If we evaluate $t_{\mathrm{relax}}(N_{\mathrm{opt}})$:

$$t_{\mathrm{relax}}(N_{\mathrm{opt}}) = \frac{0.1N_{\mathrm{opt}}(1+q)^3}{\ln\left(N_{\mathrm{opt}}(1+q)^2\right)} \frac{R^{3/2}}{\sqrt{GN_{\mathrm{opt}}m}} \tag{A.65}$$

$$= \frac{0.1e^2(1+q)^{-2}(1+q)^3}{\ln(e^2(1+q)^{-2}(1+q)^2)} \frac{R^{3/2}}{\sqrt{Ge^2(1+q)^{-2}m}} \tag{A.66}$$

$$= \frac{0.1e^2(1+q)}{2} \frac{R^{3/2}(1+q)}{\sqrt{Gme}} \tag{A.67}$$

$$= \frac{0.1e(1+q)^2 R^{3/2}}{2\sqrt{Gm}} \,. \tag{A.68}$$

Appendix B. Fourth-Order Hermite Integration Scheme

The N-body code used in this work, `ph4`, is based on a fourth-order Hermite integrator. The following subsections describe in detail the individual time step scheme (ITS) from Makino and Aarseth (1992), including the time step criterion in the N-body code.

B.1. Step 1: Particle selection

The first step is to select the next particle to be advanced. At any given time, each particle j has its own mass m_j, time t_j, time step Δt_j, coordinates $\mathbf{x}_j$ and $\mathbf{v}_j$, acceleration $\mathbf{a}_j$ and jerk $\dot{\mathbf{a}}_j$ (with $\dot{\mathbf{a}}_j = \partial \mathbf{a}_j/\partial t$). The selected particle i is the one satisfying the condition:

$$i = \arg\min_{j}(t_j + \Delta t_j) \tag{B.1}$$

Note that always $t_j \leq t_i + \Delta t_i$, for all j. The current simulation time is then set to $t = t_i + \Delta t_i$.

B.2. Step 2: Calculate the predicted position and velocities for all particles

The positions and velocities of all particles are predicted at time t, using a Taylor expansion up to third-order (jerk):

$$\mathbf{x}_{p,j} = \mathbf{x}_j + \mathbf{v}_j(t - t_j) + \frac{\mathbf{a}_{0,j}}{2!}(t - t_j)^2 + \frac{\dot{\mathbf{a}}_{0,j}}{3!}(t - t_j)^3 \tag{B.2}$$

$$\mathbf{v}_{p,j} = \mathbf{v}_j + \mathbf{a}_{0,j}(t - t_j) + \frac{\dot{\mathbf{a}}_{0,j}}{2!}(t - t_j)^2 \tag{B.3}$$

The acceleration is due only to gravitational force. The expression for each particle j is:

$$\mathbf{a}_{0,j} = \sum_{k \neq j} G m_k \frac{\mathbf{r}_{0,jk}}{(\mathbf{r}_{0,jk}^2 + \varepsilon^2)^{3/2}} \tag{B.4}$$

where ε is the softening parameter, and

$$\mathbf{r}_{0,jk} = \mathbf{r}_k - \mathbf{r}_j \tag{B.5}$$

Differentiating Eq. (B.4) with respect to time yields the jerk:

$$\dot{\mathbf{a}}_{0,j} = \frac{\partial \mathbf{a}_j}{\partial t} \tag{B.6}$$

$$= \sum_{k \neq j} G m_k \frac{\partial}{\partial t} \frac{\mathbf{r}_{0,jk}}{(\mathbf{r}_{0,jk}^2 + \varepsilon^2)^{3/2}} \tag{B.7}$$

$$\dot{\mathbf{a}}_{0,j} = \sum_{k \neq j} G m_k \left[\frac{\mathbf{v}_{0,jk}}{(\mathbf{r}_{0,jk}^2 + \varepsilon^2)^{3/2}} - \frac{3(\mathbf{v}_{0,jk} \cdot \mathbf{r}_{0,jk})\mathbf{r}_{0,jk}}{(\mathbf{r}_{0,jk}^2 + \varepsilon^2)^{5/2}} \right] \tag{B.8}$$

where

$$\mathbf{v}_{0,jk} = \mathbf{v}_k - \mathbf{v}_j \tag{B.9}$$

B.3. Step 3: Calculate acceleration and jerk for particle *i*

Similar to equations (B.4) and (B.8), the acceleration and jerk are calculated using the predicted positions and velocity:

$$\mathbf{a}_{1,i} = \sum_{j\neq i} Gm_j \frac{\mathbf{r}_{ij}}{(\mathbf{r}_{ij}^2+\varepsilon^2)^{3/2}} \tag{B.10}$$

$$\dot{\mathbf{a}}_{1,i} = \sum_{j\neq i} Gm_j \left[\frac{\mathbf{v}_{ij}}{(\mathbf{r}_{ij}^2+\varepsilon^2)^{3/2}} - \frac{3(\mathbf{v}_{ij}\cdot\mathbf{r}_{ij})\mathbf{r}_{ij}}{(\mathbf{r}_{ij}^2+\varepsilon^2)^{5/2}}\right] \tag{B.11}$$

$$\mathbf{r}_{ij} = \mathbf{r}_{p,j} - \mathbf{r}_{p,i} \tag{B.12}$$

$$\mathbf{v}_{ij} = \mathbf{v}_{p,j} - \mathbf{v}_{p,i} \tag{B.13}$$

In this notation, the subscript 0 denotes quantities evaluated at the beginning of the time step (t_j), while the subscript 1 denotes quantities evaluated at the end of the time step (t), using the predicted position and velocity ($\mathbf{x}_{p,i}$ and $\mathbf{v}_{p_i}$).

B.4. Step 4: Second and third time derivative of acceleration, using Hermite interpolation

To obtain a high-order correction of **x** and **v**, the acceleration of particle *i* is modeled as a third-order polynomial in time:

$$\mathbf{a}(\tau) = \alpha\tau^3 + \beta\tau^2 + \gamma\tau + \delta \tag{B.14}$$

The successive time derivatives of $\mathbf{a}(\tau)$ are then:

$$\dot{\mathbf{a}}(\tau) = 3\alpha\tau^2 + 2\beta\tau + \gamma \tag{B.15}$$

$$\mathbf{a}^{(2)}(\tau) = 6\alpha\tau + 2\beta \tag{B.16}$$

$$\mathbf{a}^{(3)}(\tau) = 6\alpha \tag{B.17}$$

The coefficients α, β, γ, and δ are determined by using the known values of acceleration and jerk at times t_i and t. This provides a system of four equations:

$$\mathbf{a}(t_i) = \mathbf{a}_{0,i} = \alpha t_i^3 + \beta t_i^2 + \gamma t_i + \delta \tag{B.18}$$

$$\mathbf{a}(t) = \mathbf{a}_{1,i} = \alpha t^3 + \beta t^2 + \gamma t + \delta \tag{B.19}$$

$$\dot{\mathbf{a}}(t_i) = \dot{\mathbf{a}}_{0,i} = 3\alpha t_i^2 + 2\beta t_i + \gamma \tag{B.20}$$

$$\dot{\mathbf{a}}(t) = \dot{\mathbf{a}}_{1,i} = 3\alpha t^2 + 2\beta t + \gamma \tag{B.21}$$

where $t = t_i + \Delta t_i$ is the current time. As the system of equations is linearly independent, the four coefficients can be determined. However, determining α and β is enough to get the second and third time derivative of acceleration. From (B.21):

$$\gamma = \dot{\mathbf{a}}_{1,i} - 3\alpha t^2 - 2\beta t \tag{B.22}$$

Subtracting Eq. (B.20) from (B.21):

$$\beta = \frac{(\dot{\mathbf{a}}_{1,i} - \dot{\mathbf{a}}_{0,i}) - \alpha 3(t^2 - t_i^2)}{2(t - t_i)} \tag{B.23}$$

Subtracting Eq. (B.18) from (B.19), and using results from (B.22) and (B.23) gives an expression for α:

$$\alpha = \frac{(\dot{\mathbf{a}}_{1,i} + \dot{\mathbf{a}}_{0,i})(t - t_i) - 2(\mathbf{a}_{1,i} - \mathbf{a}_{0,i})}{(t - t_i)^3} \tag{B.24}$$

Replacing this result in Eq. (B.23):

$$\beta = \frac{-\dot{\mathbf{a}}_{1,i}(t + 2t_i) - \dot{\mathbf{a}}_{0,i}(2t + t_i)}{(t - t_i)^2} + \frac{3(\mathbf{a}_{1,i} - \mathbf{a}_{0,i})(t + t_i)}{(t - t_i)^3} \tag{B.25}$$

The second and third time derivatives of the acceleration at time t_i can now be evaluated. For the Eq. (B.17), and using $t - t_i = \Delta t_i$:

$$\mathbf{a}_{0,i}^{(3)} = \mathbf{a}^{(3)}(t_i) = 6\alpha \tag{B.26}$$

$$\mathbf{a}_{0,i}^{(3)} = 6\frac{(\dot{\mathbf{a}}_{1,i} + \dot{\mathbf{a}}_{0,i})\Delta t_i - 2(\mathbf{a}_{1,i} - \mathbf{a}_{0,i})}{\Delta t_i{}^3} \tag{B.27}$$

$$\mathbf{a}_{0,i}^{(3)} = \frac{12(\mathbf{a}_{0,i} - \mathbf{a}_{1,i}) + 6(\dot{\mathbf{a}}_{0,i} + \dot{\mathbf{a}}_{1,i})\Delta t_i}{\Delta t_i{}^3} \tag{B.28}$$

Evaluating the second time derivative in t_i, and replacing the obtained for α and β:

$$\mathbf{a}_{0,i}^{(2)} = \mathbf{a}^{(2)}(t_i) = 6\alpha t_i + \beta \tag{B.29}$$

$$\mathbf{a}_{0,i}^{(2)} = \frac{-6(\mathbf{a}_{0,i} - \mathbf{a}_{1,i}) - \Delta t_i(4\dot{\mathbf{a}}_{0,i} + 2\dot{\mathbf{a}}_{1,i})}{\Delta t_i{}^2} \tag{B.30}$$

B.5. Step 5: Add corrections

The next step in the scheme is to correct position and velocity of particle i at time t. Using a higher-order Taylor expansion around t_i:

$$\mathbf{x}_i(t_i + \Delta t_i) = \mathbf{x}_i(t_i) + \dot{\mathbf{x}}_i(t_i)\Delta t_i + \frac{\mathbf{x}_i{}^{(2)}(t_i)\Delta t_i{}^2}{2!} + \ldots + \frac{\mathbf{x}_i{}^{(5)}(t_i)\Delta t_i{}^5}{5!} \tag{B.31}$$

$$= \mathbf{x}_i + \mathbf{v}_i\Delta t_i + \frac{\mathbf{a}_{0,i}\Delta t_i{}^2}{2} + \frac{\dot{\mathbf{a}}_{0,i}\Delta t_i{}^3}{6} + \frac{\mathbf{a}_{0,i}^{(2)}\Delta t_i{}^4}{24} + \frac{\mathbf{a}_{0,i}^{(3)}\Delta t_i{}^5}{120} \tag{B.32}$$

The first four terms of this expansion correspond to the predicted position $\mathbf{x}_{p,i}$ from Eq. (B.2). Therefore, the corrected position can be written as:

$$\mathbf{x}_i(t_i + \Delta t_i) = \mathbf{x}_{p,i} + \frac{\mathbf{a}_{0,i}^{(2)}\Delta t_i{}^4}{24} + \frac{\mathbf{a}_{0,i}^{(3)}\Delta t_i{}^5}{120} \tag{B.33}$$

In a similar way, the corrected velocity is:

$$\mathbf{v}_i(t_i + \Delta t_i) = \mathbf{v}_{p,i} + \frac{\mathbf{a}_{0,i}^{(2)}\Delta t_i{}^3}{6} + \frac{\mathbf{a}_{0,i}^{(3)}\Delta t_i{}^4}{24} \tag{B.34}$$

The advantage of this scheme is that achieving high integration accuracy only requires the explicit calculation of the acceleration and jerk for all particles, and the predicted acceleration and jerk for the particle being advanced. This results in an integration scheme that is fourth-order accurate in time ($\mathscr{O}(a^{(3)})$).

B.6. Step 6: Time step update

After updating the position and velocity of particle i, a new time step Δt_i must be calculated. This step is crucial for controlling the integration error. A proven and stable criterion is given by the standard formula Aarseth (1985):

$$\Delta t_i = \sqrt{\eta \frac{|\mathbf{a}_{1,i}||\mathbf{a}^{(2)}_{1,i}| + |\dot{\mathbf{a}}_{1,i}|^2}{|\dot{\mathbf{a}}_{1,i}||\mathbf{a}^{(3)}_{1,i}| + |\mathbf{a}^{(2)}_{1,i}|^2}} \tag{B.35}$$

The values for $\mathbf{a}_{1,i}$ and $\dot{\mathbf{a}}_{1,i}$ are known from the direct calculation in Step 3. The value of $\mathbf{a}^{(3)}(t)$ is constant, since a third-order polynomial interpolation is used. Only $\mathbf{a}^{(2)}_{1,i}$ should be calculated:

$$\mathbf{a}^{(2)}_{1,i} = \mathbf{a}^{(2)}_{0,i} + \Delta t_i \mathbf{a}^{(3)}_{0,i} \tag{B.36}$$

During the initialization of the algorithm, the higher-order derivatives of the acceleration are not available, so an alternative formula for time step can be used in the startup:

$$\Delta t = \eta_s \frac{|\mathbf{a}|}{|\dot{\mathbf{a}}|} \tag{B.37}$$

The suggested value for the startup parameter is $\eta_s \sim 0.01$ Makino and Aarseth (1992).

Appendix C. SPH equations

In this section, we derive the hydrodynamic equations for the SPH method. We follow the derivation of Springel and Hernquist (2002). From a particle perspective, the idea of SPH is to use a discrete representation of the Lagrangian (Rasio, 2000):

$$L_{\mathrm{SPH}} = \sum_{i=1}^{N} m_i \left(\frac{1}{2} \mathbf{v}_i^2 - u_i \right) \ . \tag{C.1}$$

The sum is over all the particles (or fluid elements), described by a mass m_i, velocity $\mathbf{v}_i = d\mathbf{r}_i/dt$ and position $\mathbf{r}_i$. For Equation C.1 to be the Lagrangian of a fluid, it is necessary to have an adequate prescription to compute the density ρ_i of any particle. In SPH, the density is calculated as

$$\rho_i = \rho(\mathbf{r}_i) = \sum_j m_j W(|\mathbf{r}_i - \mathbf{r}_j|; h_i) \ , \tag{C.2}$$

where W is the smoothing kernel, and h_i is the smoothing length parameter. In particular, the one used in FI is the cubic spline kernel (Monaghan, 1992; Pelupessy, 2005):

$$W(r,h) = \frac{1}{\pi h^3} \begin{cases} 1 - \frac{3}{2}\left(\frac{r}{h}\right)^2 + \frac{3}{4}\left(\frac{r}{h}\right)^3, & 0 \leq \frac{r}{h} \leq 1, \\ \frac{1}{4}\left(2 - \frac{r}{h}\right)^3, & 1 \leq \frac{r}{h} \leq 2, \\ 0, & \text{otherwise} \ . \end{cases} \tag{C.3}$$

In some derivations, h_i is taken as a constant. However, since different particles needs different resolutions, modern codes use a variable h_i. Requiring that a smoothing volume contains a fixed mass leads to $M_{\mathrm{SPH}} = (4\pi/3)h_i^3\rho_i$, where $M_{\mathrm{SPH}} = \bar{m}N_{\mathrm{SPH}}$, $\bar{m}$ the mean mass among SPH particles and N_{SPH} is a fixed number of particles neighbors. The condition over h_i gives the restriction:

$$\phi_i = \frac{4\pi}{3} h_i^3 (\rho_i + \rho_0) - M_{\mathrm{SPH}} = 0 \ . \tag{C.4}$$

Here, a fixed density ρ_0 is introduced to prevent h_i from diverging in low-density regions. The restrictions can be introduced on the Lagrangian via Lagrange multipliers λ_i. Then the equations of motion yields

$$\frac{d}{dt}\frac{\partial L_{\mathrm{SPH}}}{\partial \dot{q}_i} - \frac{\partial L_{\mathrm{SPH}}}{\partial q_i} = \sum_{j=1}^{N} \lambda_j \frac{\partial \phi_j}{\partial q_i} \ , \tag{C.5}$$

where q_i is replaced by the coordinate, and $\dot{q}_i = dq_i/dt$. The advantage of this approach is that by treating h_i as independent variables in the Lagrangian, the ∇h_iterms are consistently included in the equations of motion.

C.1. Solving λ_i

If we consider a polytropic equation of state, the pressure yields

$$P = A\rho(\mathbf{r})^{\gamma} \ , \tag{C.6}$$

where γ is the polytropic index, and A is the entropic function (see Horedt (2004) for a complete derivation). For SPH formalism, the pressure for particle is

$$P_i = A_i \rho_i^{\gamma} \ . \tag{C.7}$$

This is the entropic equation of state. Since we are interested in energy conservation, we replace A_i in terms of the specific internal energy $u_i = A_i \rho_i^{\gamma-1}/(\gamma-1)$. Then, the pressure is

$$P_i = u_i \rho_i (\gamma - 1) \ . \tag{C.8}$$

First, we solve the Lagrangian equations (Equation (C.5)), for the variable h_i. As there is no explicit dependency on $\dot{h}_i$, then $\partial L_{\mathrm{SPH}}/\partial \dot{h}_i = 0$. Then, for $q_i = h_i$ Equation (C.5) becomes:

$$-\frac{\partial}{\partial h_i}\sum_{j=1}^{N} m_j \left(\frac{1}{2}|\dot{\mathbf{r}}_j|^2 - u_j\right) = \sum_{k=1}^{N} \lambda_k \frac{\partial \phi_k}{\partial h_i} \tag{C.9}$$

$$-\sum_{j=1}^{N} m_j \left(\dot{\mathbf{r}}_j \frac{\partial \dot{\mathbf{r}}_j}{\partial h_i} - \frac{\partial u_j}{\partial h_i}\right) = \sum_{k=1}^{N} \lambda_k \frac{\partial}{\partial h_i}\left(\frac{4\pi}{3} h_k^3 (\rho_k + \rho_0) - M_{\mathrm{SPH}}\right) \tag{C.10}$$

$$\sum_{j=1}^{N} m_j \frac{\partial u_j}{\partial \rho_j}\frac{\partial \rho_j}{\partial h_j}\frac{\partial h_j}{\partial h_i} = \sum_{k=1}^{N} \lambda_k \frac{4\pi}{3}\left(3h_k^2(\rho_k + \rho_0) + h_k^3 \frac{\partial \rho_k}{\partial h_k}\right)\frac{\partial h_k}{\partial h_i} \ . \tag{C.11}$$

Now, we use that $\partial u/\partial \rho = P/\rho^2$, and applying $\partial h_l/\partial h_m = \delta_{lm}$ in the summations:

$$\sum_{j=1}^{N} m_j \frac{P_j}{\rho_j^2}\frac{\partial \rho_j}{\partial h_j}\delta_{ji} = \sum_{k=1}^{N} \lambda_k \frac{4\pi}{3}\left(3h_k^2(\rho_k + \rho_0) + h_k^3 \frac{\partial \rho_k}{\partial h_k}\right)\delta_{ki} \tag{C.12}$$

$$m_i \frac{P_i}{\rho_i^2}\frac{\partial \rho_i}{\partial h_i} = \lambda_i \frac{4\pi}{3}\left(3h_i^2(\rho_i + \rho_0) + h_i^3 \frac{\partial \rho_i}{\partial h_i}\right) \tag{C.13}$$

$$= \lambda_i \frac{4\pi h_i^3}{3}\frac{\partial \rho_i}{\partial h_i}\left(1 + \frac{3(\rho_i + \rho_0)}{h_i}\left[\frac{\partial \rho_i}{\partial h_i}\right]^{-1}\right) \ . \tag{C.14}$$

Then, the Lagrangian multipliers yields

$$\lambda_i = \frac{3}{4\pi}\frac{m_i}{h_i^3}\frac{P_i}{\rho_i^2}\left(1+\frac{3(\rho_i+\rho_0)}{h_i}\left[\frac{\partial\rho_i}{\partial h_i}\right]^{-1}\right)^{-1} . \tag{C.15}$$

C.2. Equation of motion for SPH particles

We now derive the equations of motion by applying Equation (C.5) to the spatial coordinates $q_i = \mathbf{r}_i$:

$$\frac{d}{dt}\frac{\partial L_{\mathrm{SPH}}}{\partial \dot{\mathbf{r}}_i} - \frac{\partial L_{\mathrm{SPH}}}{\partial \mathbf{r}_i} = \sum_{j=1}^{N}\lambda_j\frac{\partial \phi_j}{\partial \mathbf{r}_i} . \tag{C.16}$$

The first term in the left can be obtained directly:

$$\frac{d}{dt}\frac{\partial L_{\mathrm{SPH}}}{\partial \dot{\mathbf{r}}_i} = \frac{d}{dt}\sum_{j=1}^{N} m_j\mathbf{v}_j\frac{\partial \mathbf{v}_j}{\partial \mathbf{v}_i} \tag{C.17}$$

$$= m_i\frac{d\mathbf{v}_i}{dt} . \tag{C.18}$$

The second term:

$$\frac{\partial L_{\mathrm{SPH}}}{\partial \mathbf{r}_i} = \sum_{j=1}^{N} m_j\left(\dot{\mathbf{r}}_j\frac{\partial \dot{\mathbf{r}}_j}{\partial \mathbf{r}_j} - \frac{\partial u_j}{\partial \mathbf{r}_i}\right) \tag{C.19}$$

$$= -\sum_{j=1}^{N} m_j\frac{\partial u_j}{\partial \rho_j}\nabla_i\rho_j \tag{C.20}$$

$$= \sum_{j=1}^{N} m_j\frac{P_j}{\rho_j^2}\nabla_i\rho_j , \tag{C.21}$$

where $\nabla_i = \partial/\partial\mathbf{r}_i$. For the right side of Equation (C.16):

$$\sum_{j=1}^{N} \lambda_j \frac{\partial \phi_j}{\partial \mathbf{r}_i} = \sum_{j=1}^{N} \lambda_j \frac{\partial}{\partial \mathbf{r}_i} \left(\frac{4\pi}{3} h_j^3 (\rho_j + \rho_0) - M_{\mathrm{SPH}} \right) \tag{C.22}$$

$$= \sum_{j=1}^{N} \lambda_j \frac{4\pi}{3} h_j^3 \nabla_i \rho_j \ . \tag{C.23}$$

Replacing all the terms back to Equation (C.16),

$$m_i \frac{d\mathbf{v}_i}{dt} - \sum_{j=1}^{N} m_j \frac{P_j}{\rho_j^2} \nabla_i \rho_j = \sum_{j=1}^{N} \lambda_j \frac{4\pi}{3} h_j^3 \nabla_i \rho_j \tag{C.24}$$

$$m_i \frac{d\mathbf{v}_i}{dt} = \sum_{j=1}^{N} m_j \frac{P_j}{\rho_j^2} \left[1 + \lambda_j \frac{4\pi}{3} \frac{h_j^3}{m_j} \frac{\rho_j^2}{P_j} \right] \nabla_i \rho_j \ . \tag{C.25}$$

Expanding the term with λ_i,

$$\lambda_j \frac{4\pi}{3} \frac{h_j^3}{m_j} \frac{\rho_j^2}{P_j} = \left(1 + \frac{3(\rho_j + \rho_0)}{h_j} \left[\frac{\partial \rho_j}{\partial h_j} \right]^{-1} \right)^{-1} \tag{C.26}$$

$$= \left(1 + \left[\frac{h_j}{3(\rho_j + \rho_0)} \frac{\partial \rho_j}{\partial h_j} \right]^{-1} \right)^{-1} \ . \tag{C.27}$$

Defining

$$f_j = \left(1 + \frac{h_j}{3(\rho_j + \rho_0)} \frac{\partial \rho_j}{\partial h_j} \right)^{-1} \ , \tag{C.28}$$

then the term involving λ_j simplifies to a compact expression, as shown below:

$$\lambda_j \frac{4\pi}{3} \frac{h_j^3}{m_j} \frac{\rho_j^2}{P_j} = \left(1 + (f_j^{-1} - 1)^{-1} \right)^{-1} \tag{C.29}$$

$$= \left(\frac{f_j^{-1}}{f_j^{-1} - 1} \right)^{-1} = 1 - f_j \ . \tag{C.30}$$

Replacing back to Equation (C.25),

$$m_i \frac{d\mathbf{v}_i}{dt} = -\sum_{j=1}^{N} m_j \frac{P_j}{\rho_j^2} f_j \nabla_i \rho_j \ . \tag{C.31}$$

Using the notation $W_{ij}(h) = W(|\mathbf{r}_i - \mathbf{r}_j|, h)$, then we can write (Springel and Hernquist, 2002)

$$\nabla_i \rho_j = m_i \nabla_i W_{ij}(h_j) + \delta_{ij} \sum_{k=1}^{N} m_k \nabla_i W_{ki}(h_i) \ . \tag{C.32}$$

Back into Equation (C.30),

$$m_i \frac{d\mathbf{v}_i}{dt} = -\sum_{j=1}^{N} m_j \frac{P_j}{\rho_j^2} f_j \left(m_i \nabla_i W_{ij}(h_j) + \delta_{ij} \sum_{k=1}^{N} m_k \nabla_i W_{ki}(h_i) \right) \tag{C.33}$$

$$= -\sum_{j=1}^{N} m_j m_i \frac{P_j}{\rho_j^2} f_j \nabla_i W_{ij}(h_j) - m_i \frac{P_i}{\rho_i^2} f_i \sum_{k=1}^{N} m_k \nabla_i W_{ki}(h_i) \tag{C.34}$$

$$\frac{d\mathbf{v}_i}{dt} = -\sum_{j=1}^{N} m_j \frac{P_j}{\rho_j^2} f_j \nabla_i W_{ij}(h_j) - \frac{P_i}{\rho_i^2} f_i \sum_{k=1}^{N} m_k \nabla_i W_{ki}(h_i) \tag{C.35}$$

Renaming index k to j in the second summation, and using the symmetry of the kernel, which implies $\nabla_i W_{ki}(h_i) = \nabla_i W_{ik}(h_i)$, we obtain the final form of the equation of motion:

$$\frac{d\mathbf{v}_i}{dt} = -\sum_{j=1}^{N} m_j \left[\frac{P_j}{\rho_j^2} f_j \nabla_i W_{ij}(h_j) + \frac{P_i}{\rho_i^2} f_i \nabla_i W_{ij}(h_i) \right] \ . \tag{C.36}$$

C.3. Equation of energy for SPH particles

We now derive the evolution equation for the specific internal energy. The total energy of the SPH particles is given by:

$$E = \sum_{i=1}^{N} m_i u_i + \sum_{j=1}^{N} m_j \frac{\mathbf{v}_j^2}{2} \ , \tag{C.37}$$

and under the condition of energy conservation:

$$\frac{dE}{dt} = 0 \tag{C.38}$$

$$= \sum_{i=1}^{N} m_i \frac{du_i}{dt} + \sum_{j=1}^{N} m_j \mathbf{v}_j \frac{d\mathbf{v}_j}{dt} \ . \tag{C.39}$$

Now, we use Equation (C.36) on the right side:

$$\sum_{i=1}^{N} m_i \frac{du_i}{dt} = -\sum_{j=1}^{N} m_j \mathbf{v}_j \left(-\sum_{k=1}^{N} m_k \left[\frac{P_k}{\rho_k^2} f_k \nabla_j W_{jk}(h_k) + \frac{P_j}{\rho_j^2} f_j \nabla_j W_{jk}(h_j) \right] \right) \tag{C.40}$$

$$= \sum_{j,k} m_j m_k \mathbf{v}_j \frac{P_k}{\rho_k^2} f_k \nabla_j W_{jk}(h_k) + \sum_{j,k} m_j m_k \mathbf{v}_j \frac{P_j}{\rho_j^2} f_j \nabla_j W_{jk}(h_j) \ . \tag{C.41}$$

Since j and k are dummy indices, we can relabel them to symmetrize the expression. In the first sum we set $j \to i, k \to j$, and in the second sum we set $k \to i$, yielding

$$\sum_{i=1}^{N} m_i \frac{du_i}{dt} = \sum_{i,j} m_i m_j \mathbf{v}_i \frac{P_j}{\rho_j^2} f_j \nabla_i W_{ij}(h_j) + \sum_{i,j} m_j m_i \mathbf{v}_j \frac{P_j}{\rho_j^2} f_j \nabla_j W_{ji}(h_j) \tag{C.42}$$

$$= \sum_{i,j} m_i m_j \mathbf{v}_i \frac{P_j}{\rho_j^2} f_j \nabla_i W_{ij}(h_j) + \sum_{i,j} m_j m_i \mathbf{v}_j \frac{P_j}{\rho_j^2} f_j (-\nabla_i W_{ij}(h_j)) \tag{C.43}$$

$$= \sum_{i,j} m_i m_j f_j \frac{P_j}{\rho_j^2} \mathbf{v}_{ij} \cdot \nabla_i W_{ij}(h_j) \ , \tag{C.44}$$

where we have used $\nabla_j W_{ji}(h_j) = -\nabla_i W_{ij}(h_j)$ and $\mathbf{v}_{ij} = \mathbf{v}_i - \mathbf{v}_j$. Since i, j are also dummy indices, we can exchange them, having another solution for the equation of energy:

$$\sum_{i=1}^{N} m_i \frac{du_i}{dt} = \sum_{i,j} m_i m_j f_j \frac{P_j}{\rho_j^2} \mathbf{v}_{ij} \cdot \nabla_i W_{ij}(h_j) \tag{C.45}$$

$$= \sum_{i,j} m_i m_j f_i \frac{P_i}{\rho_i^2} \mathbf{v}_{ji} \cdot \nabla_j W_{ji}(h_i) \tag{C.46}$$

$$= \sum_{i,j} m_i m_j f_i \frac{P_i}{\rho_i^2} \mathbf{v}_{ij} \cdot \nabla_i W_{ij}(h_i) \ . \tag{C.47}$$

Taking the average of Equations (C.44) and (C.47) yields the symmetrized form:

$$\sum_{i=1}^{N} m_i \frac{du_i}{dt} = \sum_{i,j} m_i m_j \frac{1}{2} \left[f_j \frac{P_j}{\rho_j^2} \mathbf{v}_{ij} \cdot \nabla_i W_{ij}(h_j) + f_i \frac{P_i}{\rho_i^2} \mathbf{v}_{ij} \cdot \nabla_i W_{ij}(h_i) \right] , \tag{C.48}$$

and the solution for the internal energy:

$$\frac{du_i}{dt} = \frac{1}{2} \sum_{j=1}^{N} m_j \left[f_j \frac{P_j}{\rho_j^2} \mathbf{v}_{ij} \cdot \nabla_i W_{ij}(h_j) + f_i \frac{P_i}{\rho_i^2} \mathbf{v}_{ij} \cdot \nabla_i W_{ij}(h_i) \right] . \tag{C.49}$$